

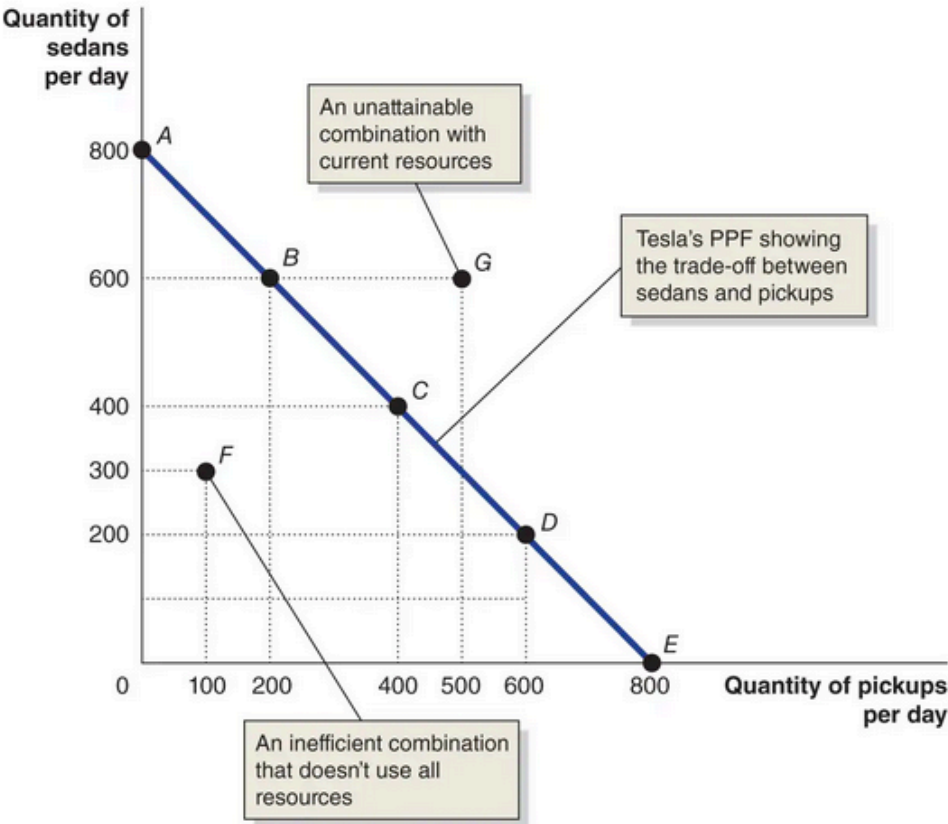
## All images and equations

Figure 2.1

Tesla’s Production Possibilities Frontier

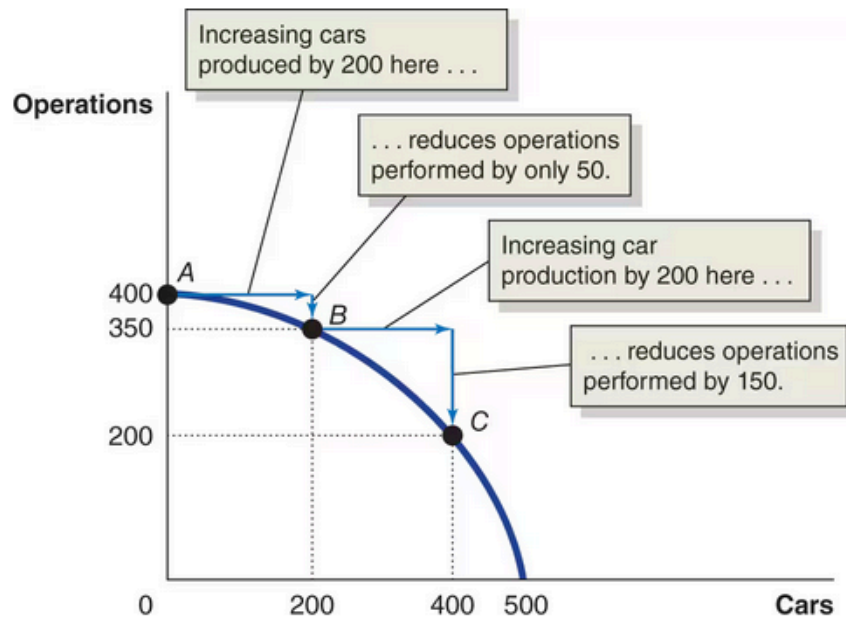


| Tesla’s Production Possibilities at Its Plant |                                     |                                         |
|-----------------------------------------------|-------------------------------------|-----------------------------------------|
| Choice                                        | Quantity of Model 3 Sedans Produced | Quantity of Cybertruck Pickups Produced |
| A                                             | 800                                 | 0                                       |
| B                                             | 600                                 | 200                                     |
| C                                             | 400                                 | 400                                     |
| D                                             | 200                                 | 600                                     |
| E                                             | 0                                   | 800                                     |
| F                                             | 300                                 | 100                                     |
| G                                             | 600                                 | 500                                     |



## Figure 2.2

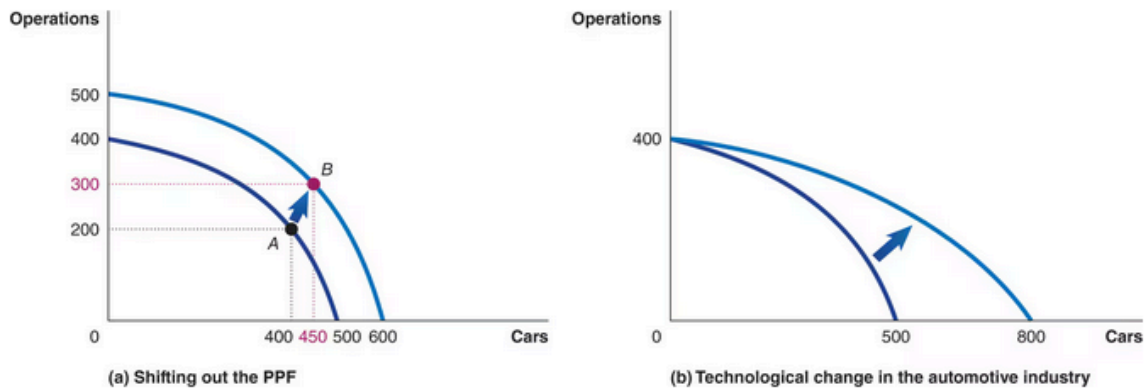
### Increasing Marginal Opportunity Costs



As the economy moves down the PPF, it experiences *increasing marginal opportunity costs* because increasing car production by a given quantity requires larger and larger decreases in the number of operations performed. For example, to increase car production from 0 to 200—moving from point A to point B—the economy has to give up only 50 operations. But to increase production of cars by another 200 vehicles—moving from point B to point C—the economy has to give up 150 operations.

## Figure 2.3

### Economic Growth

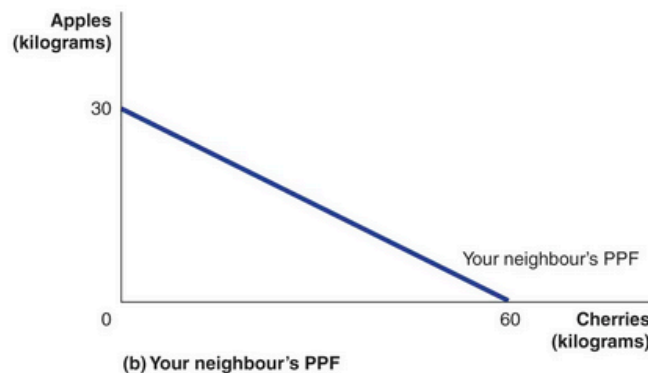
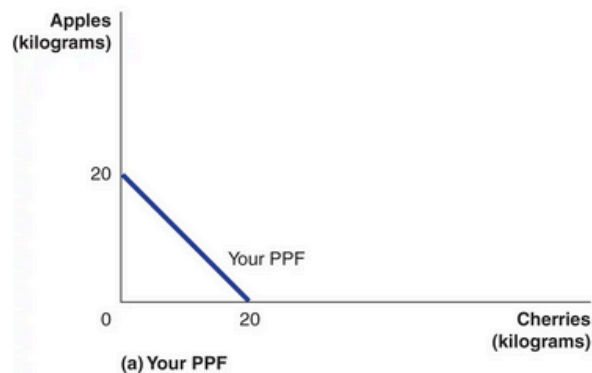


Panel (a) shows that as more economic resources become available and technological change occurs, the economy can move from point A to point B, performing more operations and producing more cars. Panel (b) shows the results of technological change in the automobile industry that increases the quantity of cars workers can produce per year while leaving unchanged the maximum quantity of operations that can be performed. Shifts in the PPF represent *economic growth*.

**Figure 2.4**

Production Possibilities for You and Your Neighbour, without Trade

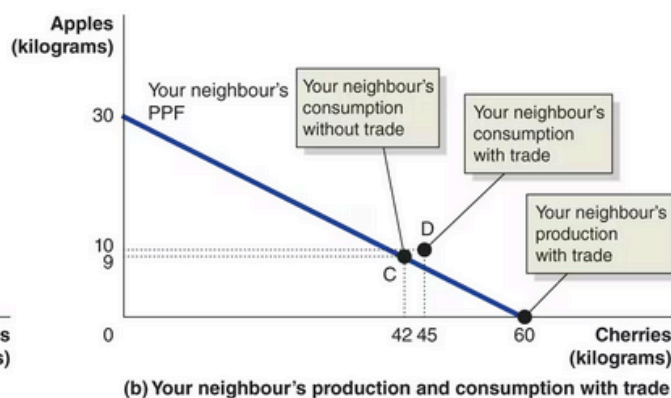
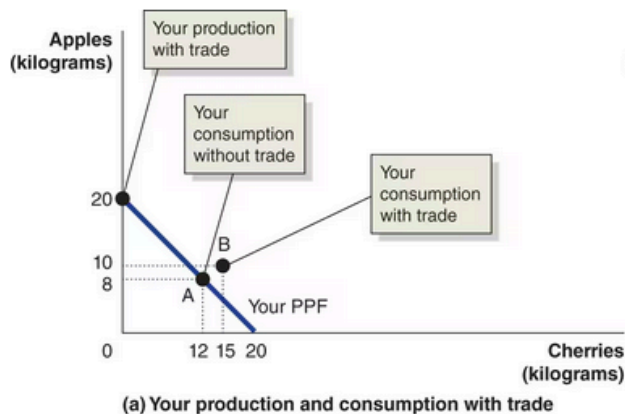
|                           | You         |               | Your Neighbour |               |
|---------------------------|-------------|---------------|----------------|---------------|
|                           | Apples (kg) | Cherries (kg) | Apples (kg)    | Cherries (kg) |
| Pick nothing but apples   | 20          | 0             | 30             | 0             |
| Pick nothing but cherries | 0           | 20            | 0              | 60            |



The table in this figure shows how many kilograms of apples and how many kilograms of cherries you and your neighbour can each pick in one week. The graphs in the figure use the data from the table to construct PPFs for you and your neighbour. Panel (a) shows your PPF. If you devote all your time to picking apples and none of it to picking cherries, you can pick 20 kilograms. If you devote all your time to picking cherries, you can pick 20 kilograms. Panel (b) shows that if your neighbour devotes all their time to picking apples, they can pick 30 kilograms. If they devote all their time to picking cherries, they can pick 60 kilograms.

**Figure 2.5**

Gains from Trade



When you don't trade with your neighbour, you pick and consume 8 kilograms of apples and 12 kilograms of cherries per week—point A in panel (a). When your neighbour doesn't trade with you, they pick and consume 9 kilograms of apples and 42 kilograms of cherries per week—point C in panel (b). If you specialize in picking apples, you can pick 20 kilograms. If your neighbour specializes in picking cherries, they can pick 60 kilograms. If you trade 10 kilograms of apples for 15 kilograms of your neighbour's cherries, you will be able to consume 10 kilograms of apples and 15 kilograms of cherries—point B in panel (a). Your neighbour can now consume 10 kilograms of apples and 45 kilograms of cherries—point D in panel (b). You and your neighbour are both better off as a result of the trade.

**Table 2.1**

A Summary of the Gains from Trade

|                                                 | You         |               | Your Neighbour |               |
|-------------------------------------------------|-------------|---------------|----------------|---------------|
|                                                 | Apples (kg) | Cherries (kg) | Apples (kg)    | Cherries (kg) |
| Production and consumption <i>without</i> trade | 8           | 12            | 9              | 42            |
| Production <i>with</i> trade                    | 20          | 0             | 0              | 60            |
| Consumption <i>with</i> trade                   | 10          | 15            | 10             | 45            |
| Increased consumption (gains from trade)        | 2           | 3             | 1              | 3             |

**Table 2.2**

Opportunity Costs of Picking Apples and Cherries

|                | Opportunity Cost of Apples | Opportunity Cost of Cherries |
|----------------|----------------------------|------------------------------|
| You            | 1 kg of cherries           | 1 kg of apples               |
| Your neighbour | 2 kg of cherries           | 0.5 kg of apples             |

**Figure 3.1**

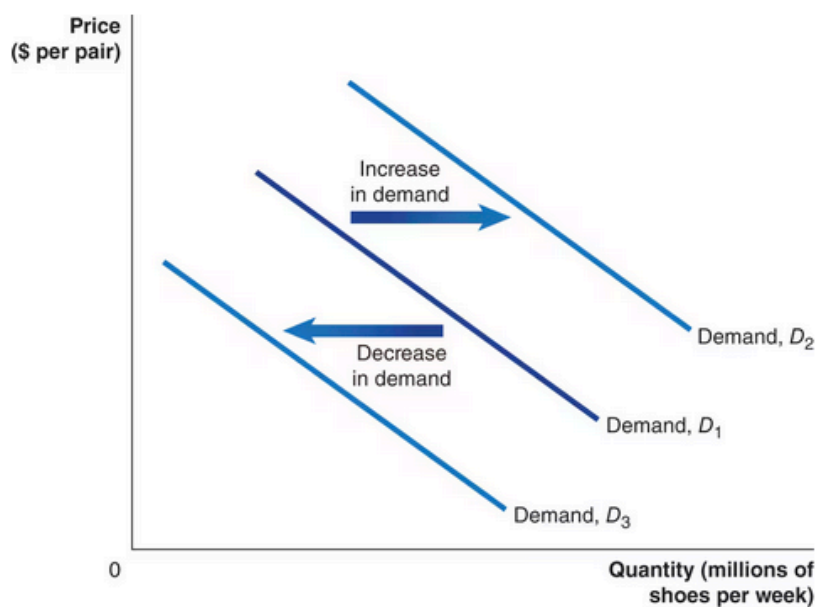
A Demand Schedule and Demand Curve



As the price changes, consumers change the quantity of athletic shoes they are willing to buy. We can show this as a *demand schedule* in a table or as a *demand curve* on a graph. The table and graph both show that as the price of athletic shoes falls, the quantity demanded increases. When the price of a pair of athletic shoes is \$150, consumers buy 8 million pairs per week. When the price drops to \$125, consumers buy 9 million pairs per week. Therefore, the demand curve for athletic shoes is downward sloping.

**Figure 3.2**

Shifting the Demand Curve



When consumers increase the quantity of a product they want to buy at a given price, the market demand curve shifts to the right, from  $D_1$  to  $D_2$ . When consumers decrease the quantity of a product they want to buy at a given price, the demand curve shifts to the left from  $D_1$  to  $D_3$ .

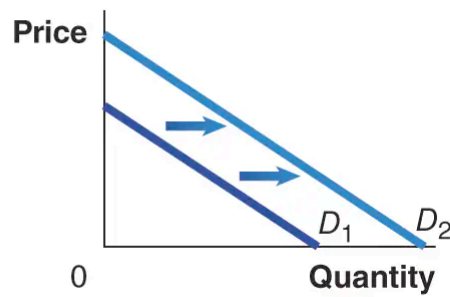
## An Increase in

## Shifts the Demand Curve

## Why?

income (and the good is normal)

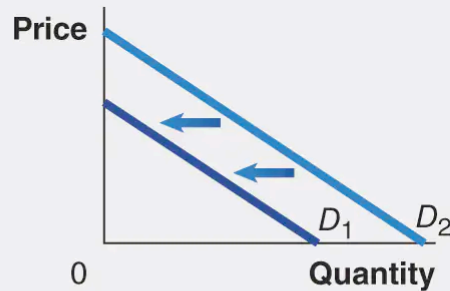
RIGHT



Consumers spend more of their higher incomes on the good.

income (and the good is inferior)

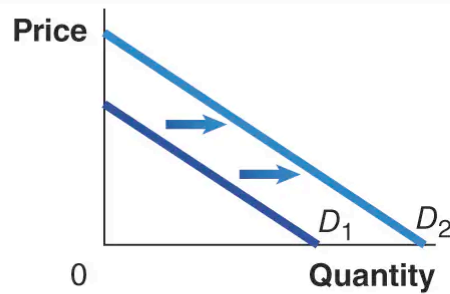
LEFT



Consumers spend less of their higher incomes on the good.

the price of a substitute good

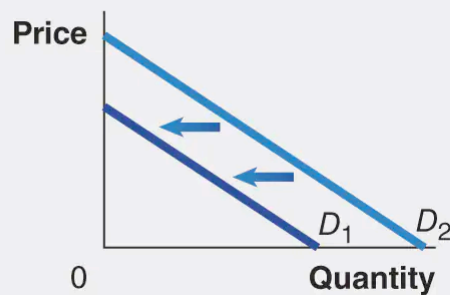
RIGHT



Consumers buy less of the substitute good and more of this good.

the price of a complementary good

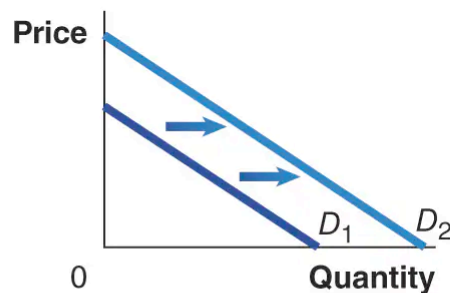
LEFT



Consumers buy less of the complementary good and less of this good too.

taste for the good

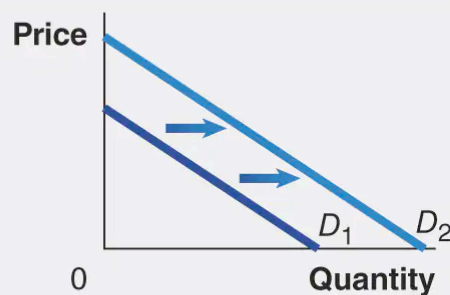
RIGHT



Consumers are willing to buy a large quantity of the good at every price.

population

RIGHT

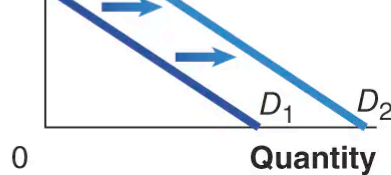


Additional consumers result in a greater demand at every price.

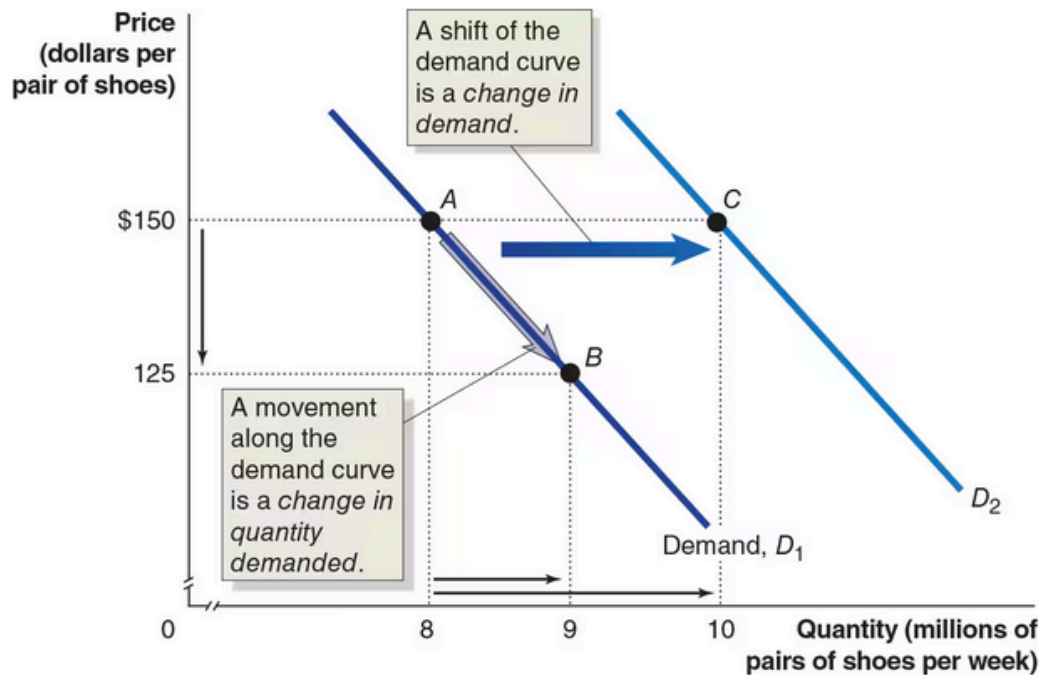
the expected future price of the good



Consumers buy more of the good today to avoid the higher future price.

**Figure 3.3**

A Change in Demand versus a Change in Quantity Demanded

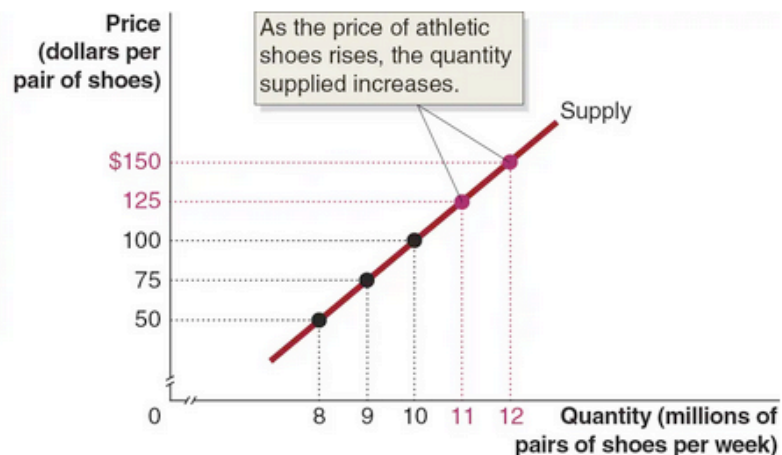


If the price of athletic shoes falls from \$150 to \$125 per pair, the result will be a movement along the demand curve from point A to point B—an increase in quantity demanded from 8 million pairs to 9 million pairs. If consumers' incomes increase, or if another factor changes that makes consumers want more athletic shoes at every price, the demand curve will shift to the right—an increase in demand. In this case, the increase in demand from  $D_1$  to  $D_2$  causes the quantity of athletic shoes demanded at a price of \$150 to increase from 8 million pairs at point A to 10 million pairs at point C.

**Figure 3.4**

A Supply Schedule and Supply Curve

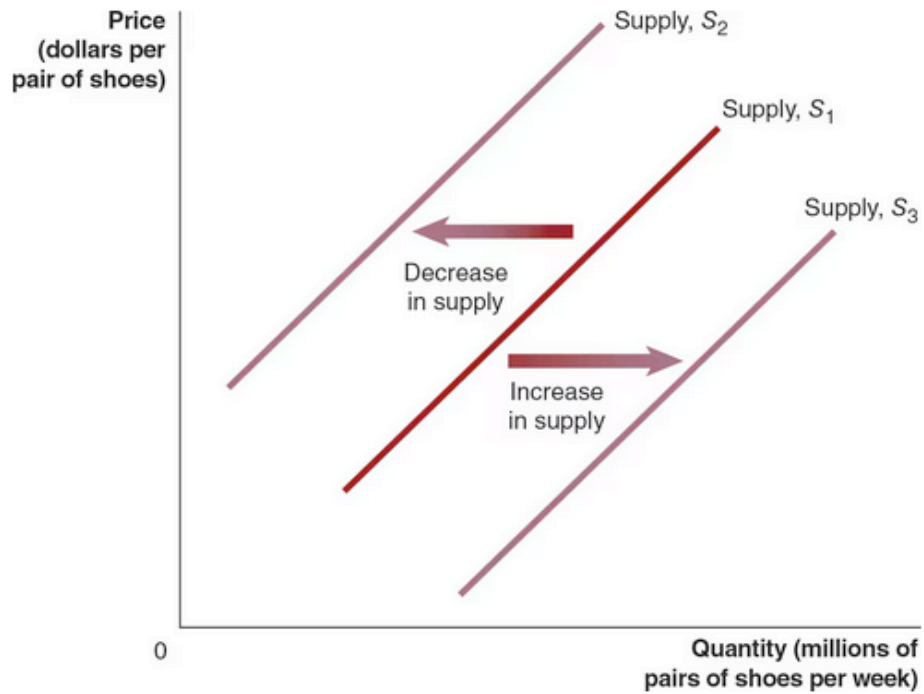
| Supply Schedule                         |                                                      |
|-----------------------------------------|------------------------------------------------------|
| Price<br>(dollars per<br>pair of shoes) | Quantity<br>(millions of pairs<br>of shoes per week) |
| \$150                                   | 12                                                   |
| 125                                     | 11                                                   |
| 100                                     | 10                                                   |
| 75                                      | 9                                                    |
| 50                                      | 8                                                    |



As the price changes, Nike, Adidas, Puma, and other firms producing athletic shoes change the quantity they are willing to supply. We can show this as a *supply schedule* in a table or a *supply curve* on a graph. The supply schedule and supply curve both show that as the price of athletic shoes rises, firms will increase the quantity they supply. At a price of \$125 per pair, firms will supply 10 million pairs. At a price of \$150 per pair, firms will supply 11 million pairs.

### Figure 3.5

#### Shifting the Supply Curve



When firms increase the quantity of a product they want to sell at a given price, the supply curve shifts to the right. The shift from  $S_1$  to  $S_3$  represents an *increase in supply*. When firms decrease the quantity of a product they want to sell at a given price, the supply curve shifts to the left. The shift from  $S_1$  to  $S_2$  represents a *decrease in supply*.

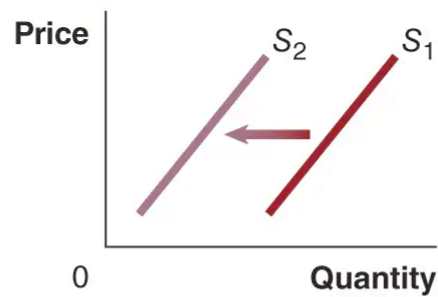
## An Increase in

## Shifts the Supply Curve

## Why?

the price of an input

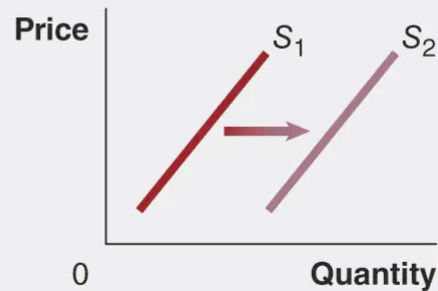
LEFT



The costs of producing the good rise.

productivity

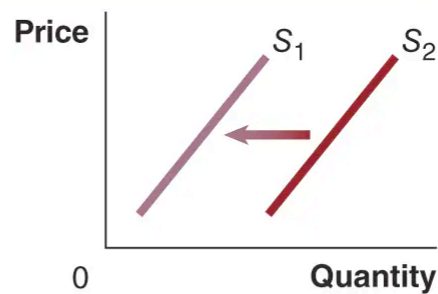
RIGHT



The costs of producing the good fall.

the price of a substitute in production

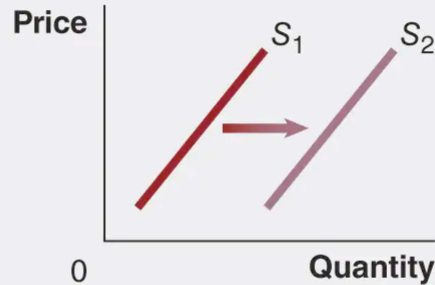
LEFT



More of the substitute is produced, and less of the good is produced.

the number of firms in the market

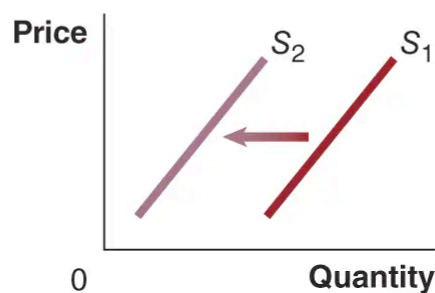
RIGHT



Additional firms result in a greater quantity supplied at every price.

the expected future price of the product

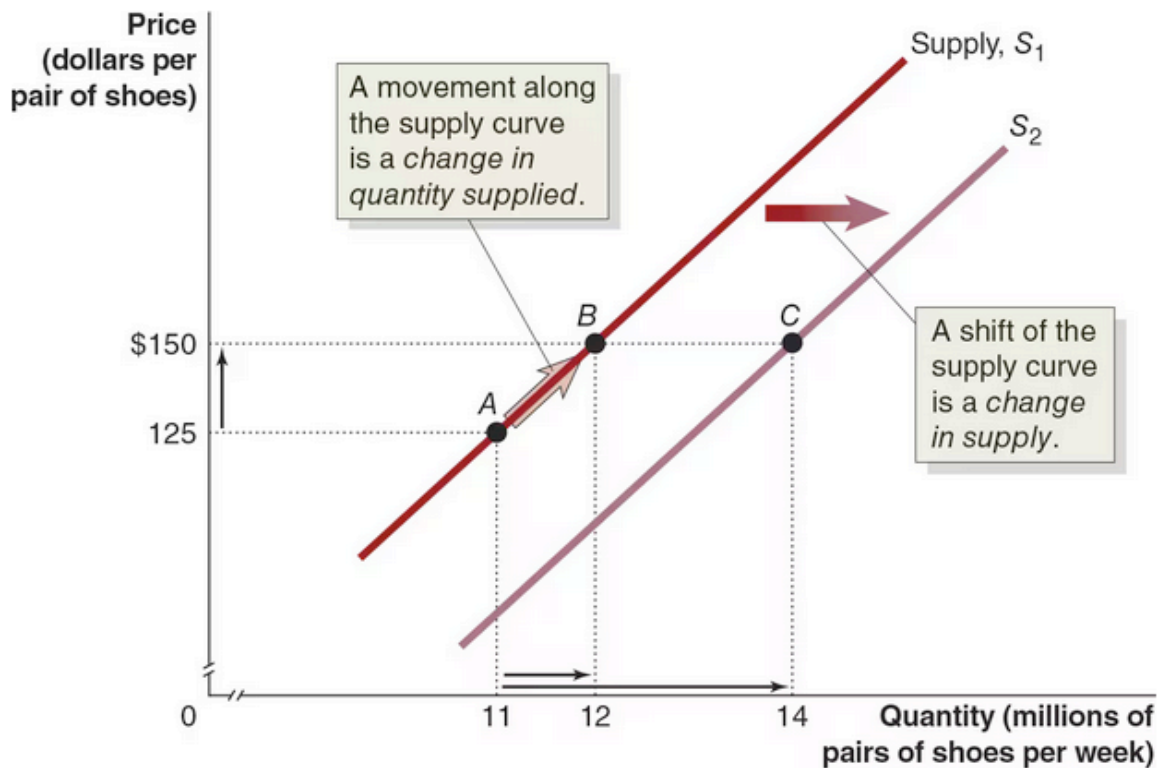
LEFT



Less of the good will be offered for sale today as firms wait to take advantage of the higher price expected in the future.

**Figure 3.6**

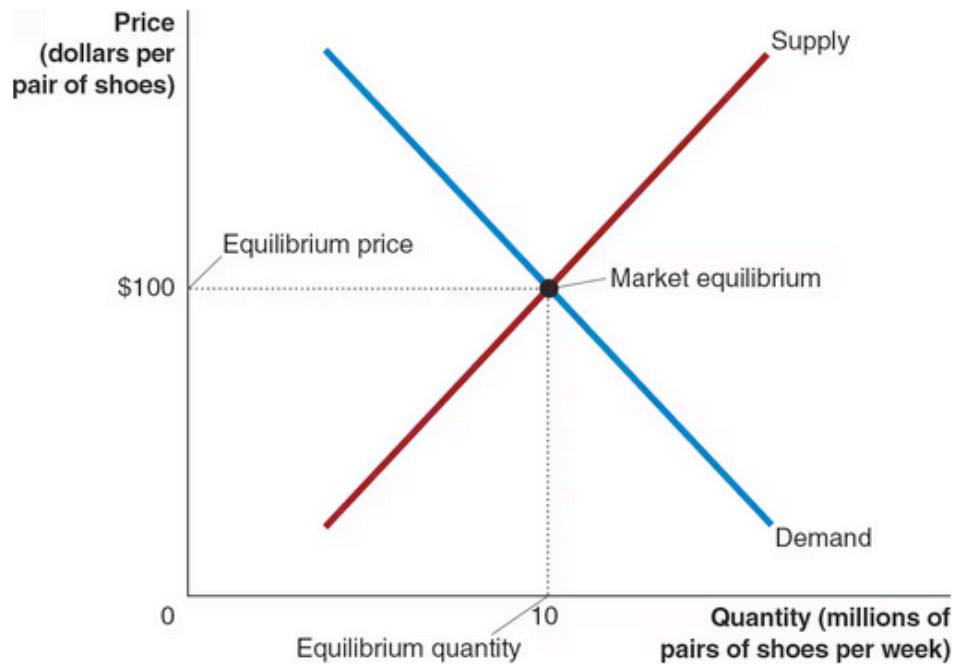
A Change in Supply versus a Change in Quantity Supplied



If the price of athletic shoes rises from \$125 to \$150, the result will be a movement along the supply curve from point A to point B—an increase in quantity supplied by Nike, Adidas, Reebok, and other firms from 11 million pairs per week to 12 million pairs. If the price of an input decreases, or if another factor changes that causes sellers to supply more of the product at every price, the supply curve will shift to the right—an increase in supply from  $S_1$  to  $S_2$  causes the quantity of athletic shoes supplied at a price of \$150 to increase from 12 million pairs at point B to 14 million pairs at point C.

**Figure 3.7**

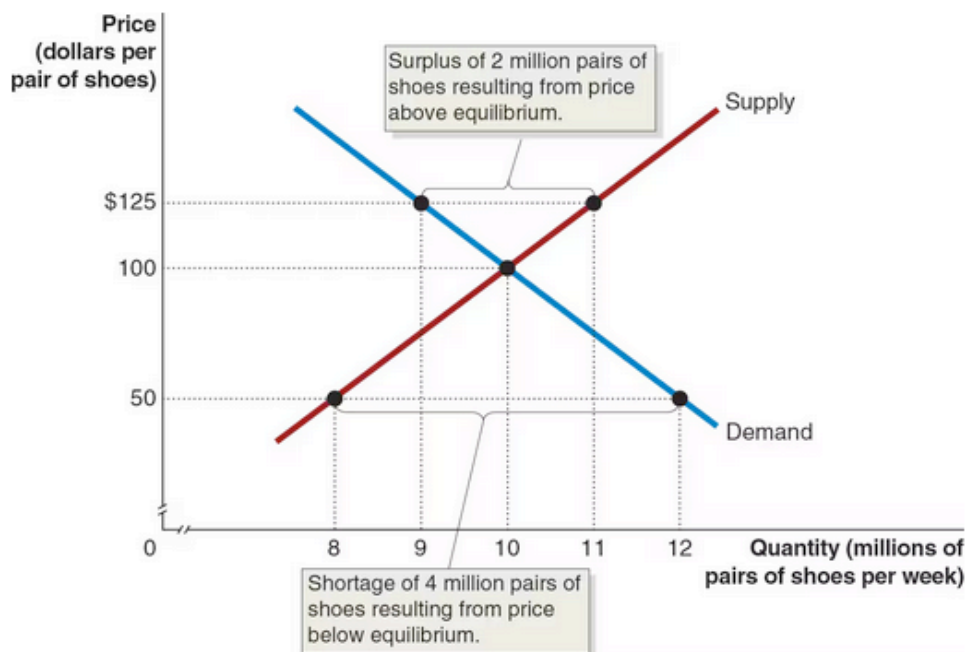
Market Equilibrium



Where the demand curve crosses the supply curve determines *market equilibrium*. In this case, the demand curve for athletic shoes crosses the supply curve at a price of \$100 and a quantity of 10 million pairs. Only at this point is the quantity of athletic shoes consumers are willing to buy equal to the quantity that Nike, Adidas, New Balance, and other firms are willing to sell: The quantity demanded is equal to the quantity supplied.

**Figure 3.8**

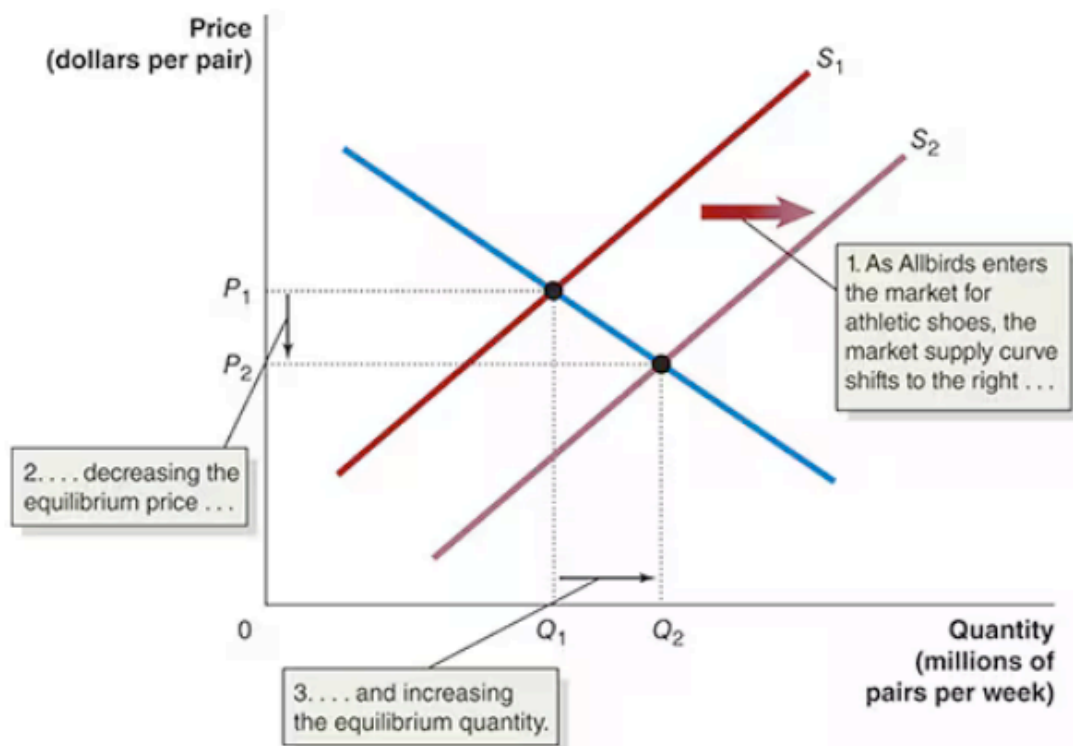
The Effect of Surpluses and Shortages on the Market



When the market price is above equilibrium, there will be a *surplus*. A price of \$125 for athletic shoes results in 11 million pairs being supplied but only 9 million pairs being demanded, or a surplus of 2 million pairs. As Nike, Adidas, New Balance, and other firms cut the price to dispose of the surplus, the price will fall to the equilibrium price of \$100. When the market price is below equilibrium, there will be a *shortage*. A price of \$50 results in 12 million pairs being demanded but only 8 million being supplied, or a shortage of 4 million. As firms find that consumers who are unable to find athletic shoes available for sale are willing to pay higher prices to get them, the price will rise to the equilibrium price of \$100.

**Figure 3.9**

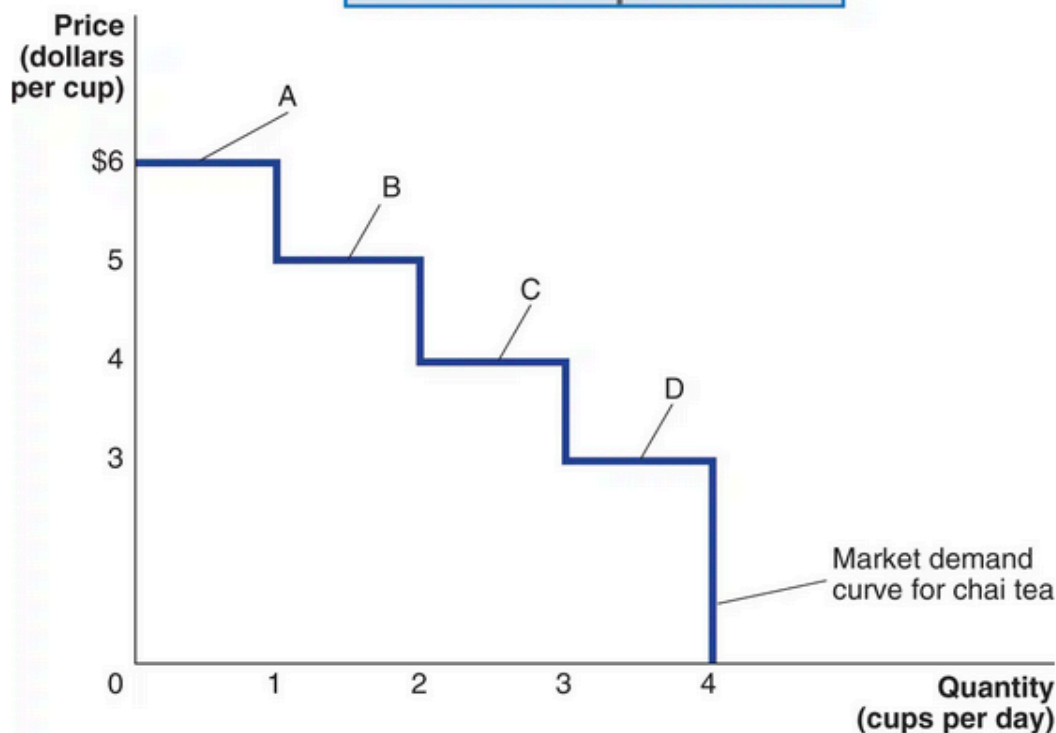
The Effect of an Increase in Supply on Equilibrium



**Figure 4.1**

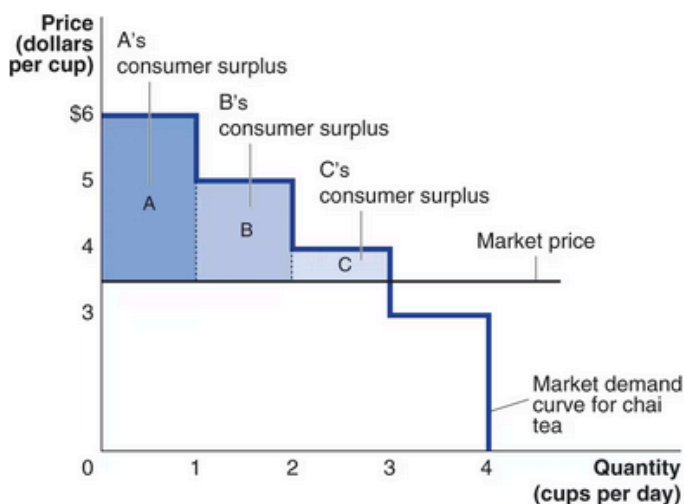
Deriving the Demand Curve for Chai Tea

| Consumer | Highest Price Willing to Pay |
|----------|------------------------------|
| A        | \$6                          |
| B        | 5                            |
| C        | 4                            |
| D        | 3                            |

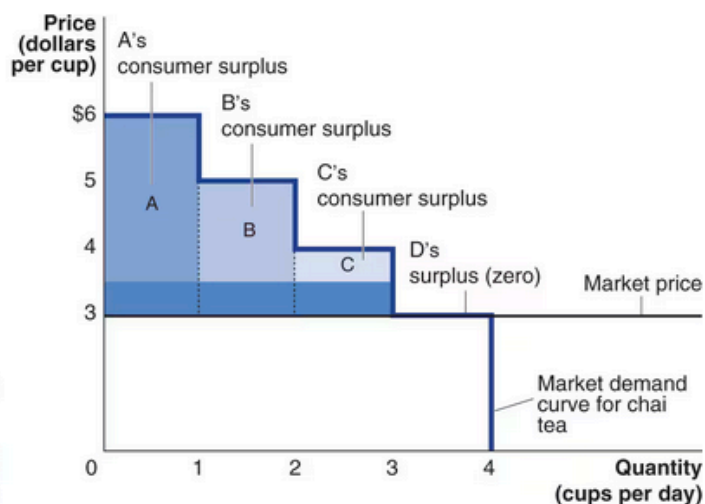


**Figure 4.2**

Measuring Consumer Surplus



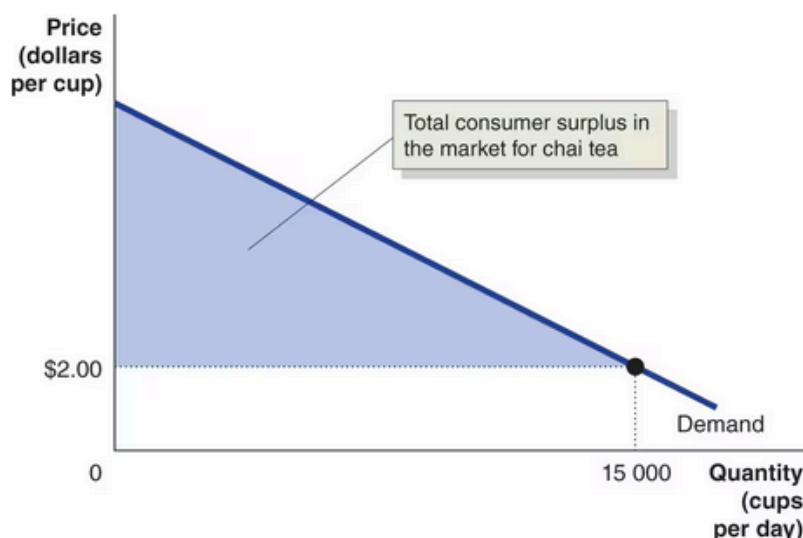
(a) Consumer surplus with a market price of \$3.50



(b) Consumer surplus with a market price of \$3.00

**Figure 4.3**

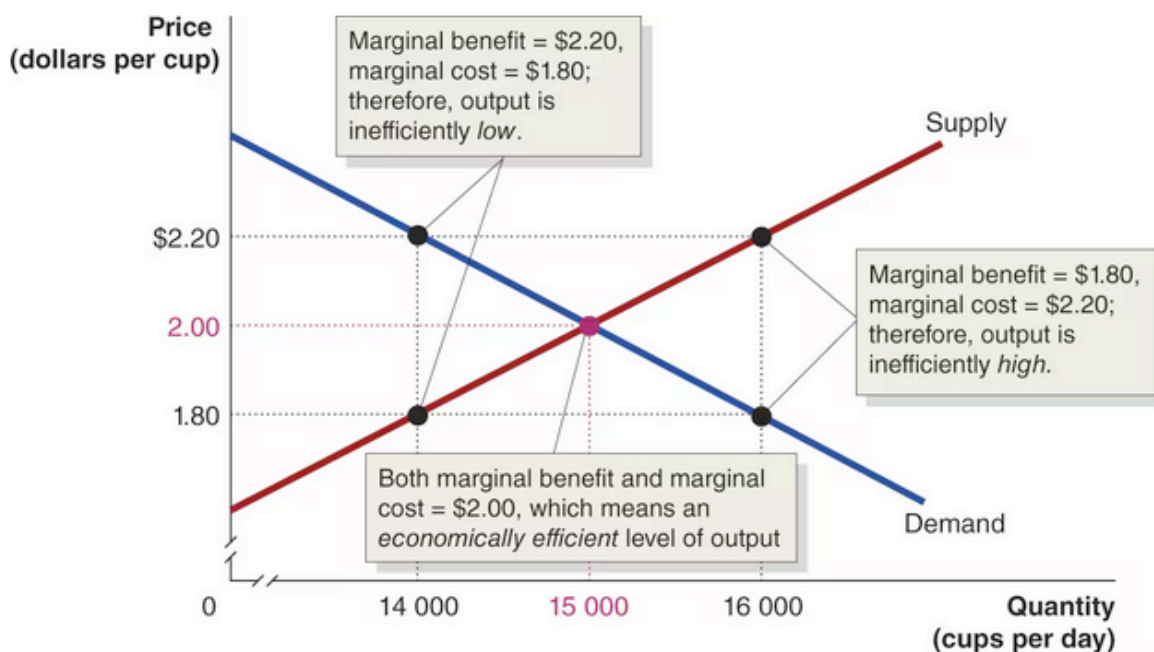
Total Consumer Surplus in the Market for Chai Tea



The demand curve tells us that most buyers of chai tea would have been willing to pay more than the market price of \$2.00. For each buyer, consumer surplus is equal to the difference between the highest price the person is willing to pay and the market price actually paid. Therefore, the total amount of consumer surplus in the market for chai tea is equal to the area below the demand curve and above the market price. Consumer surplus represents the benefit to consumers in excess of the price they paid to purchase the product.

**Figure 4.5**

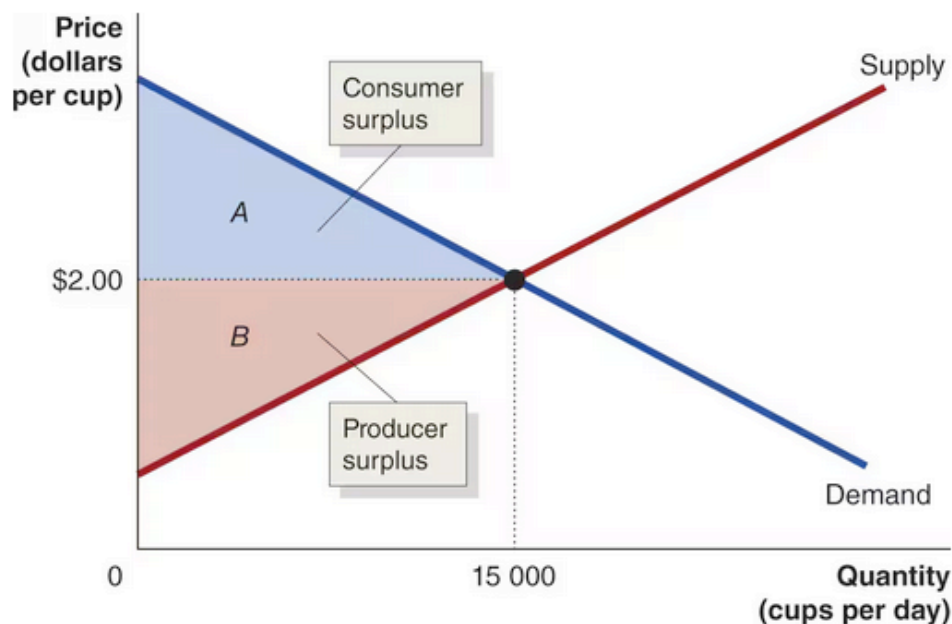
Marginal Benefit Equals Marginal Cost Only at Competitive Equilibrium



In a competitive market, equilibrium occurs at a quantity of 15 000 cups and a price of \$2.00 per cup, where marginal benefit equals marginal cost. This is the economically efficient level of output because every cup has been produced where the marginal benefit to buyers is greater than or equal to the marginal cost to producers.

**Figure 4.6**

Economic Surplus Equals the Sum of Consumer Surplus and Producer Surplus

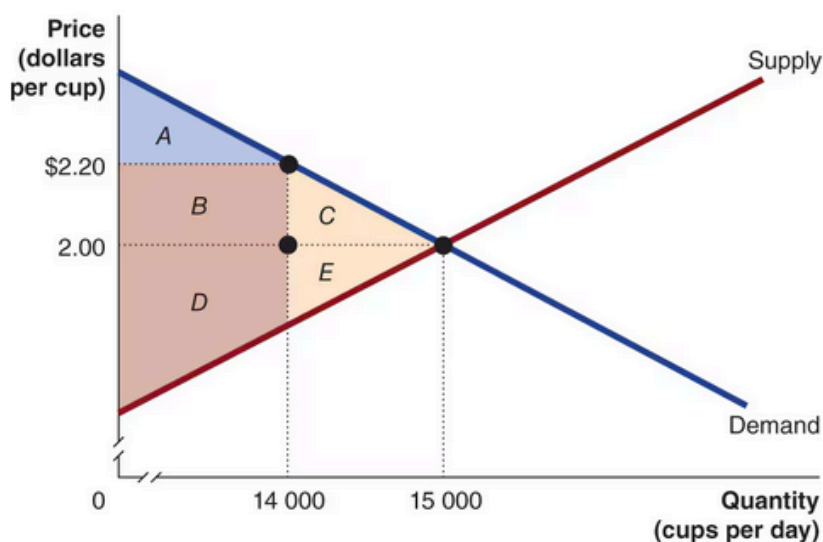


The economic surplus in a market is the sum of triangle A, representing consumer surplus, and triangle B, representing producer surplus.

**Figure 4.7**

When a Market Is Not in Equilibrium, There Is a Deadweight Loss

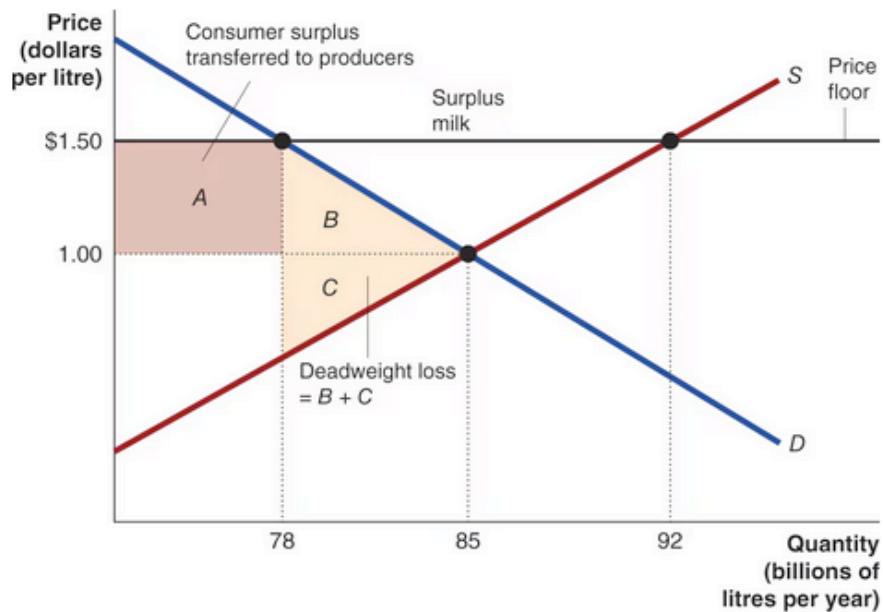
|                  | At Competitive Equilibrium | At a Price of \$2.20 |
|------------------|----------------------------|----------------------|
| Consumer Surplus | $A + B + C$                | $A$                  |
| Producer Surplus | $D + E$                    | $B + D$              |
| Deadweight Loss  | None                       | $C + E$              |



Economic surplus is maximized when a market is in competitive equilibrium. When a market is not in equilibrium, there is a deadweight loss. When the price of chai tea is \$2.20 instead of \$2.00, consumer surplus declines from an amount equal to the sum of areas A, B, and C to just area A. Producer surplus increases from the sum of areas D and E to the sum of areas B and D. At competitive equilibrium, there is no deadweight loss. At a price of \$2.20, there is a deadweight loss equal to the sum of areas C and E.

**Figure 4.8**

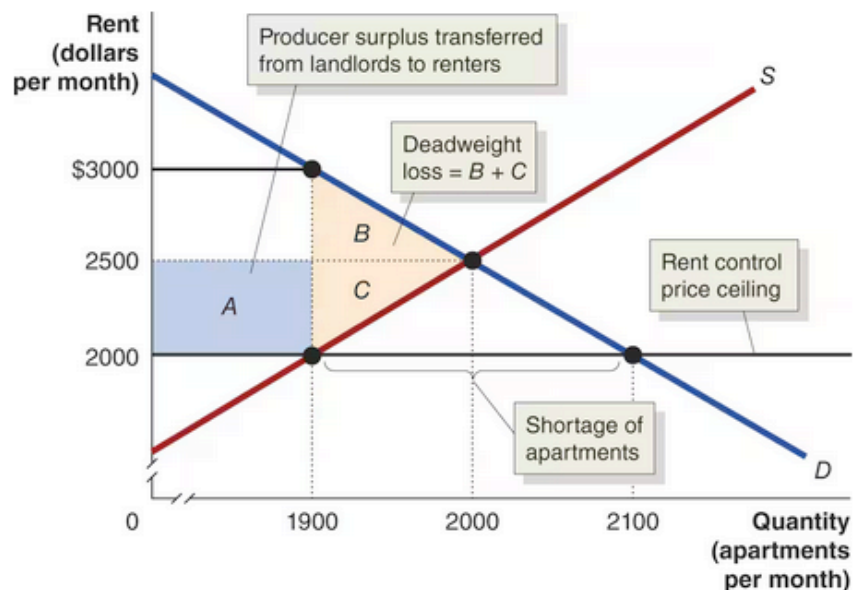
The Economic Effect of a Price Floor in the Milk Market



If dairy farmers convince the government to impose a price floor of \$1.50 per litre, the amount of milk sold will fall from 85 billion litres a year to 78 billion litres a year. If we assume that farmers produce 78 billion litres, producer surplus then increases by rectangle A—which is transferred from consumer surplus—and falls by triangle C. Consumer surplus declines by rectangle A plus triangle B. There is a deadweight loss equal to triangles B and C, representing the decline in economic efficiency due to the price floor. In reality, without further government intervention, a price floor of \$1.50 would cause farmers to expand their production from 85 billion litres to 92 billion litres, resulting in a surplus of milk.

**Figure 4.9**

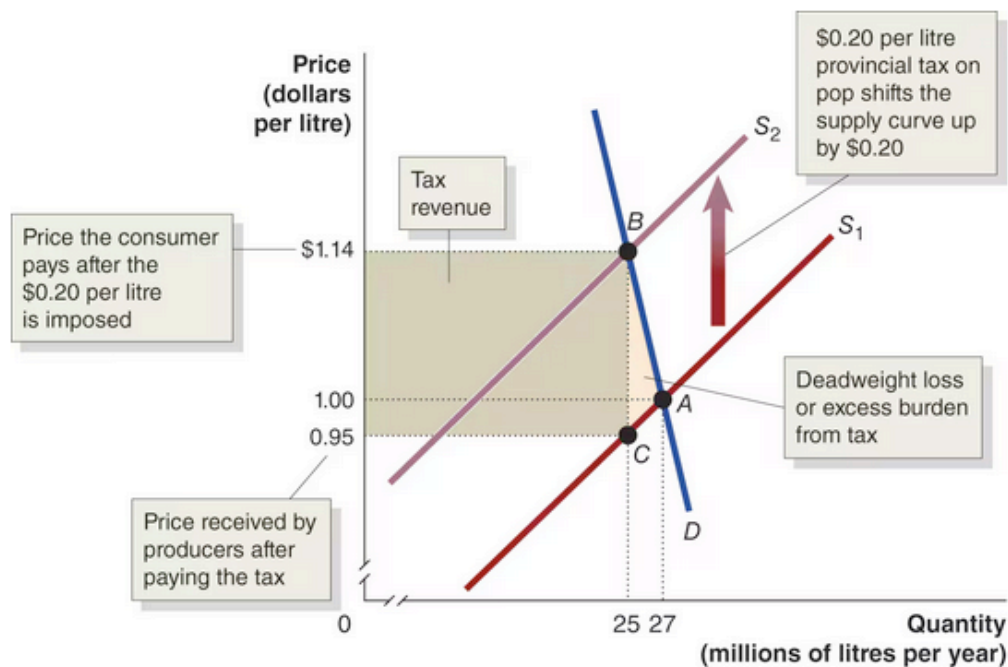
The Economic Effect of a Rent Ceiling



Without rent control, the equilibrium rent is \$2500 per month. At that price, 2000 apartments would be rented. If the government imposes a rent ceiling of \$2000, the quantity of apartments supplied falls to 1900, and the quantity of apartments demanded increases to 2100, resulting in a shortage of 200 apartments. Producer surplus equal to the area of rectangle A is transferred from landlords to renters, and there is a deadweight loss equal to the areas of triangles B and C.

**Figure 4.10**

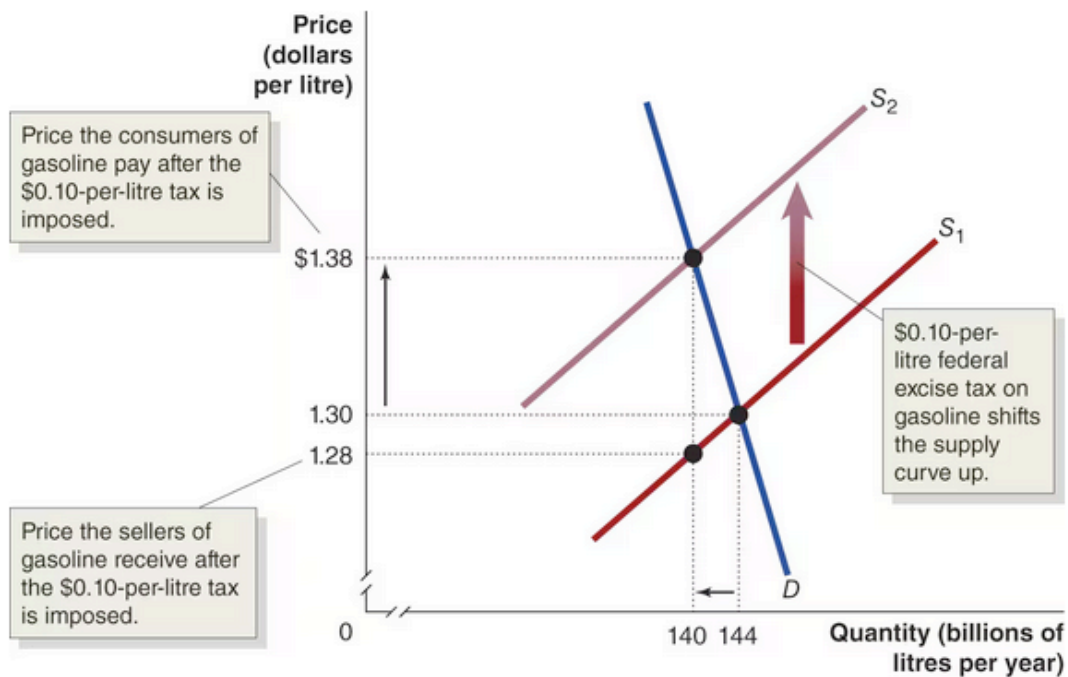
The Effect of a Tax on the Market for Pop



Without the tax, market equilibrium occurs at point A. The equilibrium price of pop is \$1.00 per litre, and 27 million litres are sold per year. A \$0.20 per litre tax on pop will cause the supply curve for pop to shift up by \$0.20, from  $S_1$  to  $S_2$ . The new equilibrium occurs at point B. The price of pop will increase by \$0.15 per litre, and the quantity sold will fall to 25 million litres. The tax on pop has increased the price paid by consumers from \$1.00 per litre to \$1.15 per litre. Producers receive a price of \$1.15 per litre (point B), but after paying the \$0.20 tax, they are left with \$0.95 (point C). The government will receive tax revenue equal to the shaded box. Some consumer surplus and some producer surplus will become tax revenue for the government, and some will become deadweight loss, shown by the shaded triangle.

**Figure 4.11**

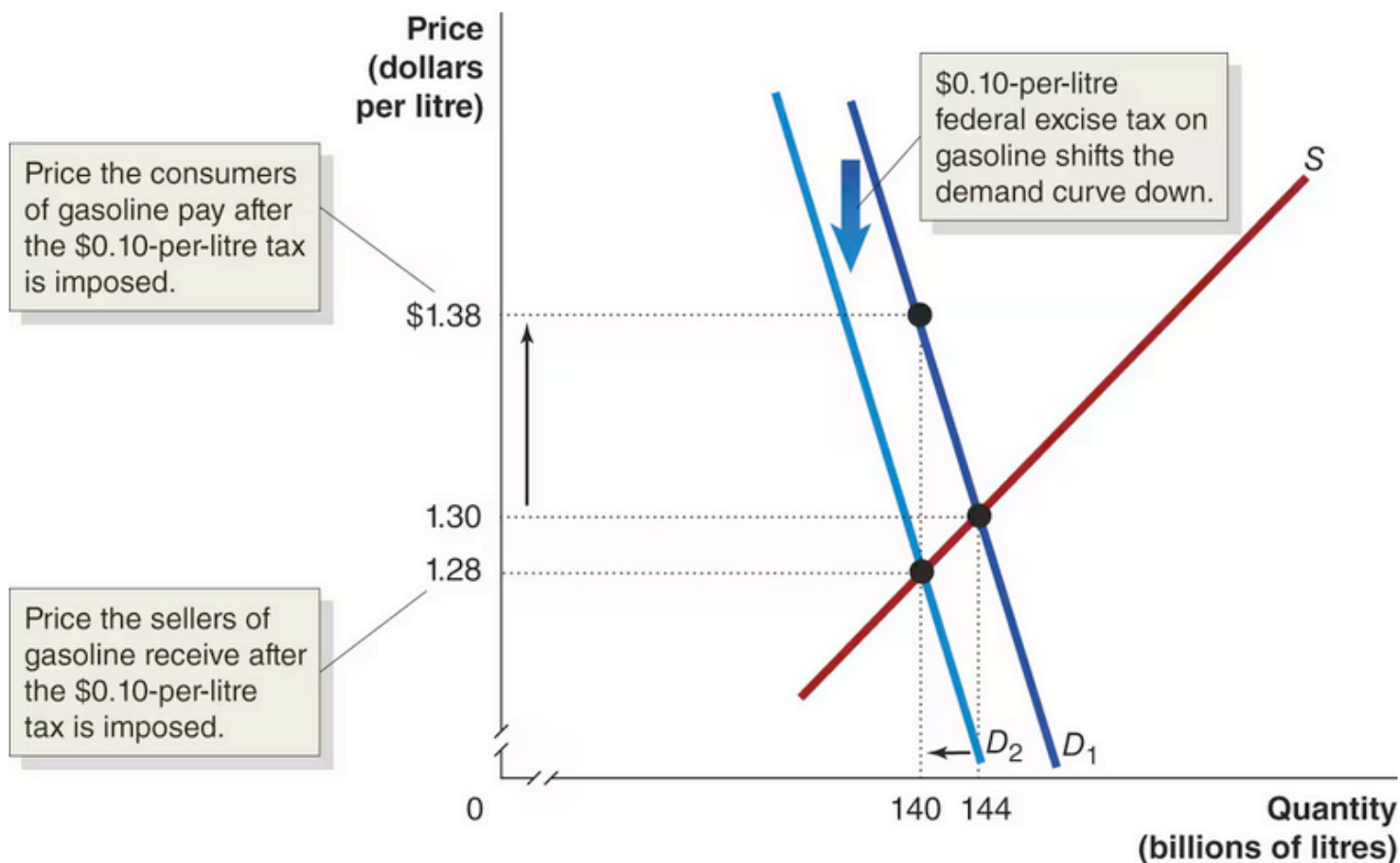
The Incidence of a Tax on Gasoline



With no tax on gasoline, the price would be \$1.30 per litre, and 144 billion litres of gasoline would be sold each year. A \$0.10-per-litre excise tax shifts up the supply curve from  $S_1$  to  $S_2$ , raises the price consumers pay from \$1.30 to \$1.38, and lowers the price sellers receive from \$1.30 to \$1.28. Therefore, consumers pay \$0.08 of the \$0.10-cents-per-litre tax on gasoline, and sellers pay \$0.02.

**Figure 4.12**

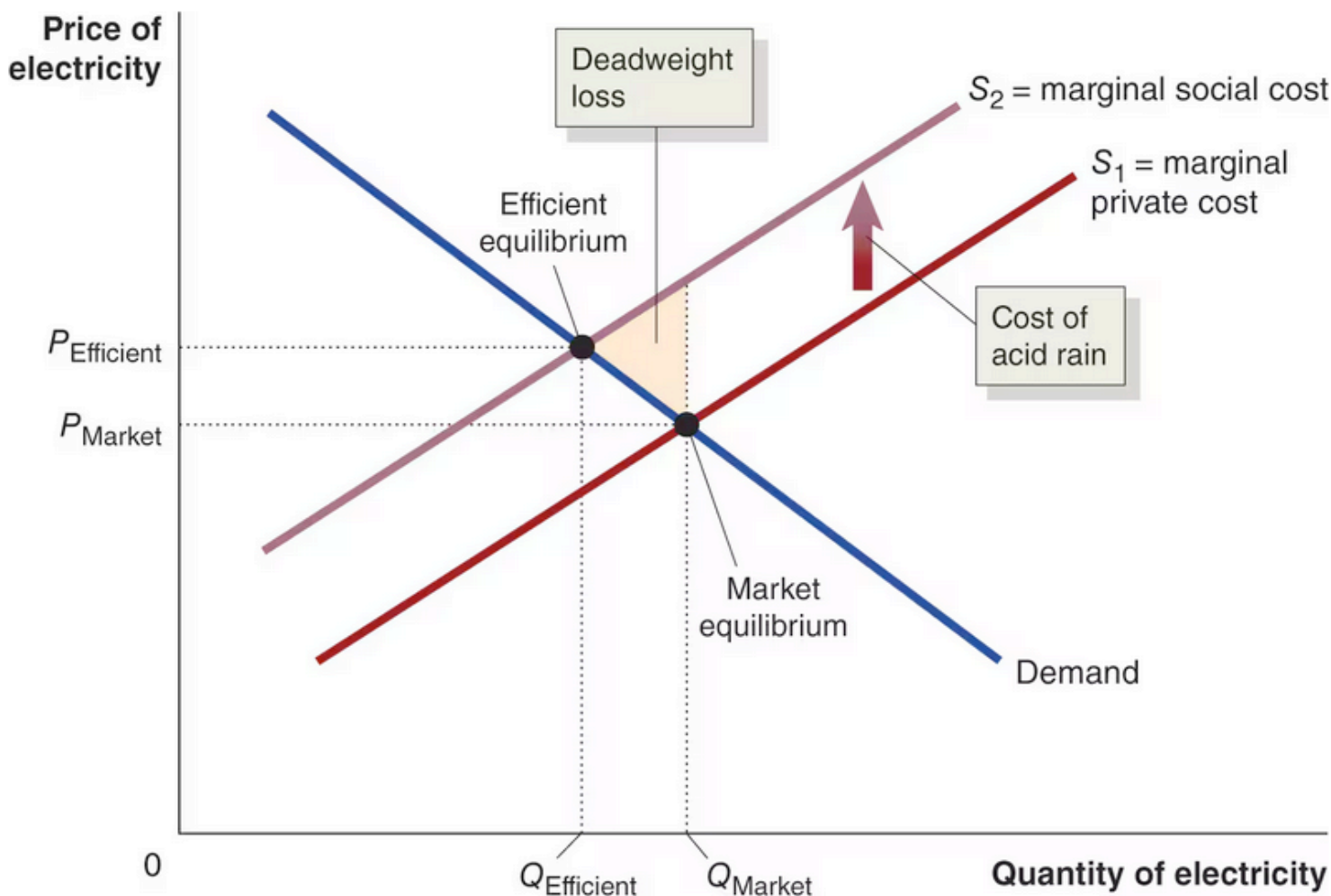
The Incidence of a Tax on Gasoline Paid by Buyers



With no tax on gasoline, the demand curve is  $D_1$ . If a \$0.10-per-litre tax is imposed that consumers are responsible for paying, the demand curve shifts down by the amount of the tax, from  $D_1$  to  $D_2$ . In the new equilibrium, consumers pay a price of \$1.38 per litre, including the tax. Producers receive \$1.28 per litre. This is the same result we saw when producers were responsible for paying the tax.

**Figure 5.1**

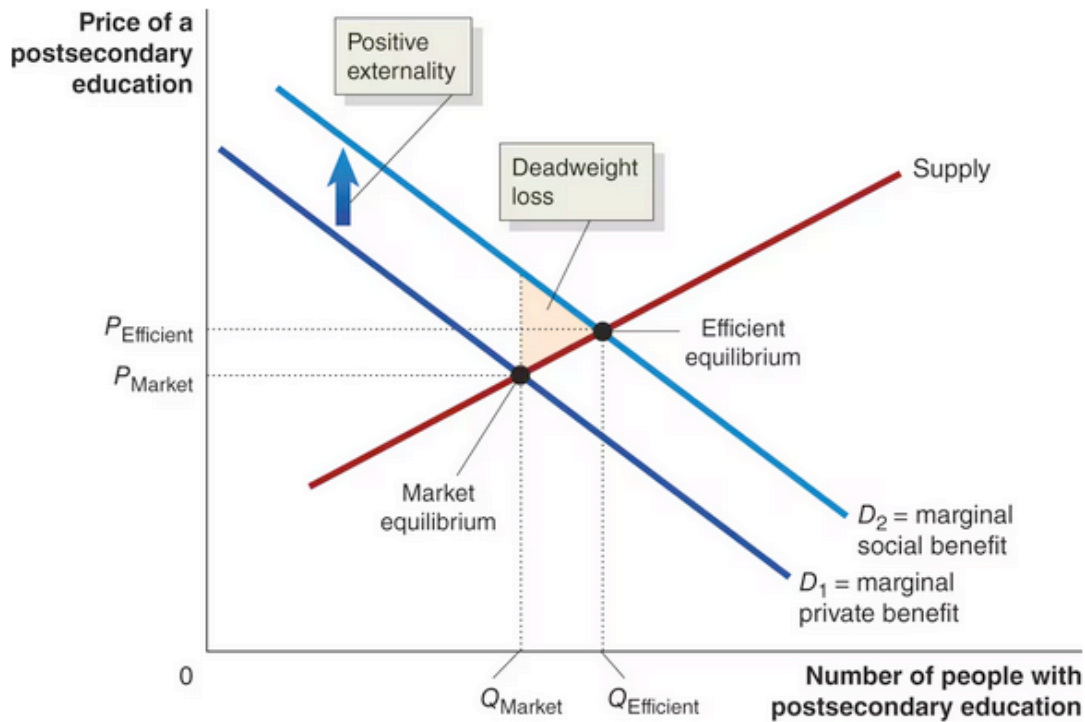
The Effect of Pollution on Economic Efficiency



Because utilities do not bear the cost of acid rain, they produce electricity beyond the economically efficient level. Supply

**Figure 5.2**

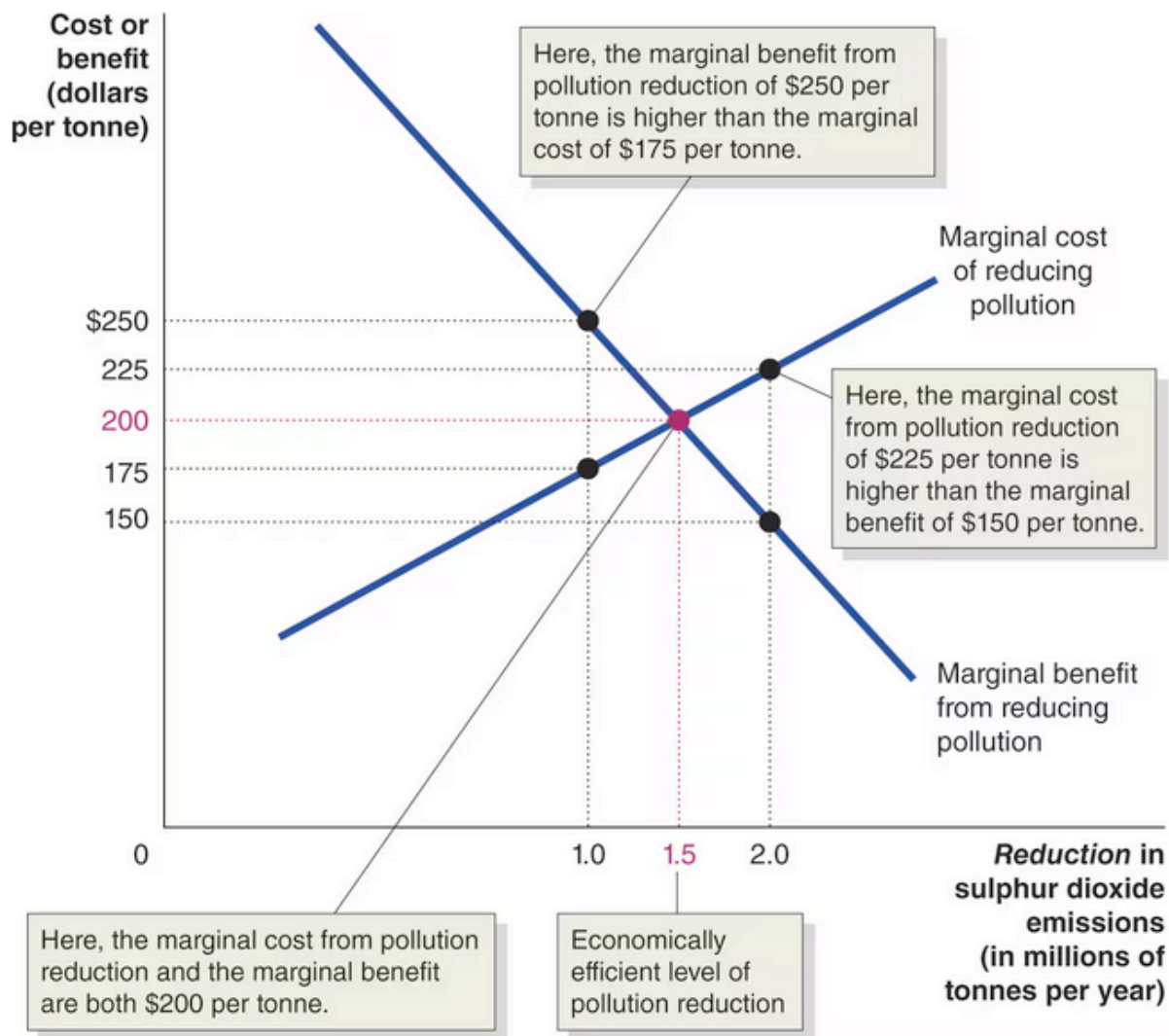
The Effect of a Positive Externality on Economic Efficiency



People who do not get a postsecondary education can still benefit from it. As a result, the marginal social benefit from a postsecondary education is greater than the marginal private benefit to postsecondary students. Because only the marginal private benefit is represented in the market demand curve  $D_1$ , the number of people with postsecondary education (quantity produced),  $Q_{\text{Market}}$ , is too low. If the market demand curve were  $D_2$  instead of  $D_1$ , the number of people with postsecondary education would be  $Q_{\text{Efficient}}$ , which is the efficient level. At the market equilibrium of  $Q_{\text{Market}}$ , there is a deadweight loss equal to the area of the shaded triangle.

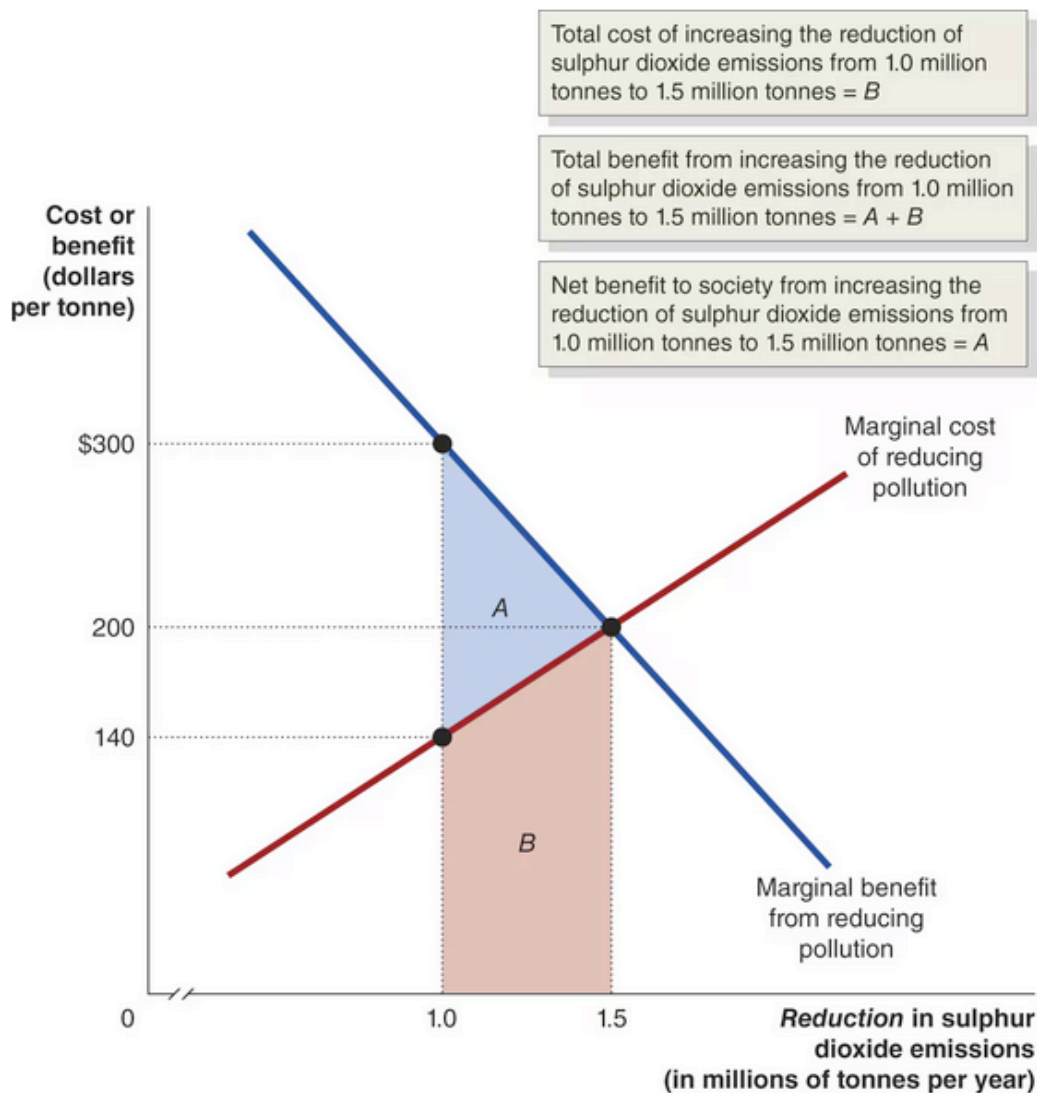
**Figure 5.3**

The Marginal Benefit from Pollution Reduction Should Equal the Marginal Cost



### Figure 5.4

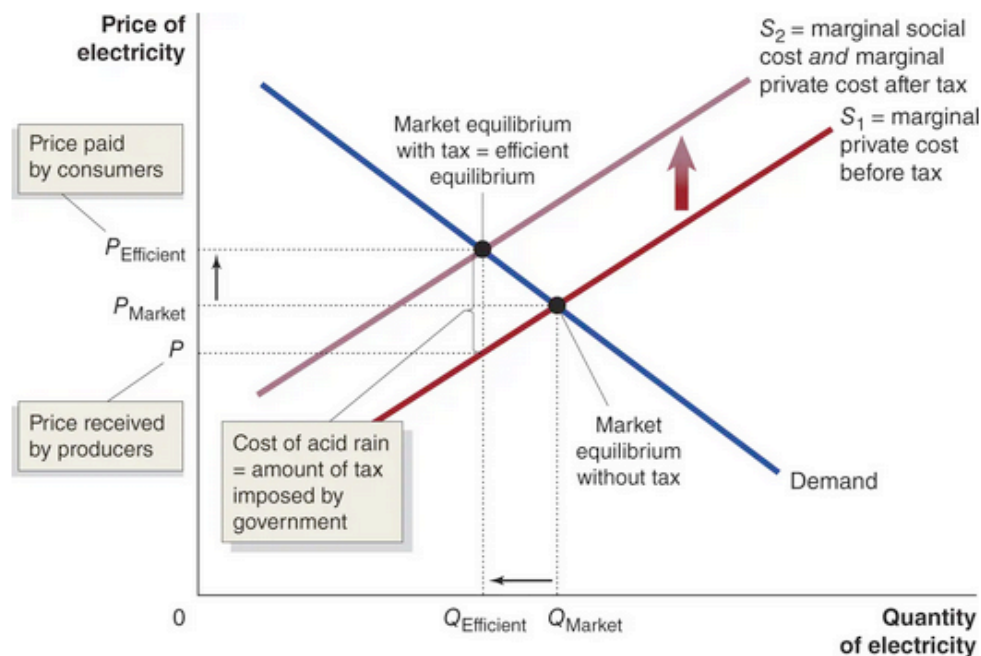
The Benefits of Reducing Pollution to the Optimal Level Are Greater than the Costs



Increasing the reduction in sulphur dioxide emissions from 1.0 million tonnes to 1.5 million tonnes results in total benefits equal to the sum of the areas  $A$  and  $B$  under the marginal benefits curve. The total cost of this decrease in pollution is equal

**Figure 5.5**

When There Is a Negative Externality, a Tax Can Lead to the Efficient Level of Output



Because utilities do not bear the cost of acid rain, they produce electricity beyond the economically efficient level. If the government imposes a tax equal to the cost of acid rain, the utilities will internalize the externality. As a consequence, the supply curve will shift up, from  $S_1$  to  $S_2$ . The market equilibrium quantity changes from  $Q_{\text{Market}}$ , where an inefficiently high level of electricity is produced, to  $Q_{\text{Efficient}}$ , the economically efficient equilibrium quantity. The price of electricity will rise from  $P_{\text{Market}}$ —which does not include the cost of acid rain—to  $P_{\text{Efficient}}$ —which does include the cost.

Consumers pay the price  $P_{\text{Efficient}}$ , while producers receive a price  $P$ , which is equal to  $P_{\text{Efficient}}$  minus the amount of the tax.

**Figure 5.7**

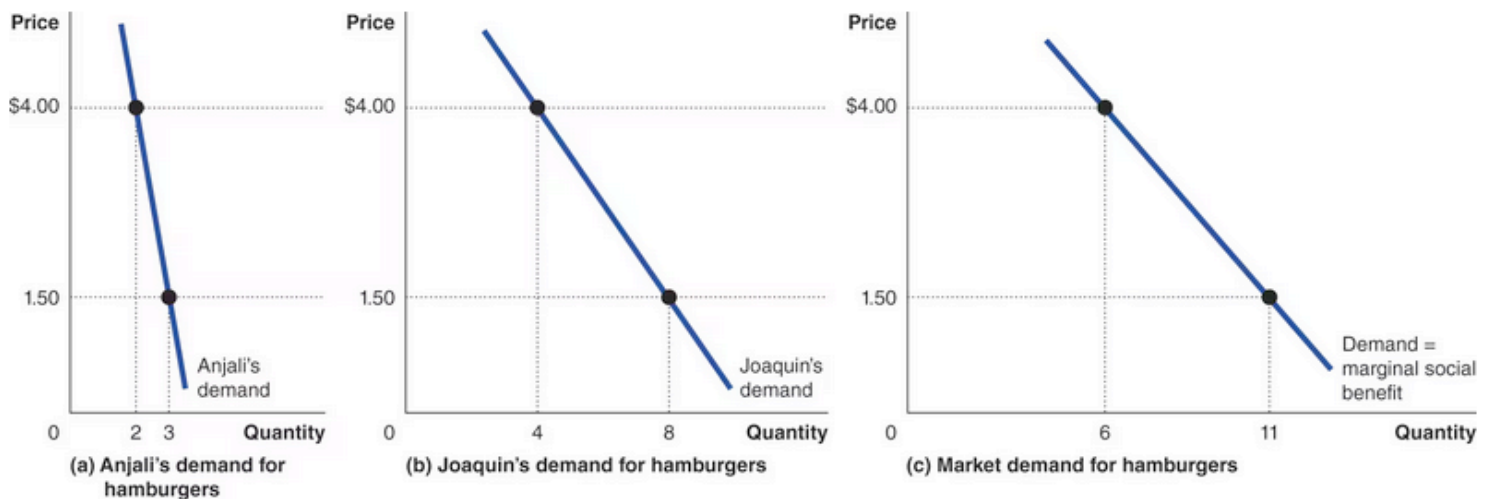
Four Categories of Goods

|           | Excludable                                                                           | Non-excludable                                                                                        |
|-----------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Rival     | <b>Private Goods</b><br><i>Examples:</i><br><i>Big Macs</i><br><i>Running shoes</i>  | <b>Common Resources</b><br><i>Examples:</i><br><i>Tuna in the ocean</i><br><i>Public pasture land</i> |
| Non-rival | <b>Quasi-Public Goods</b><br><i>Examples:</i><br><i>Cable TV</i><br><i>Toll road</i> | <b>Public Goods</b><br><i>Examples:</i><br><i>National defence</i><br><i>Court system</i>             |

Goods and services can be divided into four categories on the basis of whether people can be excluded from consuming them and whether they are rival in consumption. A good or service is rival in consumption if one person consuming a unit of a good means that another person cannot consume that unit.

## Figure 5.8

### Constructing the Market Demand Curve for a Private Good



The market demand curve for private goods is determined by adding horizontally the quantity of the good demanded at each price by each consumer. For instance, in panel (a), Anjali demands 2 hamburgers when the price is \$4.00, and in panel (b), Joaquin demands 4 hamburgers when the price is \$4.00. So, a quantity of 6 hamburgers and a price of \$4.00 is a point on the market demand curve in panel (c).

## Demand

We might measure the price elasticity of demand by using the slope of the demand curve because the slope tells us how much quantity changes as price changes, but this approach has a drawback: The measurement of slope is sensitive to the units chosen for quantity and price. For example, suppose a pop tax causes an increase in the price of pop of \$1 per two-litre bottle, which decreases the quantity demanded from 10.2 million bottles to 10.1 million bottles per day. The change in quantity is  $-0.1$  million bottles, and the change in price is \$1, so the slope is  $-0.1/1 = -0.1$ . But

- If we measure price in cents, rather than in dollars, the slope is  $-0.1/100 = -0.001$ .
- If we measure price in dollars and bottles in thousands, instead of millions, the slope is  $-100/1 = -100$ .
- If we measure price in cents and bottles in thousands, the slope is  $-100/100 = -1$ .

Clearly, the value we compute for the slope can change dramatically, depending on the units we use for quantity and price.

To avoid this confusion over units, economists use *percentage changes* when measuring the price elasticity of demand. Percentage changes are not dependent on units of measurement. (For a review of calculating percentage changes, see the appendix to Chapter 1.) No matter what units we use to measure the quantity of pop, 10 percent more pop is 10 percent more pop. Therefore,

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

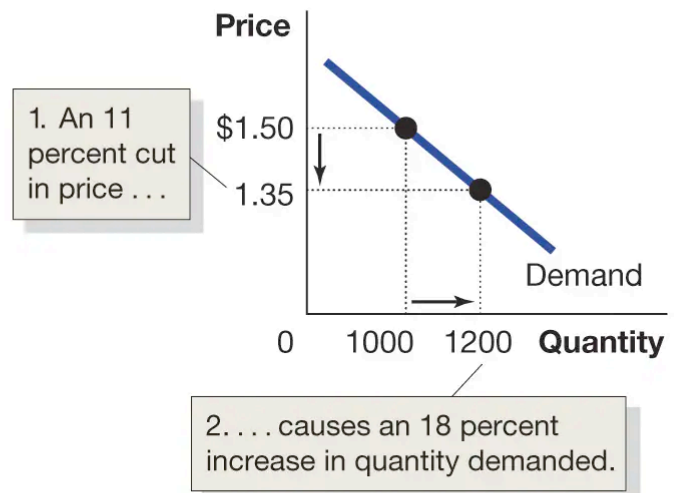
$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1)}{\left(\frac{Q_1 + Q_2}{2}\right)} \div \frac{(P_2 - P_1)}{\left(\frac{P_1 + P_2}{2}\right)}$$

If demand is ...

then the absolute value  
of price elasticity is ...

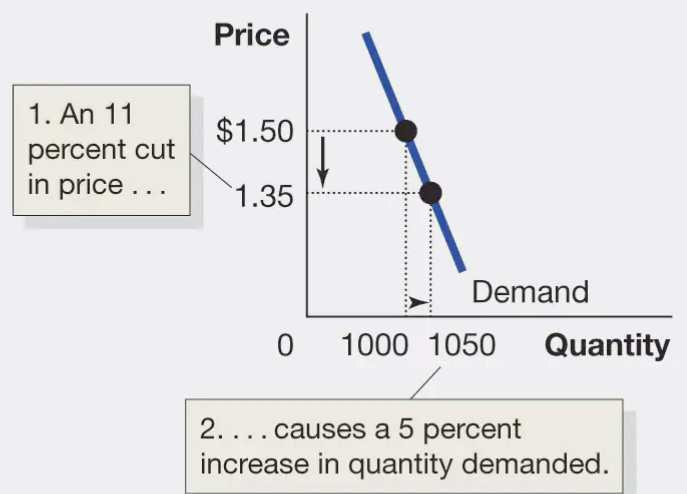
elastic

greater than 1



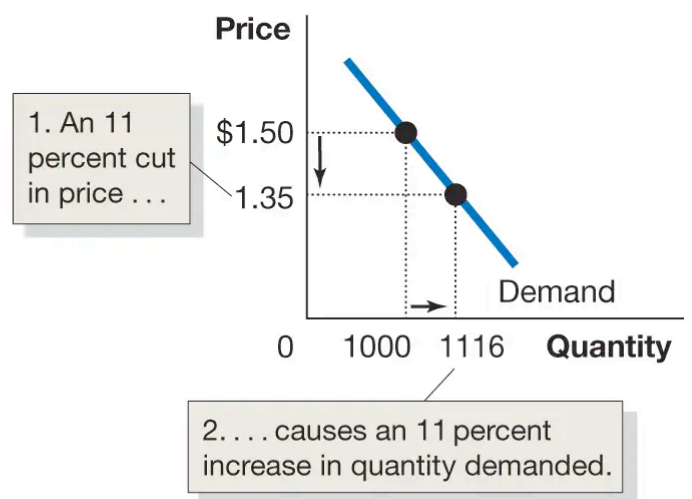
inelastic

less than 1



unit elastic

equal to 1

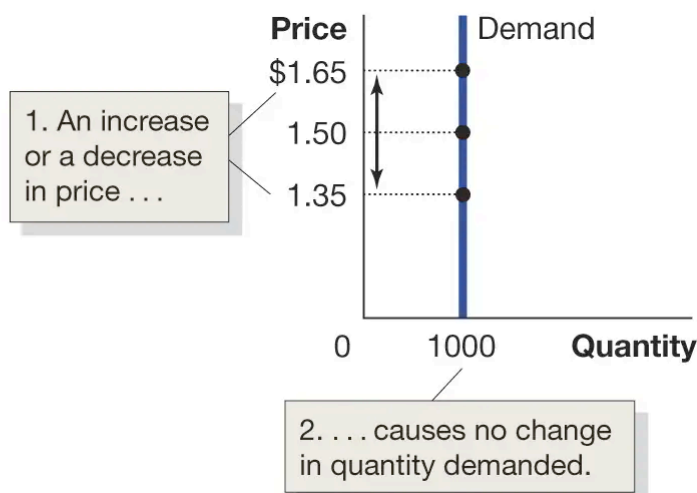


perfectly elastic

equal to infinity

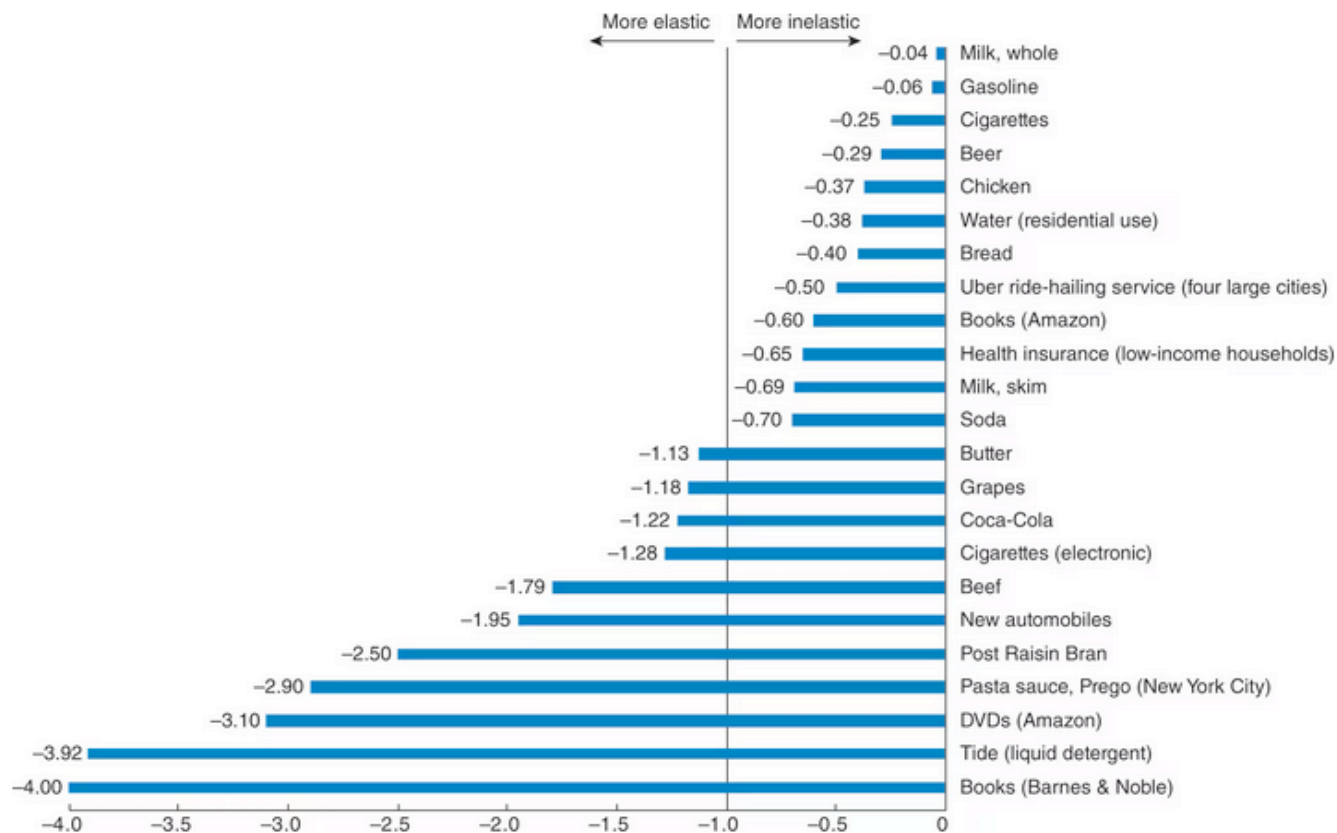


perfectly inelastic      equal to 0



## Figure 6.2

### Estimated Real-World Price Elasticities of Demand

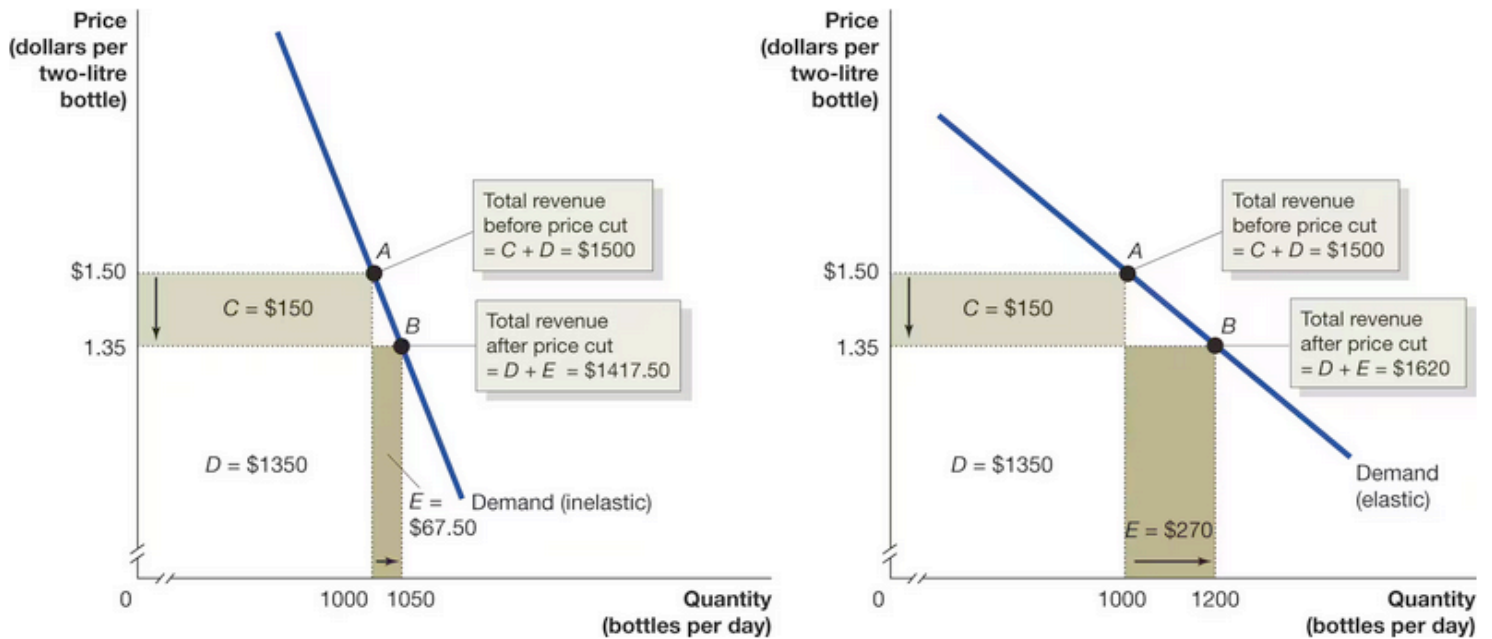


The larger (in absolute value) the price elasticity of demand, the more responsive a good is to changes in price. The smaller the price elasticity of demand, the less responsive a good is to changes in price. Goods for which there are few substitutes are price inelastic, as are broadly defined goods. Goods for which there are easily available substitutes, such as particular brand name products, are price elastic.

**SOURCE:** See Text Credits at the end of the book for a complete source list.

**Figure 6.3**

**The Relationship between Price Elasticity and Total Revenue**



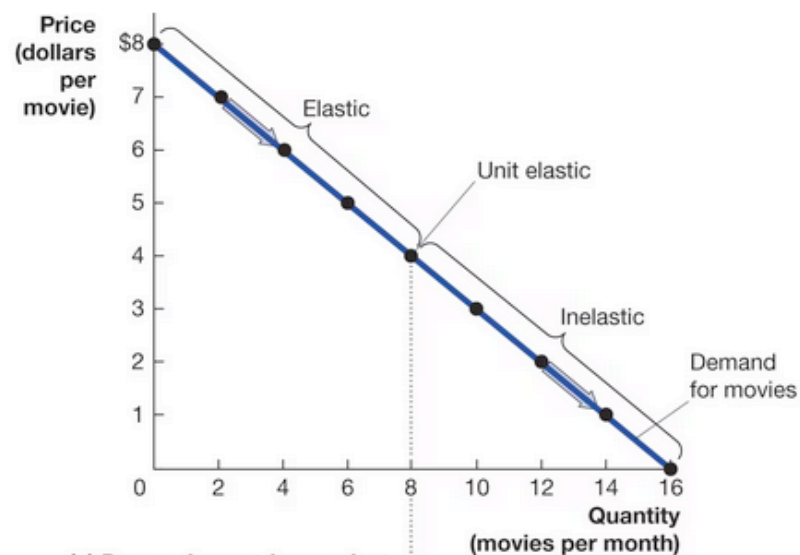
When demand is inelastic, a cut in price will decrease total revenue. In panel (a), at point A, the price is \$1.50, 1000 bottles are sold, and the total revenue you receive equals  $\$1.50 \times 1000$  bottles, or \$1500. At point B, cutting the price to \$1.35 increases the quantity demanded to 1050 bottles, but the fall in price more than offsets the increase in quantity. As a result, revenue falls to  $\$1.35 \times 1050$  gallons, or \$1417.50. When demand is elastic, a cut in the price will increase total revenue. In panel (b), at point A, the areas of rectangles C and D are still equal to \$1500. But, at point B, the areas of rectangles D and E are equal to  $\$1.35 \times 1200$  bottles, or \$1620. In this case, the increase in the quantity demanded is large enough to offset the fall in price, so total revenue increases.

**Table 6.2****The Relationship between Price Elasticity and Revenue**

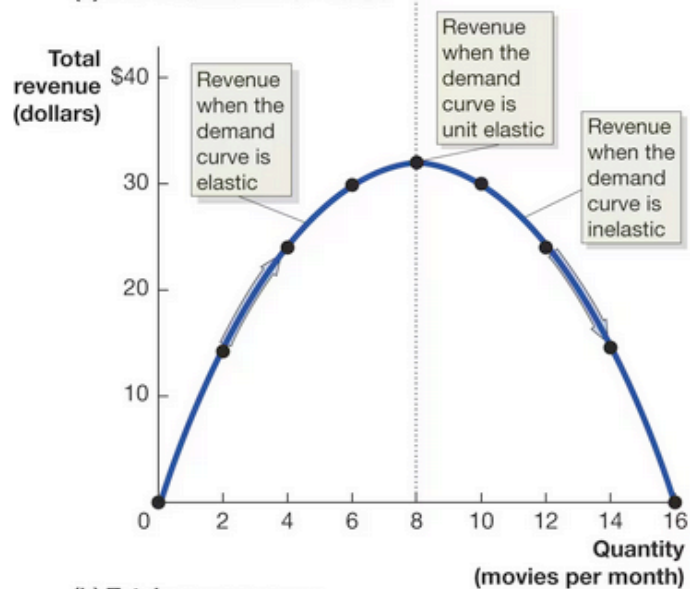
| If demand is . . . | then . . .                                   | because . . .                                                                                  |
|--------------------|----------------------------------------------|------------------------------------------------------------------------------------------------|
| elastic            | an increase in price reduces revenue         | the decrease in quantity demanded is proportionally <i>greater</i> than the increase in price. |
| elastic            | a decrease in price increases revenue        | the increase in quantity demanded is proportionally <i>greater</i> than the decrease in price. |
| inelastic          | an increase in price increases revenue       | the decrease in quantity demanded is proportionally <i>smaller</i> than the increase in price. |
| inelastic          | a decrease in price reduces revenue          | the increase in quantity demanded is proportionally <i>smaller</i> than the decrease in price. |
| unit elastic       | an increase in price does not affect revenue | the decrease in quantity demanded is proportionally <i>the same as</i> the increase in price.  |
| unit elastic       | a decrease in price does not affect revenue  | the increase in quantity demanded is proportionally <i>the same as</i> the decrease in price.  |

## Figure 6.4

Elasticity Is Not Constant along a Linear Demand Curve



(a) Demand curve for movies



(b) Total revenue curve

| Price | Quantity Demanded | Total Revenue |
|-------|-------------------|---------------|
| \$8   | 0                 | \$0           |
| 7     | 2                 | 14            |
| 6     | 4                 | 24            |
| 5     | 6                 | 30            |
| 4     | 8                 | 32            |
| 3     | 10                | 30            |
| 2     | 12                | 24            |
| 1     | 14                | 14            |
| 0     | 16                | 0             |

The data from the table are plotted in the graphs. Panel (a) shows that as we move down the demand curve for renting

**Table 6.3**

Summary of Cross-Price Elasticity of Demand

| If the products are . . . | then the cross-price elasticity of demand will be . . . | Example                                                |
|---------------------------|---------------------------------------------------------|--------------------------------------------------------|
| substitutes               | positive.                                               | Two brands of smartphones                              |
| complements               | negative.                                               | iPhones and applications downloaded from online stores |
| unrelated                 | zero.                                                   | iPhones and peanut butter                              |

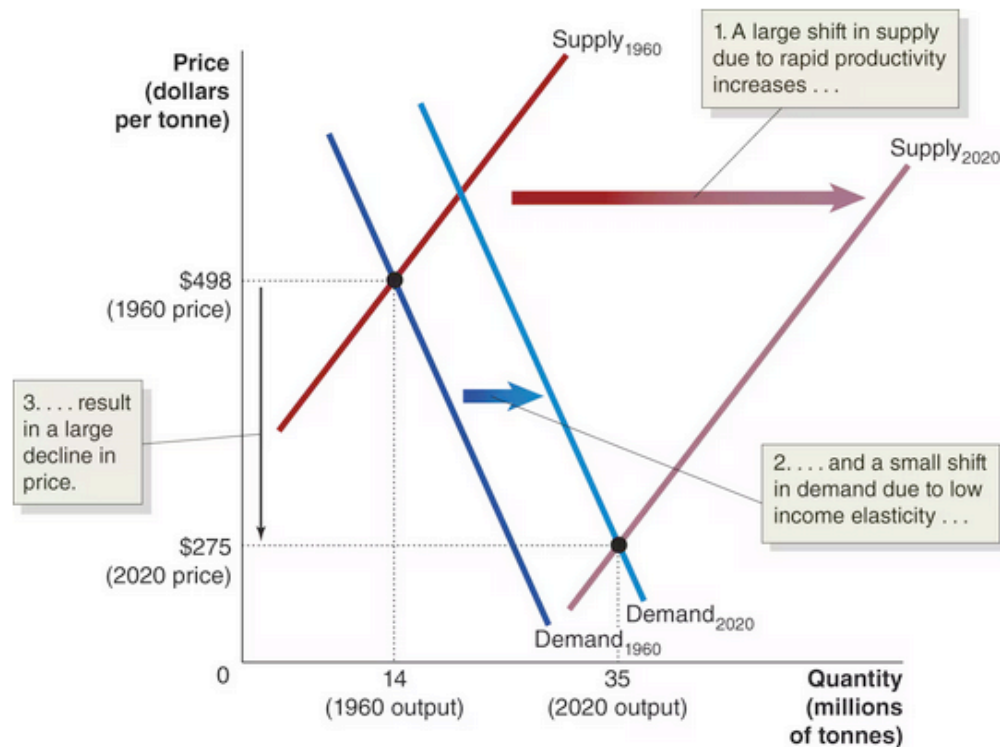
**Table 6.4**

Summary of Income Elasticity of Demand

| If the income elasticity of demand is . . . | then the good is . . .  | Example       |
|---------------------------------------------|-------------------------|---------------|
| positive but less than 1                    | normal and a necessity. | Bread         |
| positive and greater than 1                 | normal and a luxury.    | Caviar        |
| negative                                    | inferior.               | High-fat meat |

**Figure 6.5**

Elasticity and the Disappearing Family Farm



In 1960, Canadian farmers produced 14 million tonnes of wheat at a price of \$498 per tonne. Over the next 60 years, rapid increases in farm productivity caused a large shift to the right in the supply curve for wheat. Production in 2020 was at 35 million tonnes. The income elasticity of demand for wheat is low, so the demand for wheat increased relatively little over this period. Because the demand for wheat is also price inelastic, the large shift in the supply curve and the small shift in the demand curve resulted in a sharp decline in the price of wheat, from \$498 per tonne in 1960 to \$275 per tonne in 2020.

**SOURCE:** Data from Statistics Canada, Bank of Canada, and World Bank.

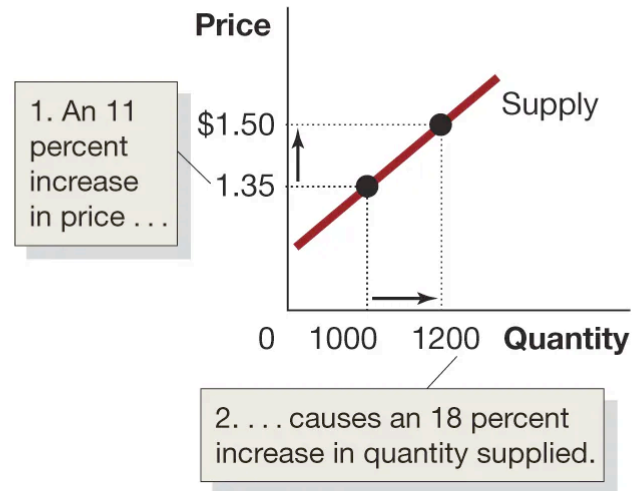
$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

If supply is . . .

then the value of  
price elasticity is . . .

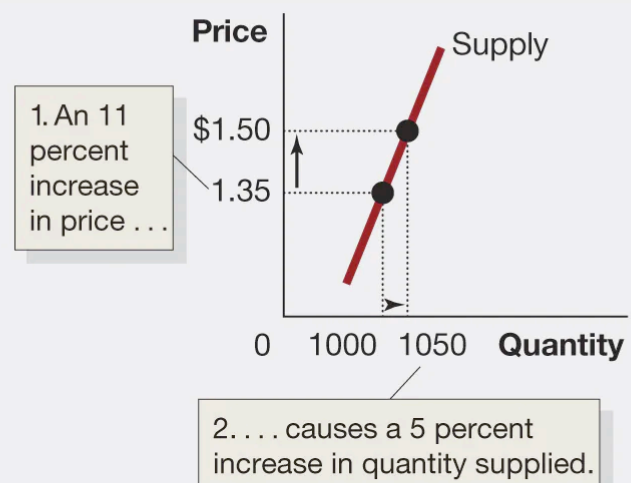
elastic

greater than 1



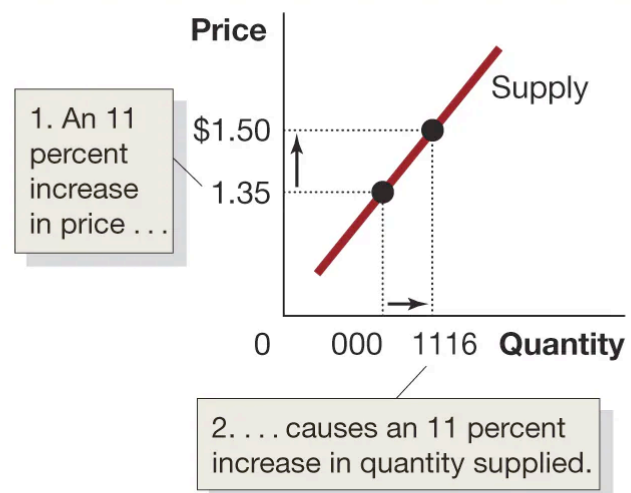
inelastic

less than 1



unit elastic

equal to 1



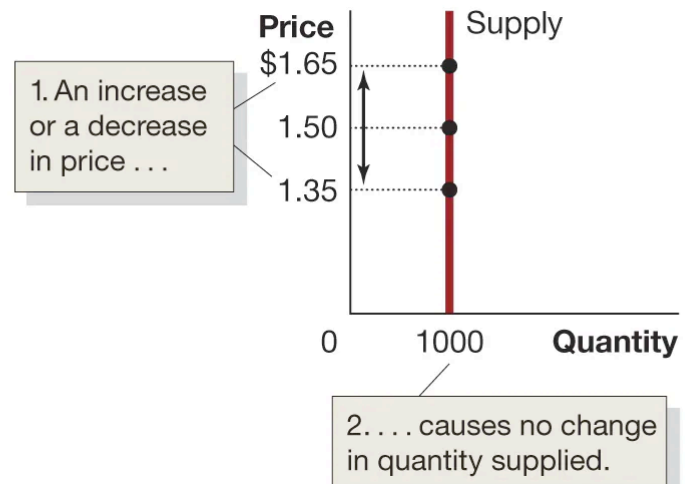
perfectly elastic

equal to infinity

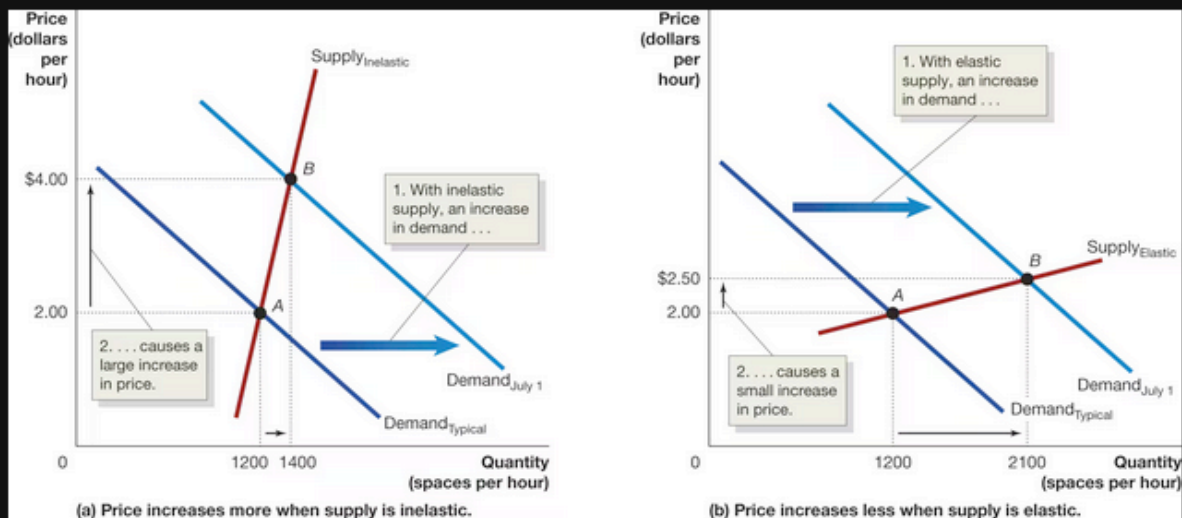


perfectly inelastic

equal to 0

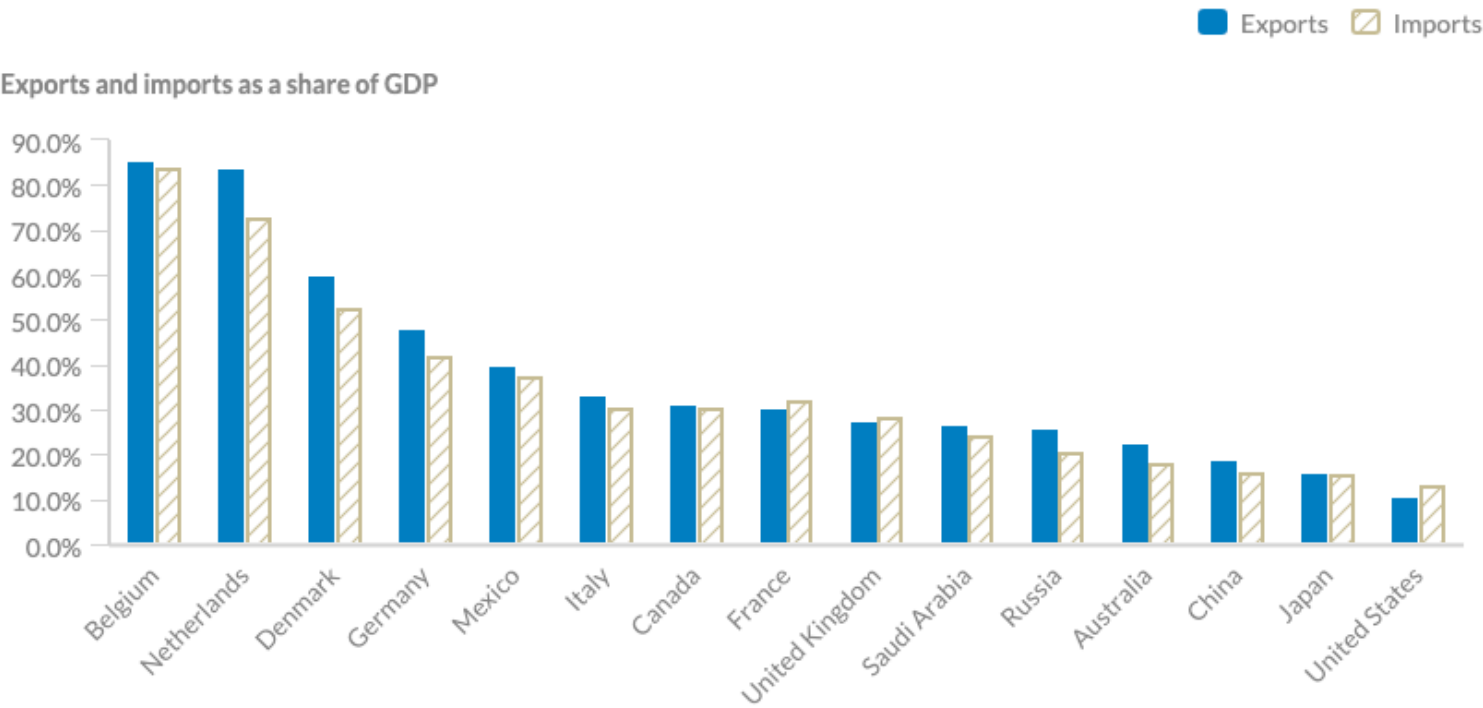
**Figure 6.6**

Changes in Price Depend on the Price Elasticity of Supply



In panel (a),  $Demand_{Typical}$  represents the typical demand for parking spaces on a summer weekend at a beach resort.  $Demand_{July 1}$  represents demand on July 1. Because supply is inelastic, the shift in equilibrium from point A to point B results in a large increase in price—from \$2.00 per hour to \$4.00—but only a small increase in the quantity of spaces supplied—from 1200 to 1400. In panel (b), supply is elastic. As a result, the change in equilibrium from point A to point B results in a smaller increase in price and a larger increase in the quantity supplied. An increase in price from \$2.00 per hour to \$2.50 is sufficient to increase the quantity of parking supplied from 1200 to 2100.

Figure 7.3 Exports and Imports as a Share of the Economy



SOURCE: Organisation for Economic Co-operation and Development, OECD.STAT, <https://stats.oecd.org/>.

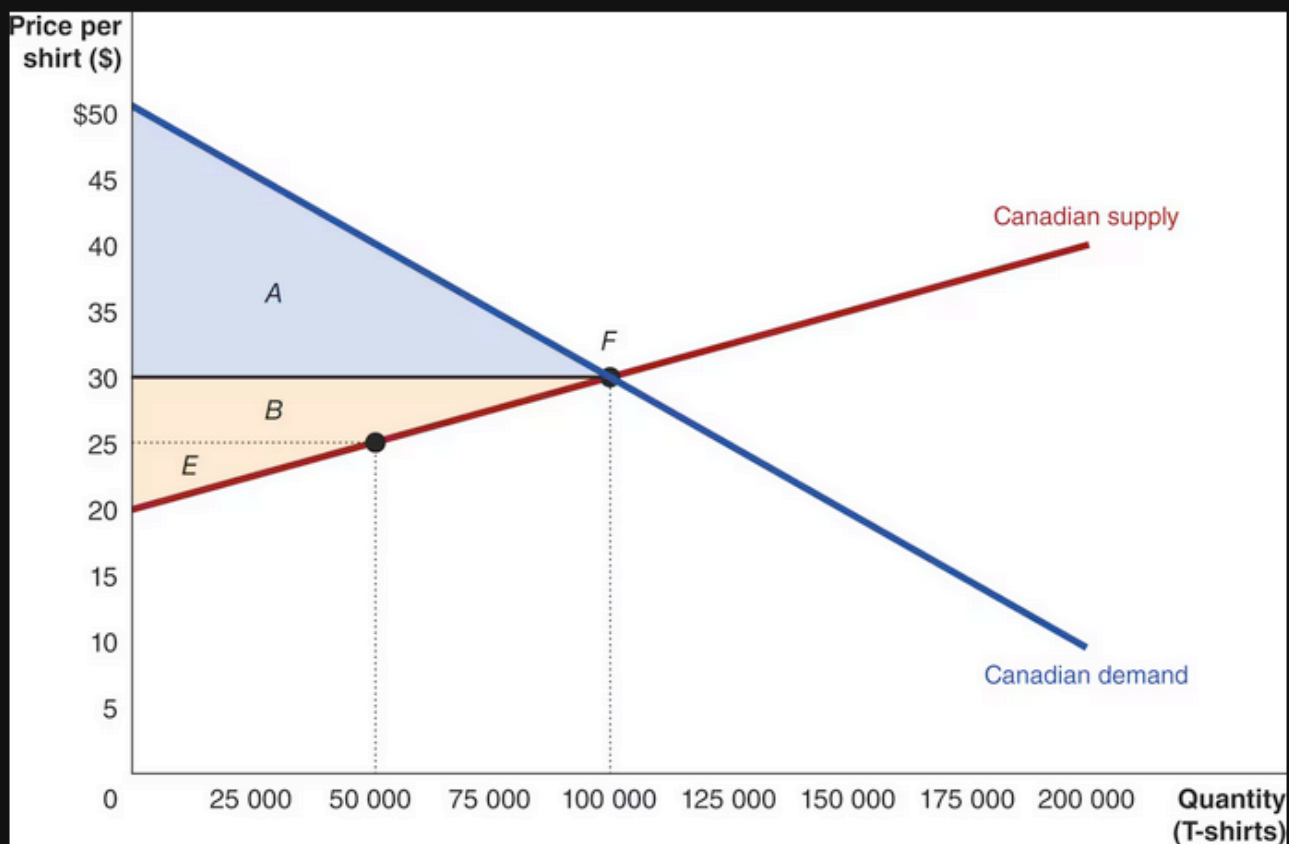
Table 7.3

Gains from Trade for Canada and South Korea

| Without Trade                           |                  |            |             |             |             |            |
|-----------------------------------------|------------------|------------|-------------|-------------|-------------|------------|
| Production and Consumption              |                  |            |             |             |             |            |
|                                         | Televisions      | Cellphones |             |             |             |            |
| Canada                                  | 1500             | 1000       |             |             |             |            |
| South Korea                             | 9000             | 1500       |             |             |             |            |
|                                         |                  |            | With Trade  |             |             |            |
|                                         | Production       |            | Trade       |             | Consumption |            |
|                                         | Televisions      | Cellphones | Televisions | Cellphones  | Televisions | Cellphones |
| Canada                                  | 0                | 4000       | Import 1500 | Export 1500 | 1500        | 2500       |
| South Korea                             | 12000            | 0          | Export 1500 | Import 1500 | 10500       | 1500       |
| Gains from Trade: Increased Consumption |                  |            |             |             |             |            |
| Canada                                  | 1500 cellphones  |            |             |             |             |            |
| South Korea                             | 1500 televisions |            |             |             |             |            |

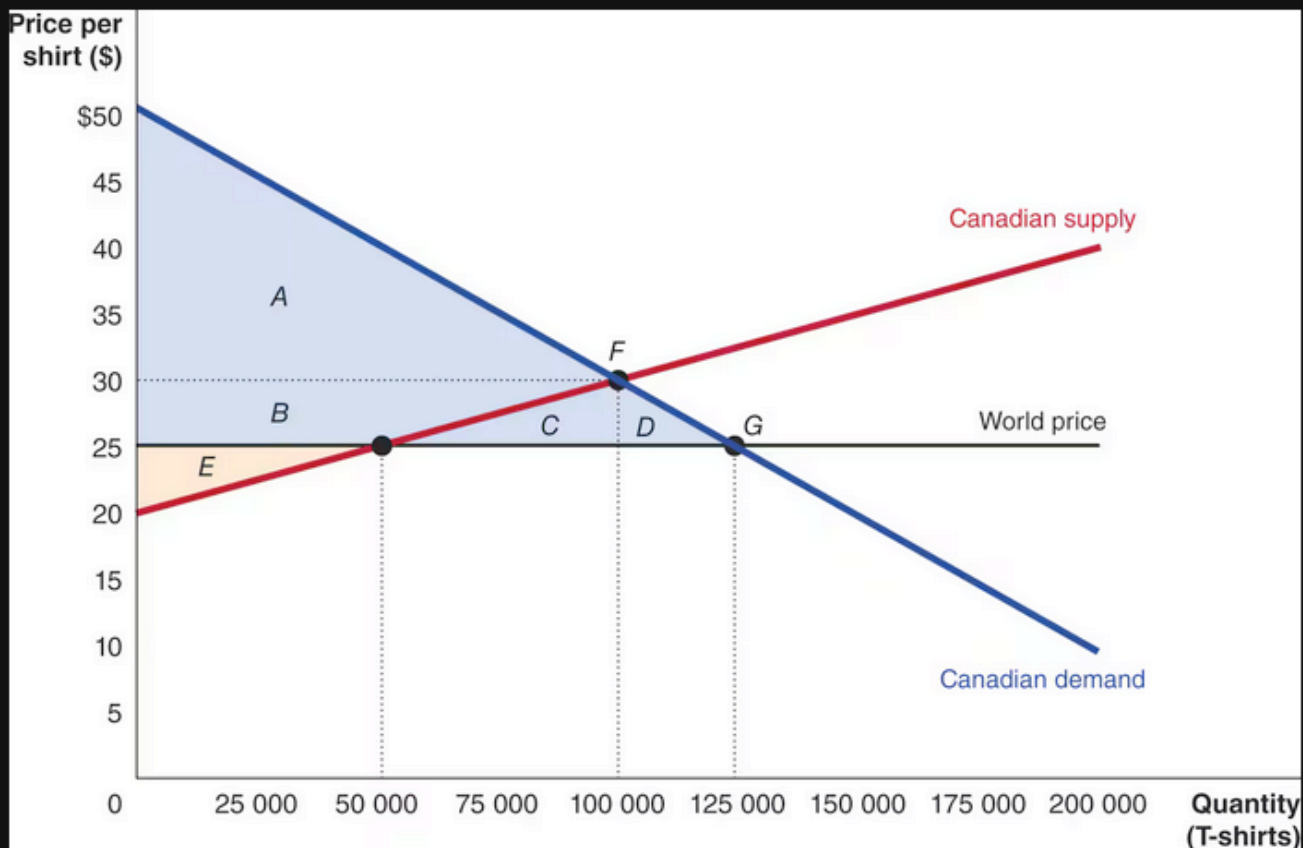
**Figure 7.4**

The Canadian Market for T-shirts in Autarky



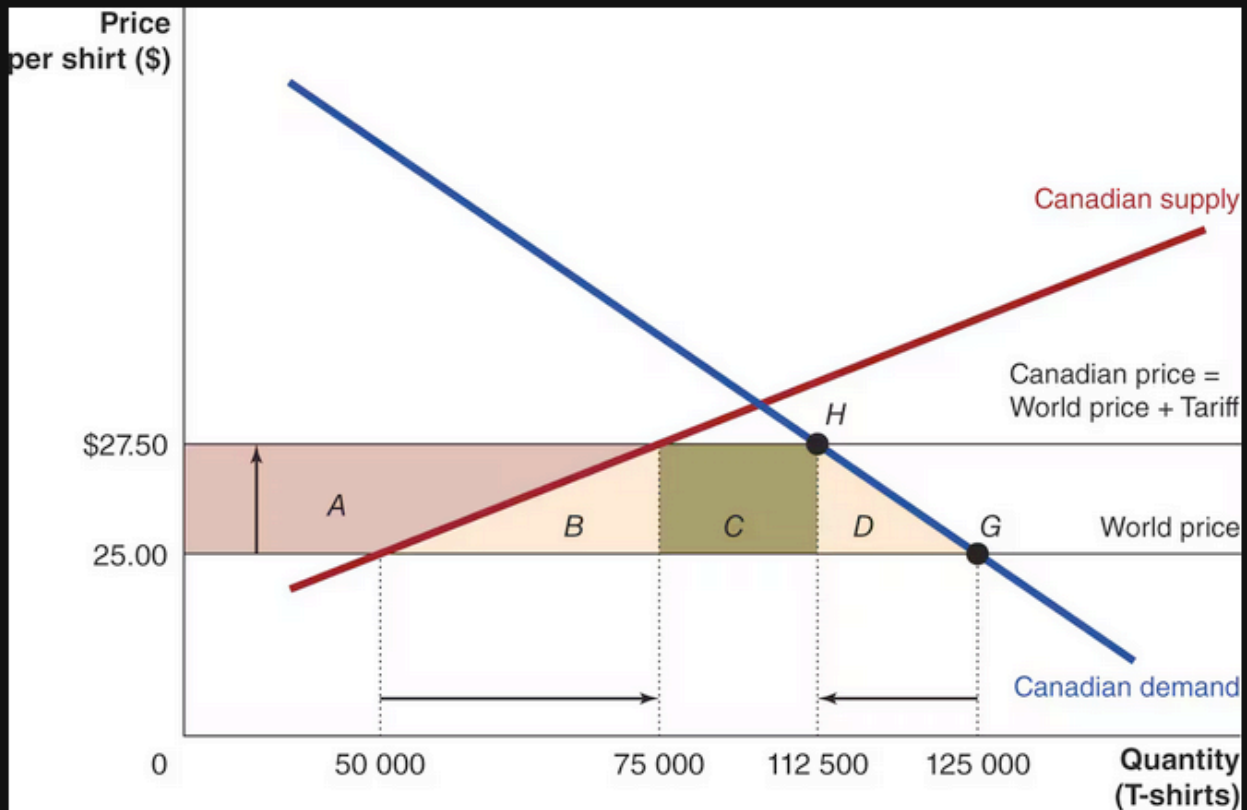
**Figure 7.5**

The Canadian Market for T-shirts after Trade



**Figure 7.8**

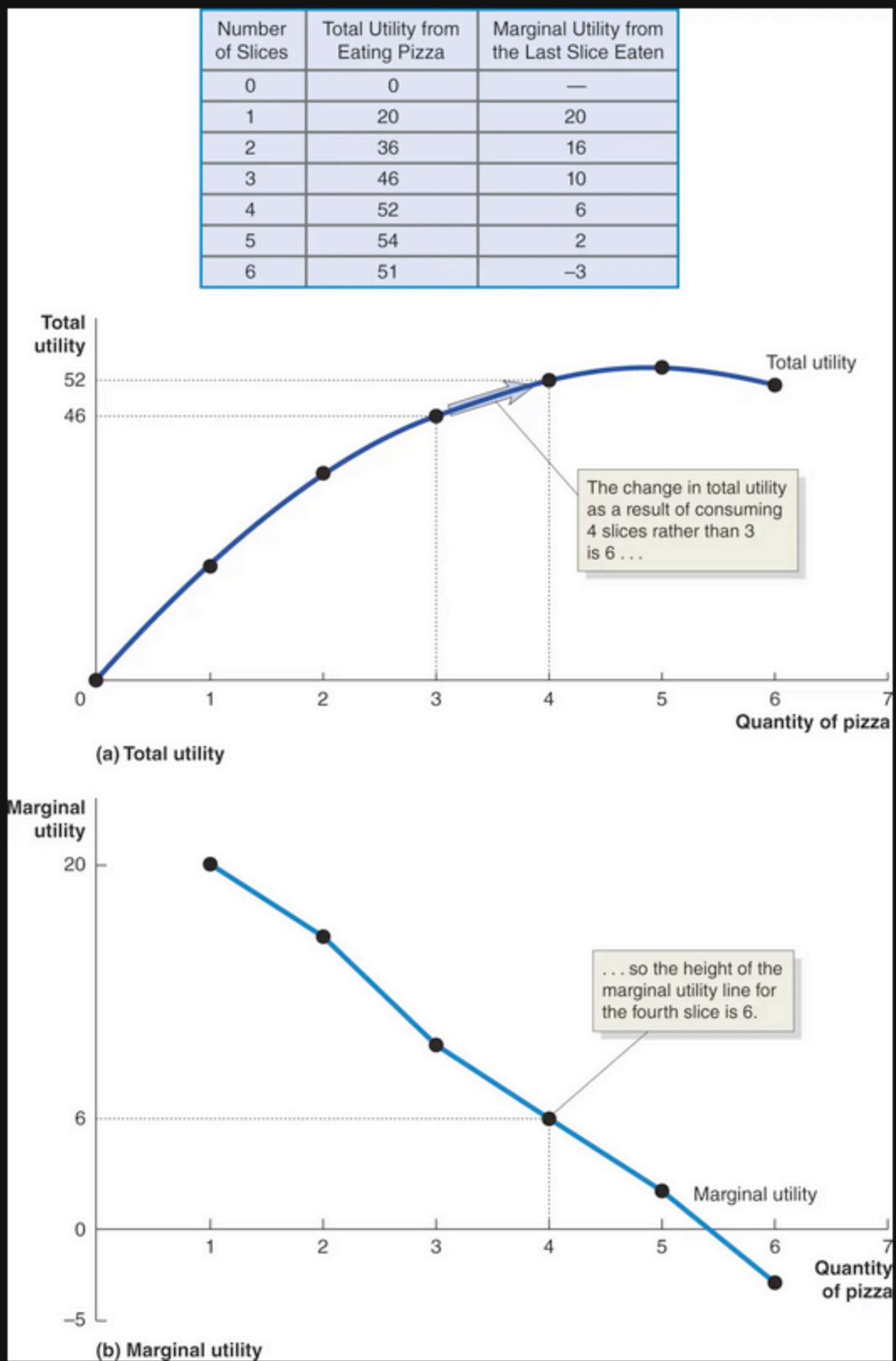
The Canadian T-shirt Market with a Tariff



Without a sugar quota, Canadians would pay the world price of \$0.35 per kilogram and purchase 3 million kilograms of sugar, importing 2.5 million kilograms. To find the price of sugar in the Canadian market after a quota of 0.5 million kilograms has been imposed, we need to look for a price at which the quantity demanded by Canadians is 0.5 million kilograms more than the quantity supplied by Canadian firms. The excess demand will be made up by imports. Thus, after the federal government imposes an import quota of 0.5 million kilograms, Canadians will have to pay \$0.6 per kilogram for sugar, and they will buy only 2 million kilograms. The price increase that results from the import quota means consumer surplus will fall by the area  $A + B + C + D$ .

**Figure 8.1**

Total and Marginal Utility from Eating at a Grey Cup Party



**Table 8.2**

Converting Marginal Utility to Marginal Utility per Dollar

| (1)<br>Slices of<br>Pizza | (2)<br>Marginal Utility<br>( $MU_{Pizza}$ ) | (3)<br>Marginal Utility per<br>Dollar ( $\frac{MU_{Pizza}}{P_{Pizza}}$ ) | (4)<br>Cups of<br>Coke | (5)<br>Marginal Utility<br>( $MU_{Coke}$ ) | (6)<br>Marginal Utility per<br>Dollar ( $\frac{MU_{Coke}}{P_{Coke}}$ ) |
|---------------------------|---------------------------------------------|--------------------------------------------------------------------------|------------------------|--------------------------------------------|------------------------------------------------------------------------|
| 1                         | 20                                          | 10                                                                       | 1                      | 20                                         | 20                                                                     |
| 2                         | 16                                          | 8                                                                        | 2                      | 15                                         | 15                                                                     |
| 3                         | 10                                          | 5                                                                        | 3                      | 10                                         | 10                                                                     |
| 4                         | 6                                           | 3                                                                        | 4                      | 5                                          | 5                                                                      |
| 5                         | 2                                           | 1                                                                        | 5                      | 3                                          | 3                                                                      |
| 6                         | -3                                          | -1.5                                                                     | 6                      | -1                                         | -1                                                                     |

**Table 8.3**

Equalizing Marginal Utility per Dollar Spent

| Combinations of Pizza and Coke with Equal<br>Marginal Utilities per Dollar | Marginal Utility per<br>Dollar ( $MU/P$ ) | Total<br>Spending  | Total Utility   |
|----------------------------------------------------------------------------|-------------------------------------------|--------------------|-----------------|
| 1 slice of pizza and 3 cups of Coke                                        | 10                                        | $\$2 + \$3 = \$5$  | $20 + 45 = 65$  |
| 3 slices of pizza and 4 cups of Coke                                       | 5                                         | $\$6 + \$4 = \$10$ | $46 + 50 = 96$  |
| 4 slices of pizza and 5 cups of Coke                                       | 3                                         | $\$8 + \$5 = \$13$ | $52 + 53 = 105$ |

We can summarize the two conditions for maximizing utility:

$$1. \frac{MU_{Pizza}}{P_{Pizza}} = \frac{MU_{Coke}}{P_{Coke}}$$

$$2. \text{Spending on pizza} + \text{Spending on Coke} = \text{Amount available to be spent}$$

The first condition shows that the marginal utility per dollar spent must be the same for both goods. The second condition is the budget constraint, which states that total spending on both goods must equal the amount available to be spent. Of course, these conditions for maximizing utility apply not just to pizza and Coke but to any two pairs of goods.

**Table 8.4****Income Effect and Substitution Effect of a Price Change**

| When price . . . | consumer purchasing power . . . | The income effect causes quantity demanded to . . .        | The substitution effect causes the opportunity cost of consuming a good to . . .               |
|------------------|---------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| decreases,       | increases.                      | increase if a normal good or decrease if an inferior good. | decrease when the price decreases, which causes the quantity of the good demanded to increase. |
| increases,       | decreases.                      | decrease if a normal good or increase if an inferior good. | increase when the price increases, which causes the quantity of the good demanded to decrease. |

## The Difference between Fixed Costs and Variable Costs

Total cost is the cost of all the inputs a firm uses in production. We have just seen that in the short run, some inputs are fixed and others are variable. The costs of the fixed inputs are called *fixed costs*, and the costs of the variable inputs are called *variable costs*. We can also think of variable costs as the costs that change as output changes. Similarly, fixed costs are costs that remain constant as output changes. A typical firm's variable costs include its labour costs, raw material costs, and costs of electricity and other utilities. Typical fixed costs include lease payments for factory or retail space, payments for fire insurance, and payments for advertising. All of a firm's costs are either fixed or variable, so we can state the following:

$$\text{Total cost} = \text{Fixed cost} + \text{Variable cost}$$

Or, using symbols

$$T\ C = F\ C + V\ C$$

Table 9.1

Ebba Kantzen’s Costs per Year

| Ebba’s Explicit Costs                            | Amount    |
|--------------------------------------------------|-----------|
| Pizza dough, tomato sauce, and other ingredients | \$20 000  |
| Wages                                            | 48 000    |
| Interest payments on her loan to buy pizza ovens | 10 000    |
| Electricity                                      | 6 000     |
| Lease payment for her restaurant                 | 24 000    |
| Ebba’s Implicit Costs                            |           |
| Forgone salary                                   | 30 000    |
| Forgone interest                                 | 3 000     |
| Economic depreciation                            | 10 000    |
| Total                                            | \$151 000 |

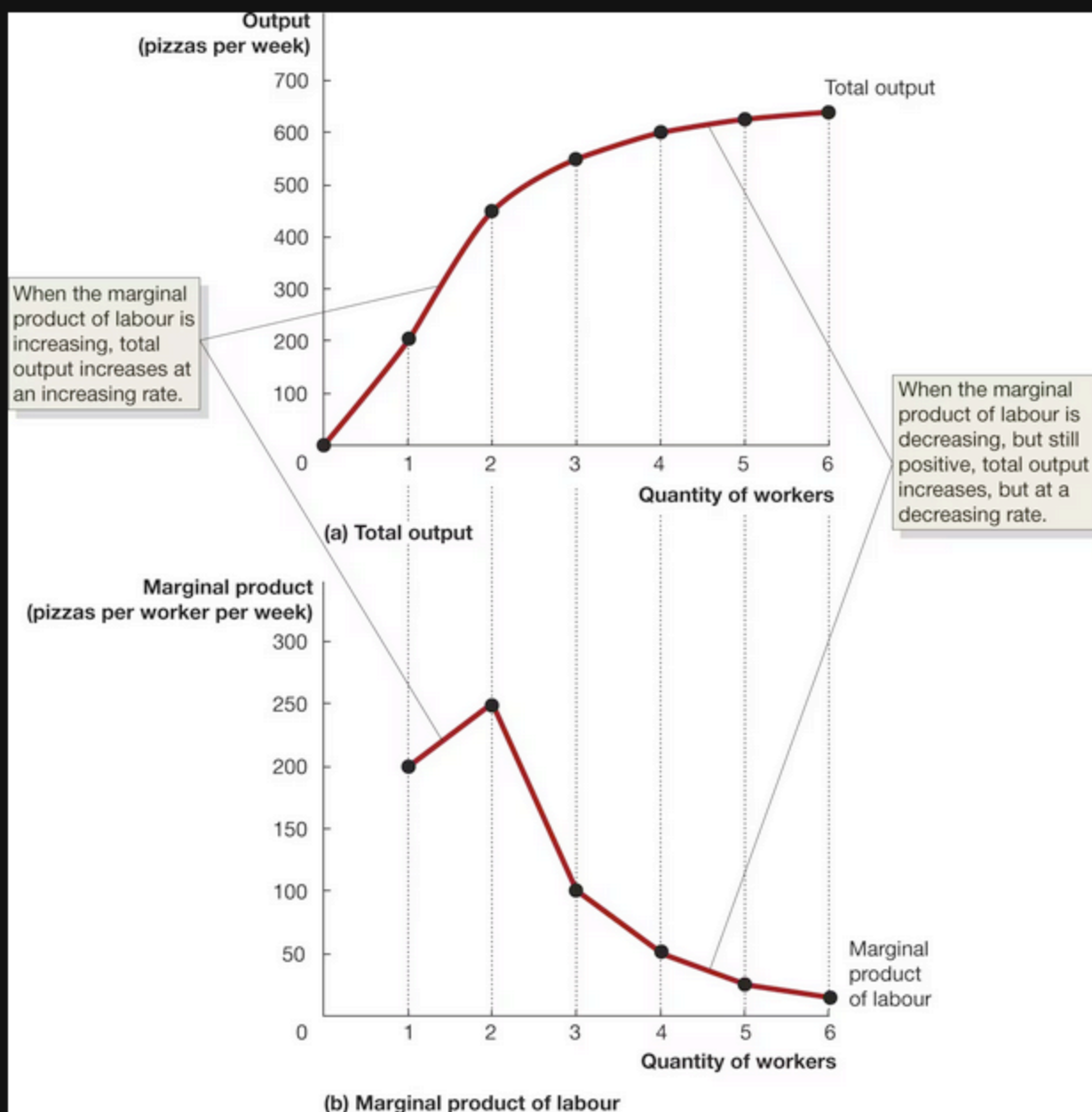
Table 9.2

Short-Run Production and Cost at Ebba Kantzen’s Restaurant

| (1)<br>Quantity of<br>Workers | (2) Quantity<br>of Pizza<br>Ovens | (3) Quantity<br>of Pizzas per<br>Week | (4) Cost of<br>Pizza<br>Ovens<br>(fixed cost) | (5) Cost of<br>Workers<br>(variable<br>cost) | (6) Total<br>Cost of<br>Pizzas per<br>Week | (7) Cost per<br>Pizza<br>(average total<br>cost) |
|-------------------------------|-----------------------------------|---------------------------------------|-----------------------------------------------|----------------------------------------------|--------------------------------------------|--------------------------------------------------|
| 0                             | 2                                 | 0                                     | \$800                                         | \$0                                          | \$800                                      | —                                                |
| 1                             | 2                                 | 200                                   | 800                                           | 650                                          | 1450                                       | \$7.25                                           |
| 2                             | 2                                 | 450                                   | 800                                           | 1300                                         | 2100                                       | 4.67                                             |
| 3                             | 2                                 | 550                                   | 800                                           | 1950                                         | 2750                                       | 5.00                                             |
| 4                             | 2                                 | 600                                   | 800                                           | 2600                                         | 3400                                       | 5.67                                             |
| 5                             | 2                                 | 625                                   | 800                                           | 3250                                         | 4050                                       | 6.48                                             |
| 6                             | 2                                 | 640                                   | 800                                           | 3900                                         | 4700                                       | 7.34                                             |

**Figure 9.2**

**Total Output and the Marginal Product of Labour**



In panel (a), output increases as more workers are hired, but the increase in output does not occur at a constant rate. Because of specialization and the division of labour, output at first increases at an increasing rate, with each additional worker hired causing production to increase by a *greater* amount than did the hiring of the previous worker. When the point of diminishing returns is reached, production increases at a decreasing rate. Each additional worker Ebba hires after the second worker causes production to increase by a *smaller* amount than did the hiring of the previous worker. In panel (b), the *marginal product of labour* is the additional output produced as a result of hiring one more worker. The marginal product of labour rises initially because of the effects of specialization and division of labour, and then it falls because of the effects of diminishing returns.

workers hired changes. We can also calculate the average product of labour, which is the total output produced by a firm divided by the quantity of workers. For example, using the numbers in [Table 9.3](#), if Ebba hires four workers to produce 600 pizzas, the average product of labour is  $600/4 = 150$ .

Note the relationship between the marginal and average products: *The average product of labour is the average of the marginal products of labour.* For example, the numbers from [Table 9.3](#) show that the marginal product of the first worker Ebba hires is 200, the marginal product of the second worker is 250, and the marginal product of the third worker is 100. Therefore, the average product of labour for three workers is 183.3:

$$183.3 = (200 + 250 + 100)/3$$

|                                             |                                            |                                             |                                            |
|---------------------------------------------|--------------------------------------------|---------------------------------------------|--------------------------------------------|
| Average product of labour for three workers | Marginal product of labour of first worker | Marginal product of labour of second worker | Marginal product of labour of third worker |
|---------------------------------------------|--------------------------------------------|---------------------------------------------|--------------------------------------------|

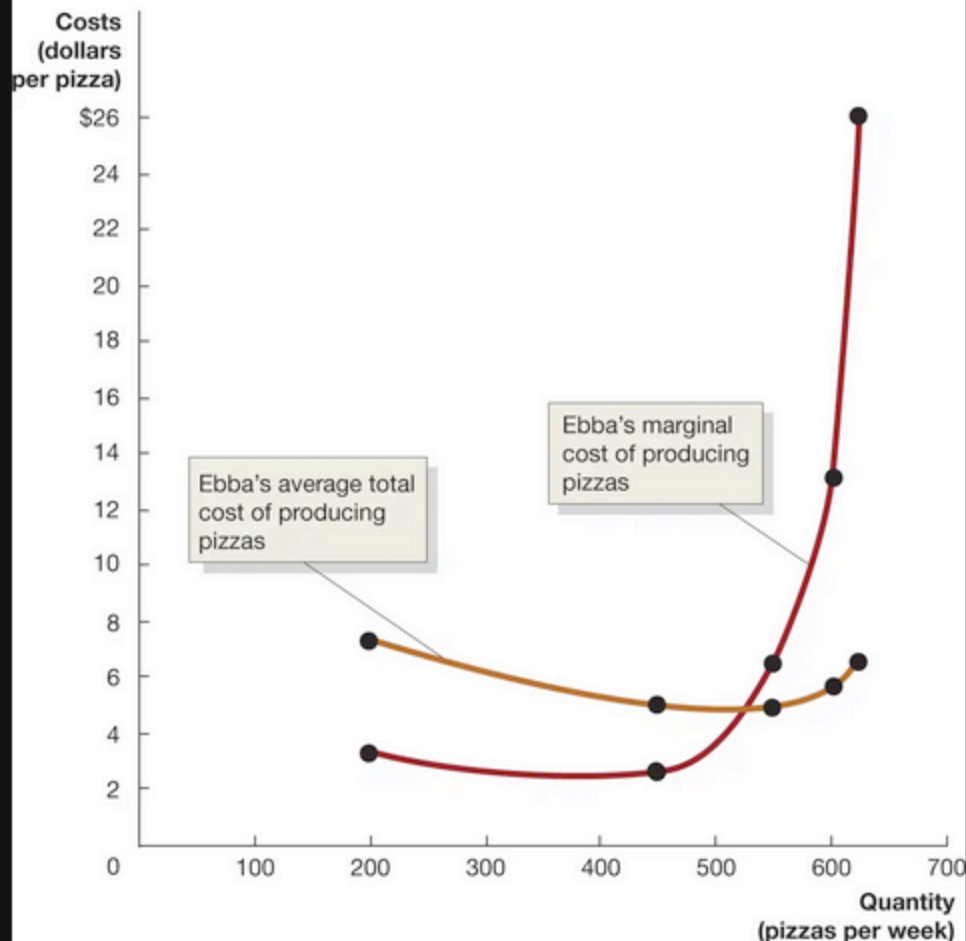
By taking the average of the marginal products of the first three workers, we have the average product of the three workers.

Whenever the marginal product of labour is greater than the average product of labour, the average product of labour must be increasing. This statement is true for the same reason that a person 188 cm tall entering a room where the average height is 175 cm raises the average height of people in the room. Whenever the marginal product of labour is less than the average product of labour, the average product of labour must be decreasing. The marginal product of labour equals the average product of labour at the quantity of workers for which the average product of labour is at its maximum.

**Figure 9.4**

Ebba Kantzen's Marginal Cost and Average Total Cost of Producing Pizzas

| Quantity of Workers | Quantity of Pizzas | Marginal Product of Labour | Total Cost of Pizzas | Marginal Cost of Pizzas | Average Total Cost of Pizzas |
|---------------------|--------------------|----------------------------|----------------------|-------------------------|------------------------------|
| 0                   | 0                  | —                          | \$800                | —                       | —                            |
| 1                   | 200                | 200                        | 1450                 | \$3.25                  | \$7.25                       |
| 2                   | 450                | 250                        | 2100                 | 2.60                    | 4.67                         |
| 3                   | 550                | 100                        | 2750                 | 6.50                    | 5.00                         |
| 4                   | 600                | 50                         | 3400                 | 13.00                   | 5.67                         |
| 5                   | 625                | 25                         | 4050                 | 26.00                   | 6.48                         |
| 6                   | 640                | 15                         | 4700                 | 43.33                   | 7.34                         |



We can use the information in the table to calculate Ebba's marginal cost and average total cost of producing pizzas. For the first two workers hired, the marginal product of labour is increasing, which causes the marginal cost of production to fall. For the last four workers hired, the marginal product of labour is falling, which causes the marginal cost of production to increase. Therefore, the marginal cost curve falls and then rises—that is, it has a U shape—because the marginal product of labour rises and then falls. As long as marginal cost is below average total cost, average total cost will be falling. When marginal cost is above average total cost, average total cost will be rising. The relationship between marginal cost and

$$\text{Average total cost} = ATC = \frac{TC}{Q}$$

$$\text{Average fixed cost} = AFC = \frac{FC}{Q}$$

$$\text{Average variable cost} = AVC = \frac{VC}{Q}$$

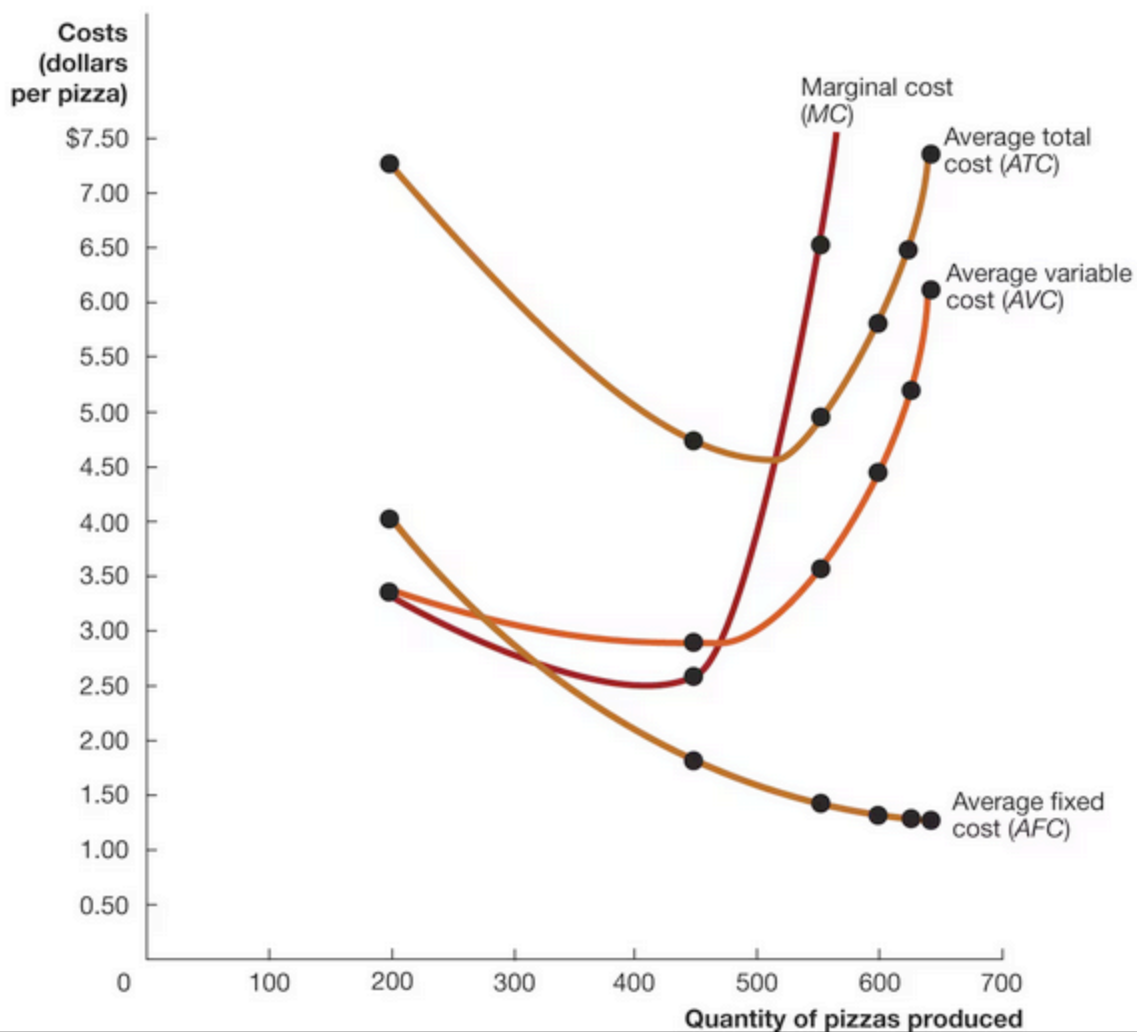
Finally, notice that average total cost is the sum of average fixed cost plus average variable cost:

$$ATC = AFC + AVC$$

**Figure 9.5**

Costs at Ebba Kantzen's Restaurant

| Quantity of Workers | Quantity of Ovens | Quantity of Pizzas | Cost of Ovens (fixed cost) | Cost of Workers (variable cost) | Total Cost of Pizzas | ATC    | AFC    | AVC    | MC     |
|---------------------|-------------------|--------------------|----------------------------|---------------------------------|----------------------|--------|--------|--------|--------|
| 0                   | 2                 | 0                  | \$800                      | \$0                             | \$800                | —      | —      | —      | —      |
| 1                   | 2                 | 200                | 800                        | 650                             | 1450                 | \$7.25 | \$4.00 | \$3.25 | \$3.25 |
| 2                   | 2                 | 450                | 800                        | 1300                            | 2100                 | 4.67   | 1.78   | 2.89   | 2.60   |
| 3                   | 2                 | 550                | 800                        | 1950                            | 2750                 | 5.00   | 1.45   | 3.54   | 6.50   |
| 4                   | 2                 | 600                | 800                        | 2600                            | 3400                 | 5.67   | 1.33   | 4.33   | 13.00  |
| 5                   | 2                 | 625                | 800                        | 3250                            | 4050                 | 6.48   | 1.28   | 5.20   | 26.00  |
| 6                   | 2                 | 640                | 800                        | 3900                            | 4700                 | 7.34   | 1.25   | 6.09   | 43.33  |

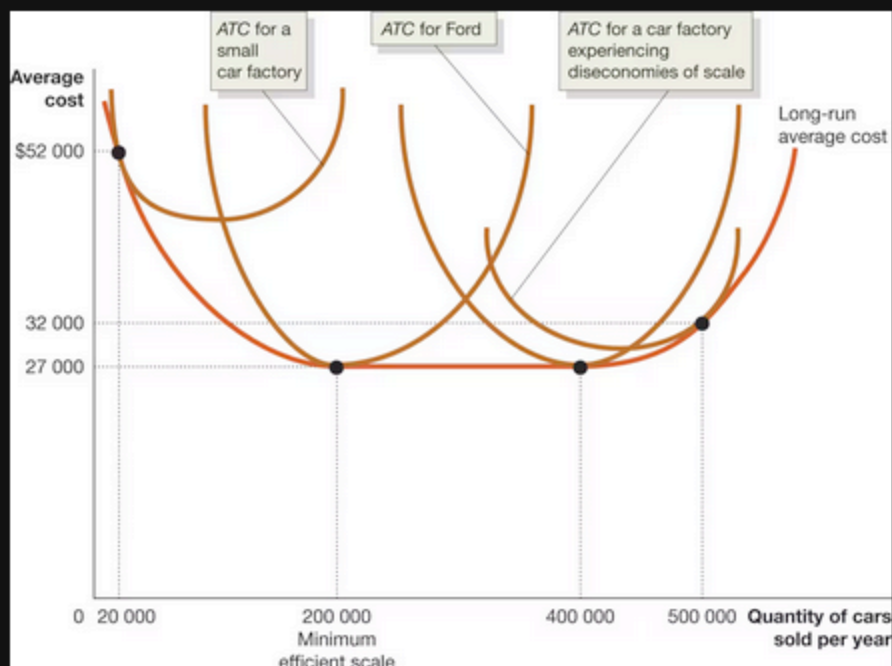


Ebba's costs of making pizzas are shown in the table and plotted in the graph. Notice three important facts about the graph: (1) The marginal cost (MC), average total cost (ATC), and average variable cost (AVC) curves are all U shaped, and the

The marginal cost (MC), average total cost (ATC), and average variable cost (AVC) curves are all U shaped, and the marginal cost curve intersects both the average variable cost curve and the average total cost curve at their minimum points. When marginal cost is below either average variable cost or average total cost, it causes them to decrease. When marginal cost is above average variable cost or average total cost, it causes them to increase. Therefore, when marginal cost equals average variable cost or average total cost, they must be at their minimum points.

**Figure 9.6**

The Relationship between Short-Run Average Cost and Long-Run Average Cost



If a small car company expects to sell only 20 000 cars per year, it will be able to produce cars at the lowest average cost of \$52 000 per car if it builds the small factory represented by the ATC curve on the left of the figure. A larger factory will be able to produce 200 000 cars per year at a lower cost of \$27 000 per car. An automobile factory producing 200 000 cars per year and a factory producing 400 000 cars per year will experience constant returns to scale and will have the same average cost. An automobile factory assembling 200 000 cars per year will have reached minimum efficient scale. Very large automobile factories will experience diseconomies of scale, and their average costs will rise as production increases beyond 400 000 cars per year.

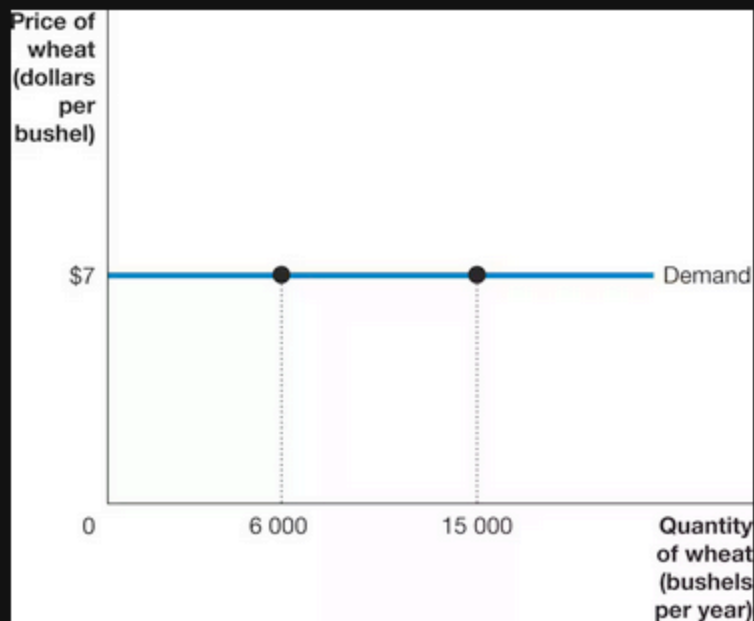
**Table 9.4**

A Summary of Definitions of Cost

| Term                  | Definition                                                                | Symbols and Equations             |
|-----------------------|---------------------------------------------------------------------------|-----------------------------------|
| Total cost            | Cost of all inputs used by a firm, or fixed costs plus variable costs     | $TC$                              |
| Fixed costs           | Costs that remain constant as a firm's level of output changes            | $FC$                              |
| Variable costs        | Costs that change as a firm's level of output changes                     | $VC$                              |
| Marginal cost         | An increase in total cost resulting from producing another unit of output | $MC = \frac{\Delta TC}{\Delta Q}$ |
| Average total cost    | Total cost divided by the quantity of output produced                     | $ATC = \frac{TC}{Q}$              |
| Average fixed cost    | Fixed cost divided by the quantity of output produced                     | $AFC = \frac{FC}{Q}$              |
| Average variable cost | Variable cost divided by the quantity of output produced                  | $AVC = \frac{VC}{Q}$              |
| Implicit cost         | A non-monetary opportunity cost                                           | —                                 |
| Explicit cost         | A cost that involves spending money                                       | —                                 |

**Figure 10.1**

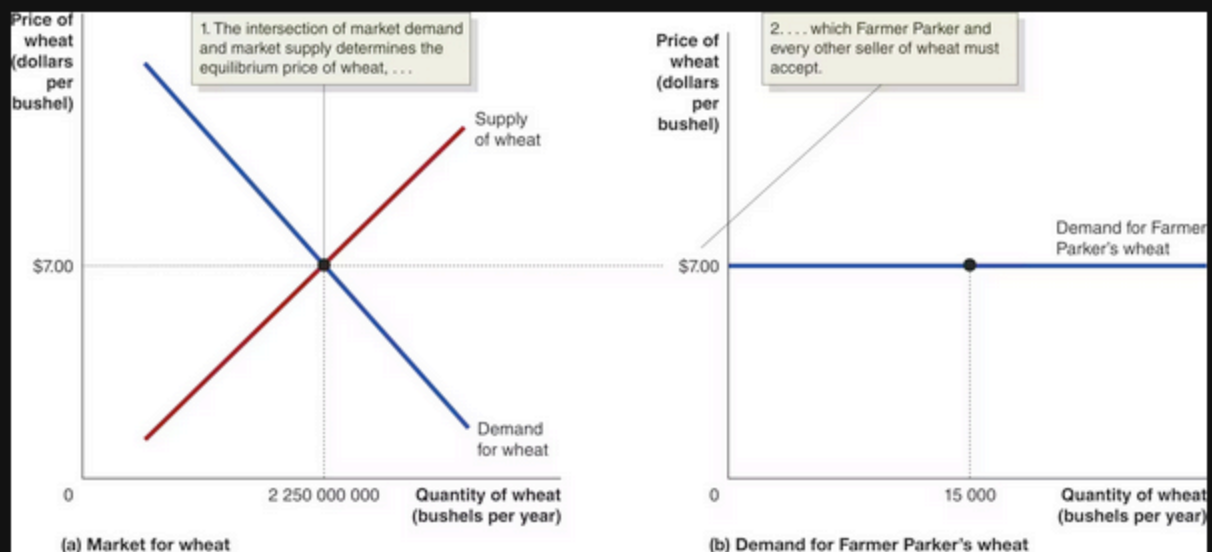
A Perfectly Competitive Firm Faces a Horizontal Demand Curve



A firm in a perfectly competitive market is selling exactly the same product as all of the other firms in the market. Therefore, it can sell as much as it wants at the current market price, but it cannot sell anything at all if it raises the price by even \$0.01. As a result, the demand curve for a perfectly competitive firm's output is a horizontal line. In the figure, whether the wheat farmer sells 6000 bushels or 15 000 bushels per year has no effect on the market price of \$7.

**Figure 10.2**

The Market Demand for Wheat versus the Demand for One Farmer's Wheat



In a perfectly competitive market, price is determined by the intersection of market demand and market supply. In panel (a), the demand and supply curves for wheat intersect at a price of \$7 per bushel. An individual wheat farmer like Farmer Parker cannot affect the market price for wheat. Therefore, as panel (b) shows, the demand curve for Farmer Parker's wheat is a horizontal line. To understand this figure, it is important to notice that the scales on the horizontal axes in the two panels are very different. In panel (a), the equilibrium quantity of wheat is 2.25 billion bushels, and in panel (b), Farmer Parker is producing only 15 000 bushels of wheat.

Farmer Parker cannot control the price of his wheat, so how does he decide how much wheat to produce? We assume that Farmer Parker's objective is to maximize profit. This assumption is reasonable for most firms most of the time. Remember that [profit](#) is the difference between total revenue ( $TR$ ) and total cost ( $TC$ ):

$$\text{Profit} = TR - TC$$

To maximize his profit, Farmer Parker should produce the quantity of wheat where the difference between the total revenue he receives and his total cost is as large as possible.

**Table 10.2**

Farmer Parker's Revenue from Wheat Farming

| (1)<br>Number of<br>Bushels ( $Q$ ) | (2)<br>Market Price (per<br>bushel) ( $P$ ) | (3)<br>Total Revenue<br>( $TR$ ) | (4)<br>Average Revenue<br>( $AR$ ) | (5)<br>Marginal Revenue<br>( $MR$ ) |
|-------------------------------------|---------------------------------------------|----------------------------------|------------------------------------|-------------------------------------|
| 0                                   | \$7                                         | \$0                              | —                                  | —                                   |
| 1                                   | 7                                           | 7                                | \$7                                | \$7                                 |
| 2                                   | 7                                           | 14                               | 7                                  | 7                                   |
| 3                                   | 7                                           | 21                               | 7                                  | 7                                   |
| 4                                   | 7                                           | 28                               | 7                                  | 7                                   |
| 5                                   | 7                                           | 35                               | 7                                  | 7                                   |
| 6                                   | 7                                           | 42                               | 7                                  | 7                                   |
| 7                                   | 7                                           | 49                               | 7                                  | 7                                   |
| 8                                   | 7                                           | 56                               | 7                                  | 7                                   |
| 9                                   | 7                                           | 63                               | 7                                  | 7                                   |
| 10                                  | 7                                           | 70                               | 7                                  | 7                                   |

The third column in [Table 10.2](#) shows that Farmer Parker's *total revenue* rises by \$7 for every additional bushel he sells because he can sell as many bushels as he wants at the market price of \$7 per bushel. The fourth and fifth columns in the table show Farmer Parker's *average revenue* and *marginal revenue* from selling wheat. His [average revenue](#) ( $AR$ ) equals his total revenue divided by the quantity of bushels he sells. For example, if he sells 5 bushels for a total of \$35, his average revenue is  $\$35/5 = \$7$ . Notice that his average revenue is also equal to the market price of \$7. In fact, for any level of output, a firm's average revenue is always equal to the market price. To see why, first note that total revenue equals price times quantity ( $TR = P \times Q$ ), and average revenue equals total revenue divided by quantity ( $AR = TR/Q$ ). Therefore,  $AR = TR/Q = (P \times Q)/Q = P$ .

Farmer Parker's [marginal revenue](#) ( $MR$ ) is the change in his total revenue from selling 1 more bushel:

$$\text{Marginal revenue} = \frac{\text{Change in total revenue}}{\text{Change in quantity}}, \text{ or } MR = \frac{\Delta TR}{\Delta Q}$$

Each additional bushel Farmer Parker sells always adds \$7 to his total revenue, so his marginal revenue is \$7. Farmer Parker's marginal revenue is \$7 per bushel because he is selling wheat in a perfectly competitive market and can sell as much as he wants at the market price. In fact, Farmer Parker's marginal revenue and average revenue are both equal to the market price. This point is important: *For a firm in a perfectly competitive market, price is equal to both average revenue and marginal revenue.*

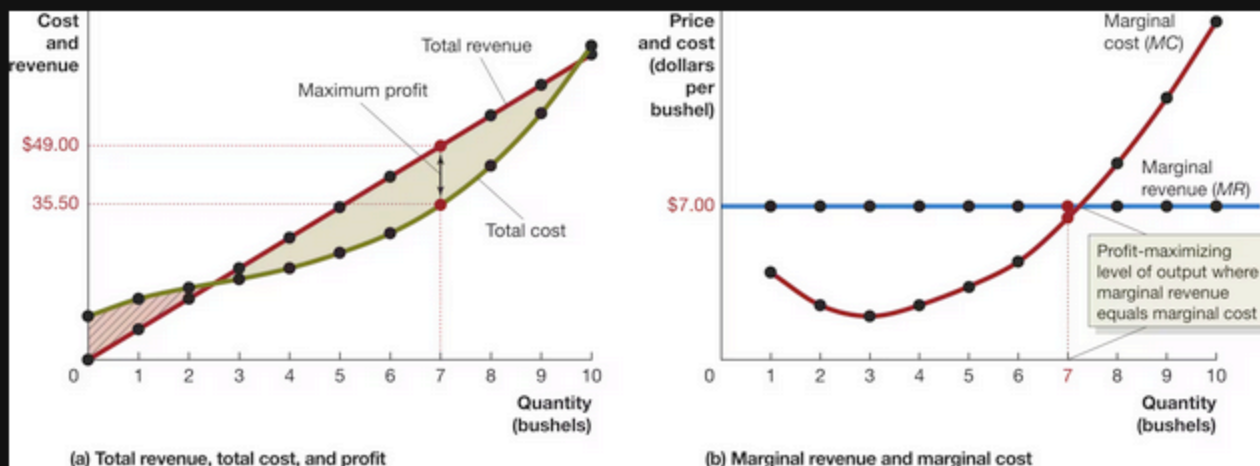
Table 10.3

Farmer Parker’s Profit from Wheat Farming

| (1)<br>Quantity<br>(bushels) ( $Q$ ) | (2)<br>Total Revenue<br>( $TR$ ) | (3)<br>Total Cost<br>( $TC$ ) | (4)<br>Profit<br>( $TR - TC$ ) | (5)<br>Marginal<br>Revenue ( $MR$ ) | (6)<br>Marginal Cost<br>( $MC$ ) |
|--------------------------------------|----------------------------------|-------------------------------|--------------------------------|-------------------------------------|----------------------------------|
| 0                                    | \$0.00                           | \$10.00                       | −\$10.00                       | —                                   | —                                |
| 1                                    | 7.00                             | 14.00                         | −7.00                          | \$7.00                              | \$4.00                           |
| 2                                    | 14.00                            | 16.50                         | −2.50                          | 7.00                                | 2.50                             |
| 3                                    | 21.00                            | 18.50                         | 2.50                           | 7.00                                | 2.00                             |
| 4                                    | 28.00                            | 21.00                         | 7.00                           | 7.00                                | 2.50                             |
| 5                                    | 35.00                            | 24.50                         | 10.50                          | 7.00                                | 3.50                             |
| 6                                    | 42.00                            | 29.00                         | 13.00                          | 7.00                                | 4.50                             |
| 7                                    | 49.00                            | 35.50                         | 13.50                          | 7.00                                | 6.50                             |
| 8                                    | 56.00                            | 44.50                         | 11.50                          | 7.00                                | 9.00                             |
| 9                                    | 63.00                            | 56.50                         | 6.50                           | 7.00                                | 12.00                            |
| 10                                   | 70.00                            | 72.00                         | −2.00                          | 7.00                                | 15.50                            |

## Figure 10.3

### The Profit-Maximizing Level of Output



In panel (a), Farmer Parker maximizes his profit at a quantity of 7 bushels, where the positive vertical distance between total revenue (\$49.00) and total cost (\$35.50) is the largest. Panel (b) shows that Farmer Parker's marginal revenue ( $MR$ ) is equal to a constant \$7 per bushel. He maximizes profit by producing wheat up to the point where the marginal revenue of the last bushel produced is equal to its marginal cost, or  $MR = MC$ . In this case, at no level of output does marginal revenue exactly equal marginal cost. The closest Farmer Parker can come is to produce 7 bushels of wheat. He will not want to continue to produce once marginal cost is greater than marginal revenue because doing so would reduce his profit. Panels (a) and (b) show alternative ways of thinking about how Farmer Parker can determine the profit-maximizing quantity of wheat to produce.

We have seen that profit is the difference between total revenue and total cost. We can also express profit in terms of *average total cost* ( $ATC$ ). This approach allows us to show profit on the cost curve graph we developed in Chapter 9, Section 9.5.

We need to work through a few steps to determine the relationship between profit and average total cost. Because profit is equal to total revenue ( $TR$ ) minus total cost ( $TC$ ), and total revenue is price multiplied by quantity, we can write the following:

$$\text{Profit} = (P \times Q) - TC$$

If we divide both sides of this equation by  $Q$ , we have

$$\frac{\text{Profit}}{Q} = \frac{(P \times Q)}{Q} - \frac{TC}{Q}$$

or, because  $TC/Q$  equals  $ATC$

or, because  $TC/Q$  equals  $ATC$

$$\frac{\text{Profit}}{Q} = P - ATC$$

This equation tells us that profit per unit (or average profit) equals price minus average total cost. Finally, we obtain the equation for the relationship between total profit and average total cost by multiplying by  $Q$ :

$$\text{Profit} = (P - ATC) \times Q$$

This equation tells us that a firm's total profit is equal to the difference between price and average total cost multiplied by the quantity produced.

We have already seen that to maximize profit, a firm produces the quantity of output where marginal revenue equals marginal cost. But will the firm actually make a profit producing that quantity? The answer depends on the relationship of price to average total cost at that quantity. There are three possibilities:

1.  $P > ATC$ , which means the firm makes a profit
2.  $P = ATC$ , which means the firm *breaks even* (its total cost equals its total revenue)
3.  $P < ATC$ , which means the firm experiences a loss

[Figure 10.4](#) shows the first possibility, where the firm makes a profit. [Figure 10.5](#) shows the situations where a firm breaks even or suffers a loss. In panel (a) of [Figure 10.5](#), at the quantity of output at which  $MR = MC$ , price is equal to average total cost. Therefore, total revenue is equal to total cost, and the firm will break even, making zero economic profit. In panel (b), at the quantity of output at which  $MR = MC$ , price is less than average total cost. Therefore, total revenue is less than total cost, and the firm suffers a loss. In this case, maximizing profit amounts to *minimizing* loss.

Remember that the supply curve for a firm tells us how many units of a product the firm is willing to sell at any given price. Notice that the marginal cost curve for a firm in a perfectly competitive market tells us the same thing. The firm will produce the quantity of output where  $MR = MC$ . Because price equals marginal revenue for a firm in a perfectly competitive market, the firm will produce the quantity where  $P = MC$ . For any given price, we can determine from the marginal cost curve the quantity of output the firm will supply. *Therefore, a perfectly competitive firm's marginal cost curve is also its supply curve.* But there's an important qualification to this fact. We have seen that if a firm is experiencing a loss, it will shut down if its total revenue is less than its variable cost:

$$\text{Total revenue} < \text{Variable cost}$$

or, in symbols

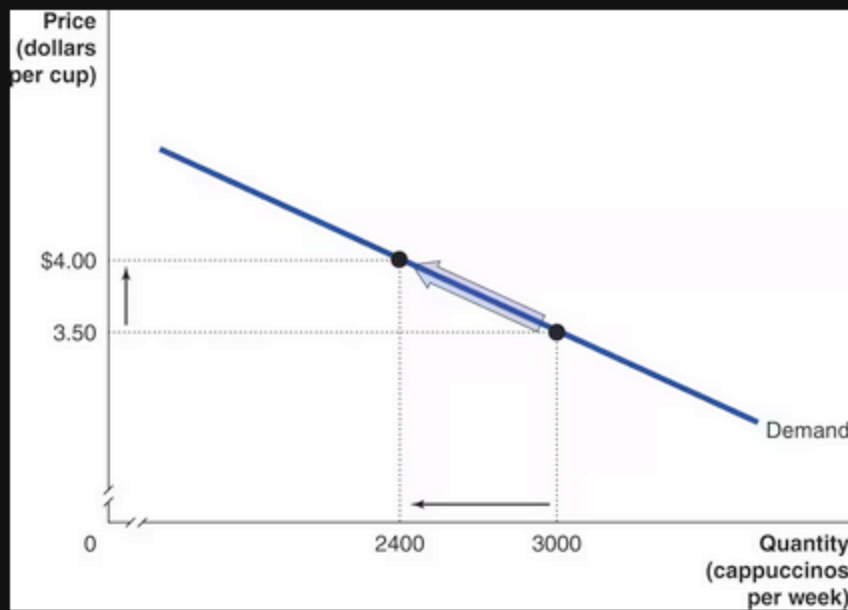
$$(P \times Q) < VC$$

If we divide both sides by  $Q$ , we have the result that the firm will shut down if

$$P < AVC$$

## Figure 11.1

### The Downward-Sloping Demand Curve for Cappuccinos at Blue Bottle



If a Blue Bottle coffee house increases the price of its cappuccinos, it will lose some, but not all, of its customers. In this case, raising the price from \$3.50 to \$4.00 reduces the quantity of cappuccinos consumers demand from 3000 per week to 2400. Therefore, unlike a perfect competitor, a Blue Bottle coffee house faces a downward-sloping demand curve.

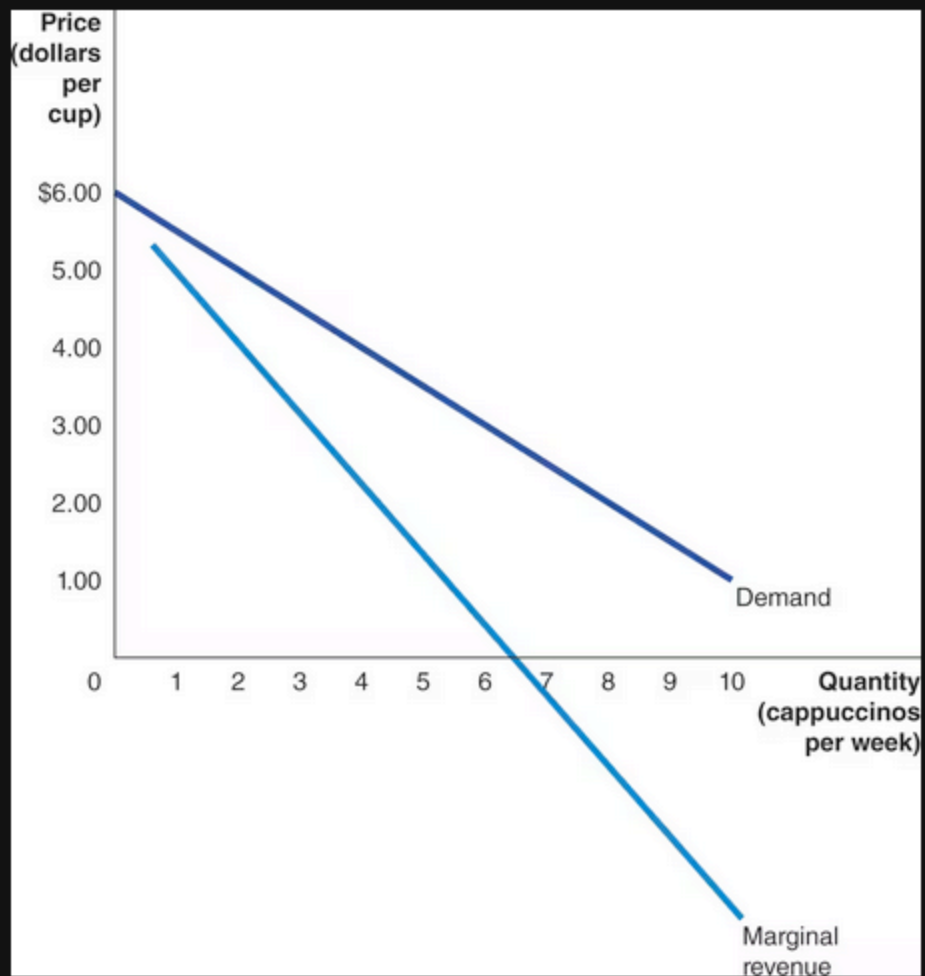
Table 11.1

Demand and Marginal Revenue at a Blue Bottle Coffee House

| Cappuccinos Sold per Week ( $Q$ ) | Price ( $P$ ) | Total Revenue ( $TR = P \times Q$ ) | Average Revenue ( $AR = \frac{TR}{Q}$ ) | Marginal Revenue ( $MR = \frac{\Delta TR}{\Delta Q}$ ) |
|-----------------------------------|---------------|-------------------------------------|-----------------------------------------|--------------------------------------------------------|
| 0                                 | \$6.00        | \$0.00                              | –                                       | –                                                      |
| 1                                 | 5.50          | 5.50                                | \$5.50                                  | \$5.50                                                 |
| 2                                 | 5.00          | 10.00                               | 5.00                                    | 4.50                                                   |
| 3                                 | 4.50          | 13.50                               | 4.50                                    | 3.50                                                   |
| 4                                 | 4.00          | 16.00                               | 4.00                                    | 2.50                                                   |
| 5                                 | 3.50          | 17.50                               | 3.50                                    | 1.50                                                   |
| 6                                 | 3.00          | 18.00                               | 3.00                                    | 0.50                                                   |
| 7                                 | 2.50          | 17.50                               | 2.50                                    | –0.50                                                  |
| 8                                 | 2.00          | 16.00                               | 2.00                                    | –1.50                                                  |
| 9                                 | 1.50          | 13.50                               | 1.50                                    | –2.50                                                  |
| 10                                | 1.00          | 10.00                               | 1.00                                    | –3.50                                                  |

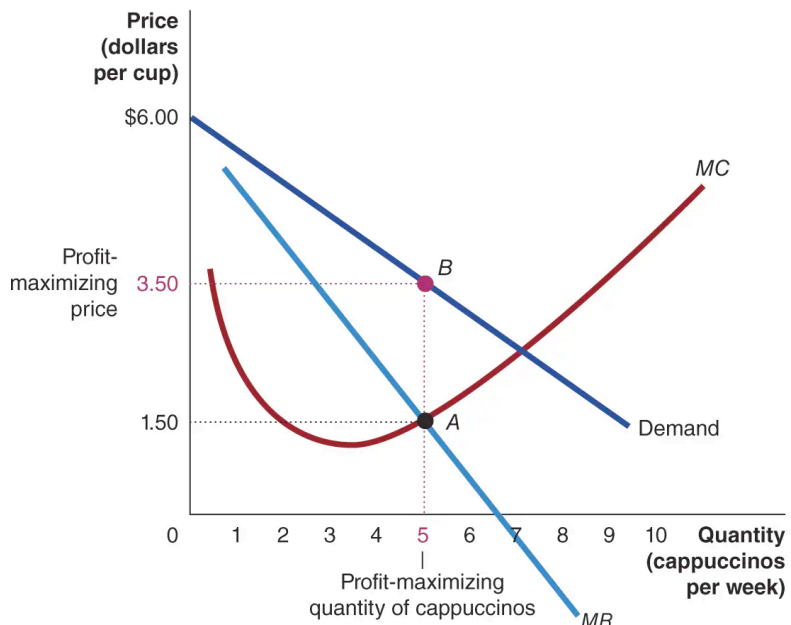
### Figure 11.3

The Demand and Marginal Revenue Curves for a Monopolistically Competitive Firm

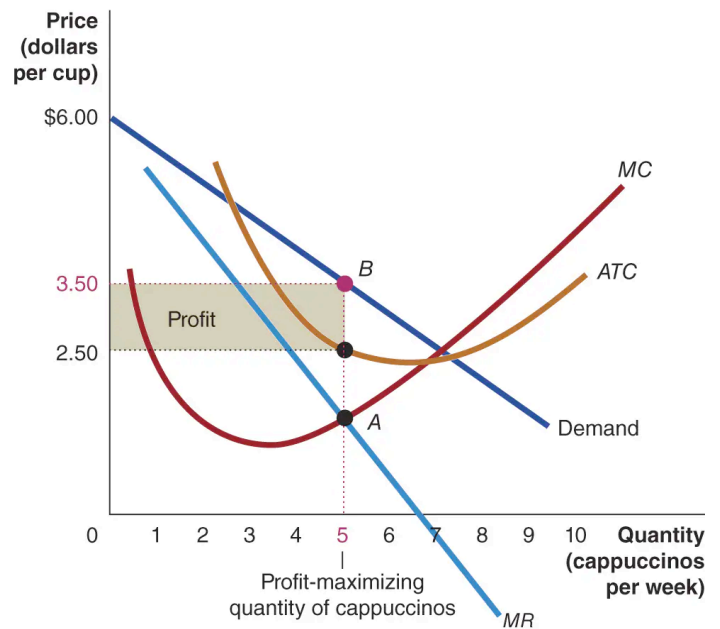


Any firm that has the ability to affect the price of the product it sells will have a marginal revenue curve that is below its demand curve. We plot the data from [Table 11.1](#) to create the demand and marginal revenue curves for a Blue Bottle coffee house. After the sixth cappuccino, marginal revenue becomes negative because the additional revenue received from selling 1 more cappuccino is less than the revenue lost from receiving a lower price on the cappuccinos that could have been sold at the original price.

| Cappuccinos Sold per Week (Q) | Price (P) | Total Revenue (TR) | Marginal Revenue (MR) | Total Cost (TC) | Marginal Cost (MC) | Average Total Cost (ATC) | Profit  |
|-------------------------------|-----------|--------------------|-----------------------|-----------------|--------------------|--------------------------|---------|
| 0                             | \$6.00    | \$0.00             | —                     | \$5.00          | —                  | —                        | −\$5.00 |
| 1                             | 5.50      | 5.50               | \$5.50                | 8.00            | \$3.00             | \$8.00                   | −2.50   |
| 2                             | 5.00      | 10.00              | 4.50                  | 9.50            | 1.50               | 4.75                     | 0.50    |
| 3                             | 4.50      | 13.50              | 3.50                  | 10.00           | 0.50               | 3.33                     | 3.50    |
| 4                             | 4.00      | 16.00              | 2.50                  | 11.00           | 1.00               | 2.75                     | 5.00    |
| 5                             | 3.50      | 17.50              | 1.50                  | 12.50           | 1.50               | 2.50                     | 5.00    |
| 6                             | 3.00      | 18.00              | 0.50                  | 14.50           | 2.00               | 2.42                     | 3.50    |
| 7                             | 2.50      | 17.50              | −0.50                 | 17.00           | 2.50               | 2.43                     | 0.50    |
| 8                             | 2.00      | 16.00              | −1.50                 | 20.00           | 3.00               | 2.50                     | −4.00   |
| 9                             | 1.50      | 13.50              | −2.50                 | 23.50           | 3.50               | 2.61                     | −10.00  |
| 10                            | 1.00      | 10.00              | −3.50                 | 27.50           | 4.00               | 2.75                     | −17.50  |

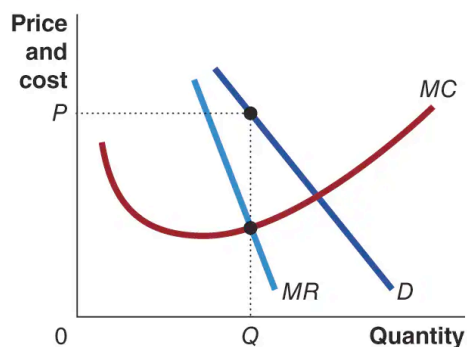


(a) Profit-maximizing quantity and price for a monopolistic competitor



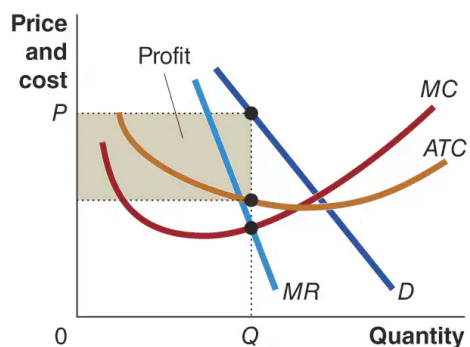
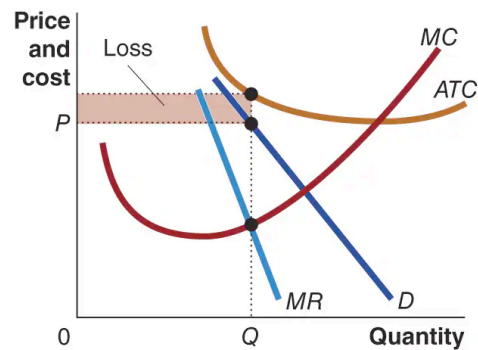
(b) Short-run profits for a monopolistic competitor

### Short Run

 $P > MC$ 


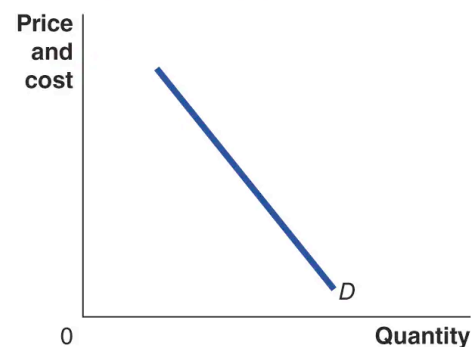
### Short Run

Economic profit

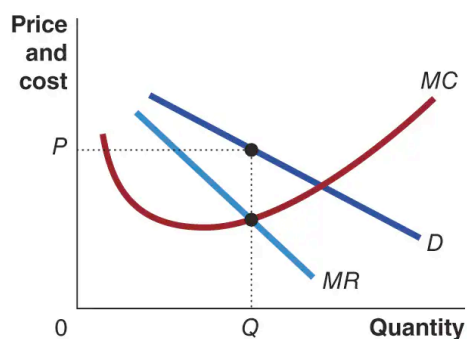

or  
Economic loss


### Short Run (Less Competition)

Less elastic demand curve

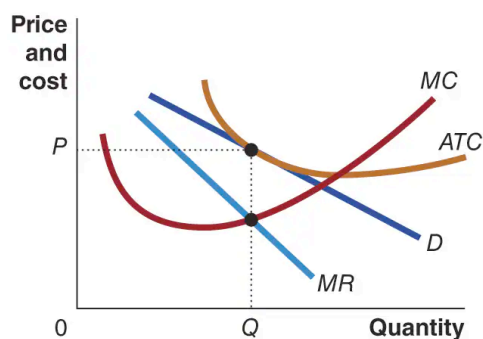


### Long Run

 $P > MC$ 


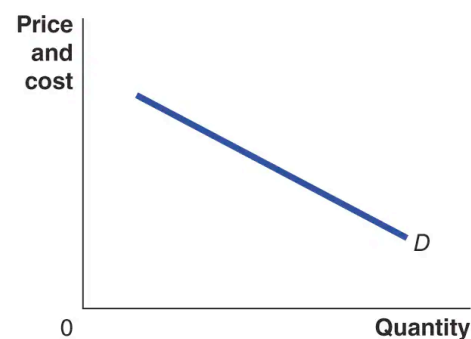
### Long Run

Zero economic profit



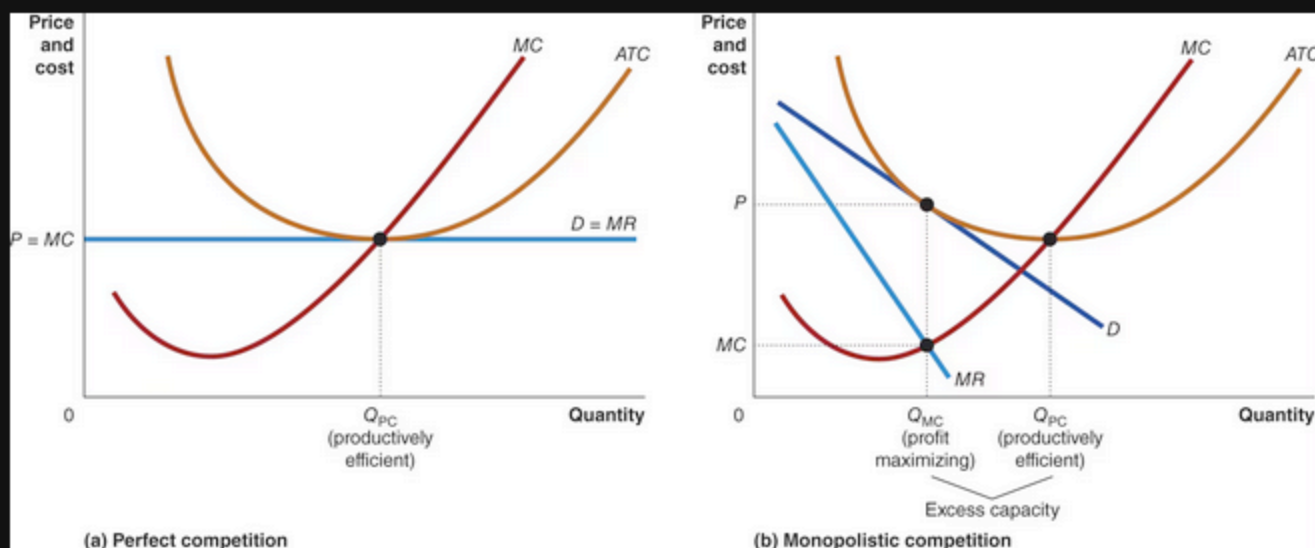
### Long Run (More Competition)

More elastic demand curve



**Figure 11.6**

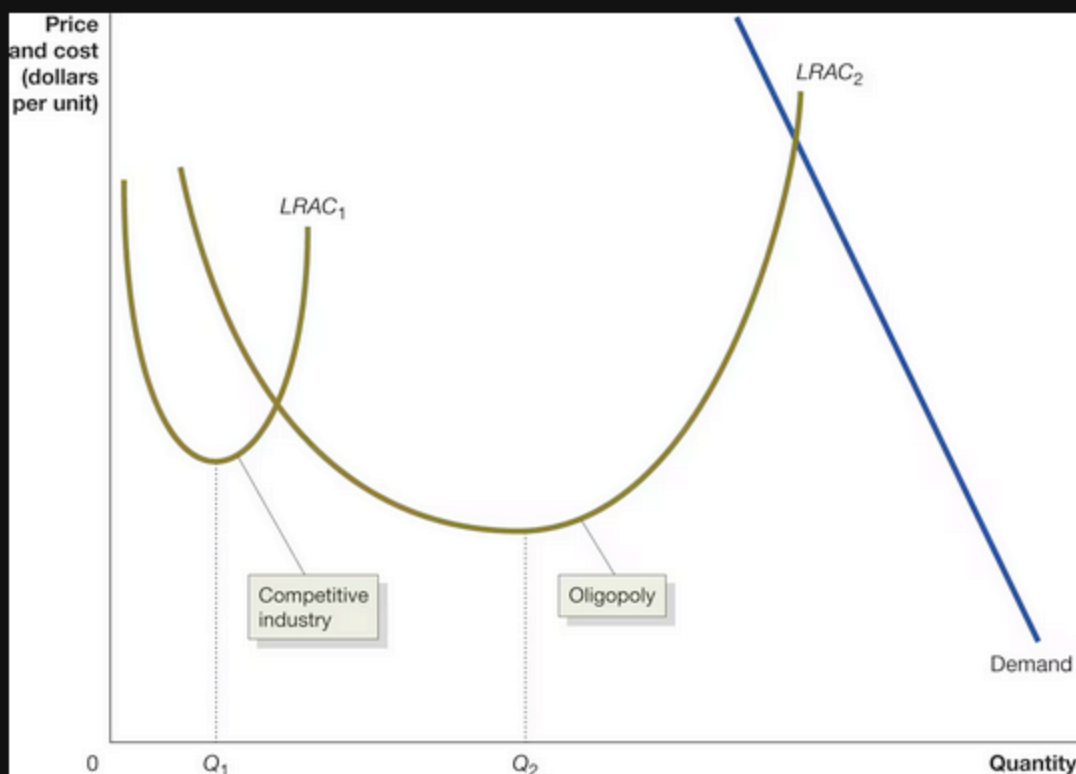
Comparing Long-Run Equilibrium under Perfect Competition and Monopolistic Competition



In panel (a), a perfectly competitive firm in long-run equilibrium produces at  $Q_{PC}$ , where price equals marginal cost, and average total cost is at a minimum. The perfectly competitive firm is both allocatively efficient and productively efficient. In panel (b), a monopolistically competitive firm produces at  $Q_{MC}$ , where price is greater than marginal cost, and average total cost is not at a minimum. As a result, the monopolistically competitive firm is neither allocatively efficient nor productively efficient. The monopolistically competitive firm has excess capacity equal to the difference between its profit-maximizing level of output and the productively efficient level of output.

**Figure 12.2**

Economies of Scale Help Determine the Extent of Competition in an Industry



An industry will be competitive if the minimum point on the typical firm's long-run average cost curve (LRAC<sub>1</sub>) occurs at a level of output that is a small fraction of total industry sales, such as Q<sub>1</sub>. An industry will be an oligopoly if the minimum point on the typical firm's long-run average cost curve (LRAC<sub>2</sub>) occurs at a level of output that is a large fraction of industry sales, such as Q<sub>2</sub>.

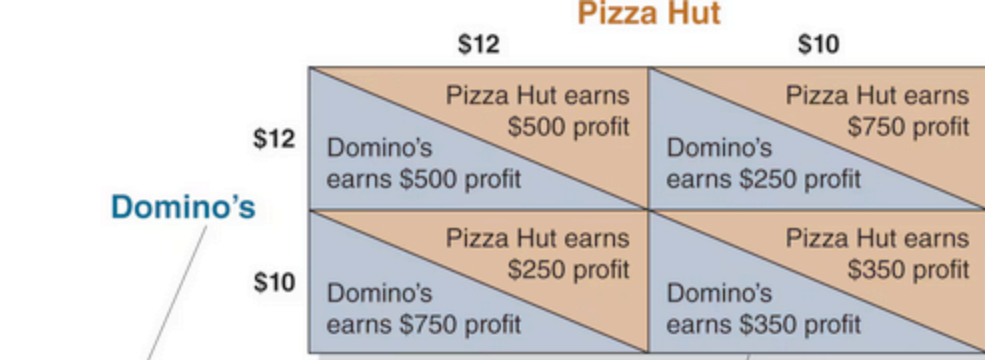
\*\*\*\*

**Figure 12.3**

A Duopoly Game

|         |         | Apple                                                                |                                                                        |
|---------|---------|----------------------------------------------------------------------|------------------------------------------------------------------------|
|         |         | \$14.99                                                              | \$9.99                                                                 |
| Spotify | \$14.99 | Apple earns \$10 million profit<br>Spotify earns \$10 million profit | Apple earns \$15 million profit<br>Spotify earns \$5 million profit    |
|         | \$9.99  | Apple earns \$5 million profit<br>Spotify earns \$15 million profit  | Apple earns \$7.5 million profit<br>Spotify earns \$7.5 million profit |

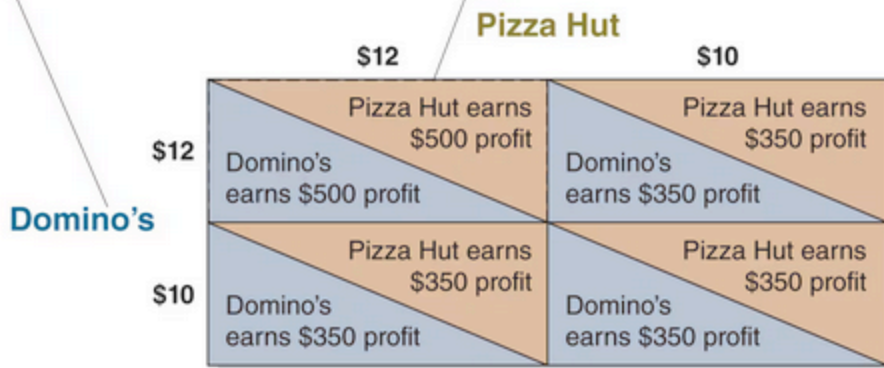
Spotify's profit is shown in blue, and Apple's profit is shown in gold. Spotify and Apple will each earn a profit of \$10 million per month on sales of streaming music subscriptions if they both charge \$14.99. However, each firm has an incentive to undercut the other by charging a lower price. If both firms charge \$9.99, they will each earn a profit of only \$7.5 million per month.



(a) The payoff matrix with no offer to match prices

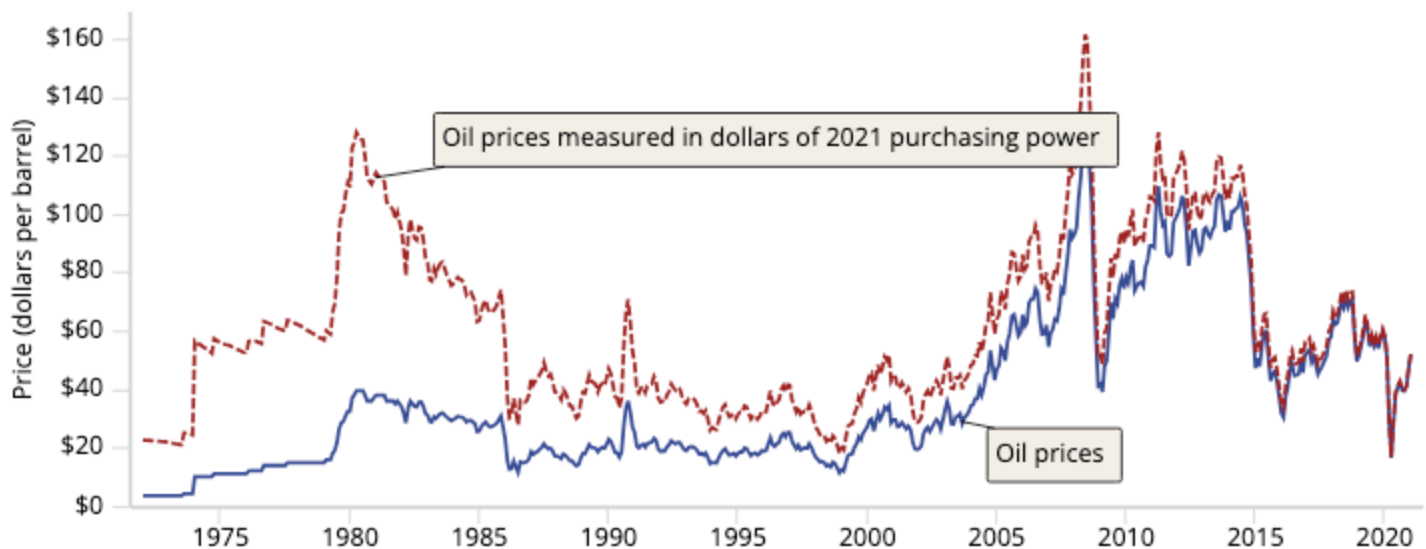
If Domino's and Pizza Hut each advertise that they will match their competitor's price, the payoff matrix changes from the matrix in panel (a) to the matrix in panel (b).

Equilibrium changes from this outcome in panel (a) to this outcome in panel (b).



(b) The payoff matrix with an offer to match prices

Figure 12.5: Oil Prices from January 1972

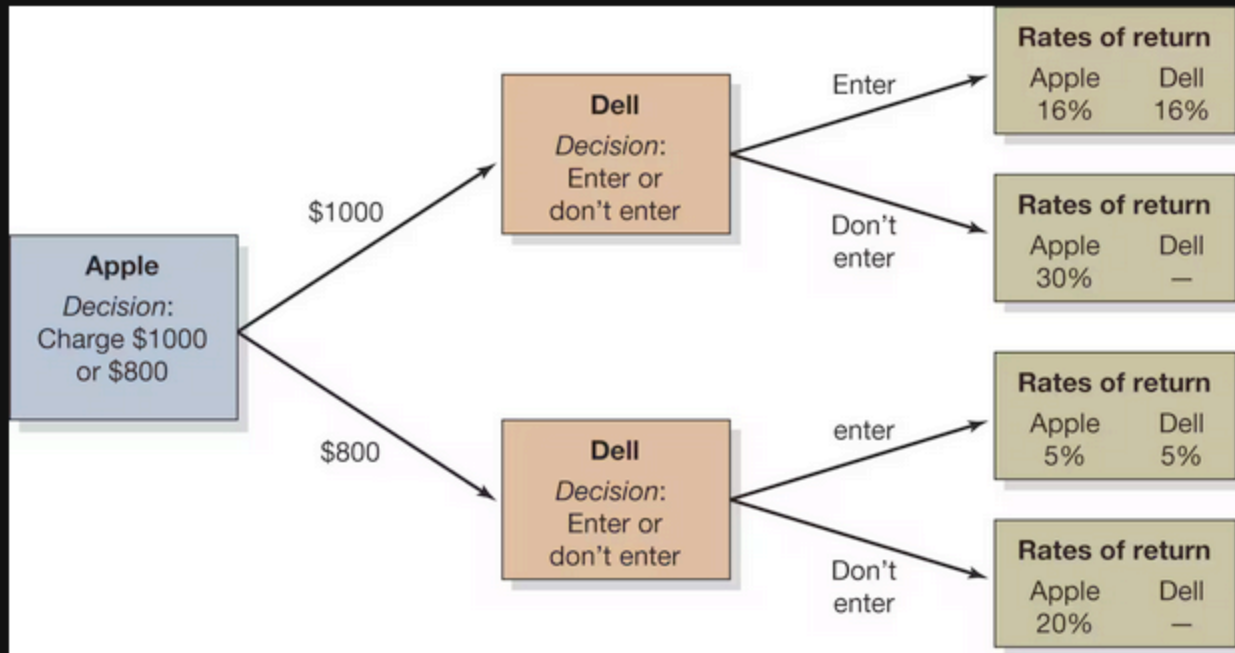


Source: Federal Reserve Bank of St. Louis.

|              |             | Nigeria                                                                      |                                                                             |
|--------------|-------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
|              |             | Low output                                                                   | High output                                                                 |
| Saudi Arabia | Low output  | Saudi Arabia earns \$100 million profit<br>Nigeria earns \$10 million profit | Saudi Arabia earns \$75 million profit<br>Nigeria earns \$15 million profit |
|              | High output | Saudi Arabia earns \$80 million profit<br>Nigeria earns \$7 million profit   | Saudi Arabia earns \$60 million profit<br>Nigeria earns \$10 million profit |

**Figure 12.7**

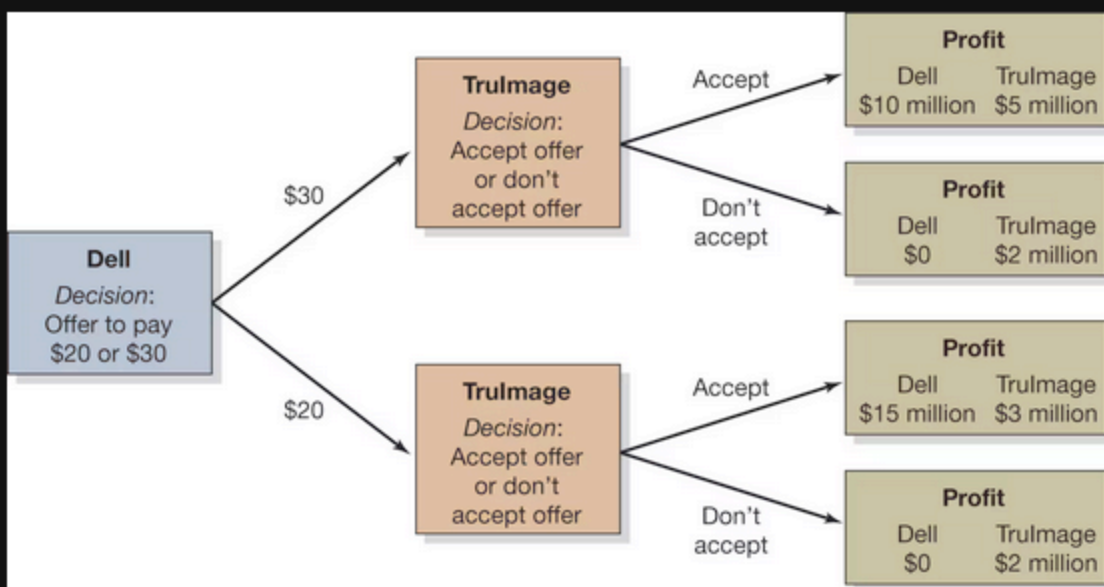
The Decision Tree for an Entry Game



Apple earns its highest return if it charges \$1000 for its very thin, light laptop and Dell does not enter the market. But at that price, Dell will enter the market, and Apple will earn only 16 percent. If Apple charges \$800, Dell will not enter because Dell will suffer an economic loss by earning only a 5 percent return on its investment. Therefore, Apple's best decision is to deter Dell's entry by charging \$800. Apple will earn an economic profit by earning a 20 percent return on its investment. Note that the dashes (–) indicate the situation where Dell does not enter the market and so makes no investment and earns nothing in this market.

**Figure 12.8**

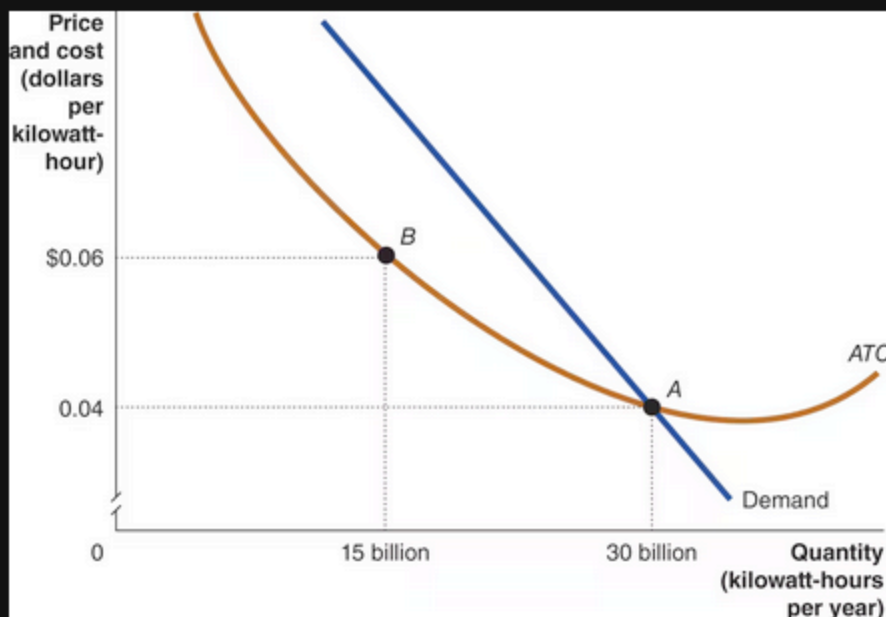
The Decision Tree for a Bargaining Game



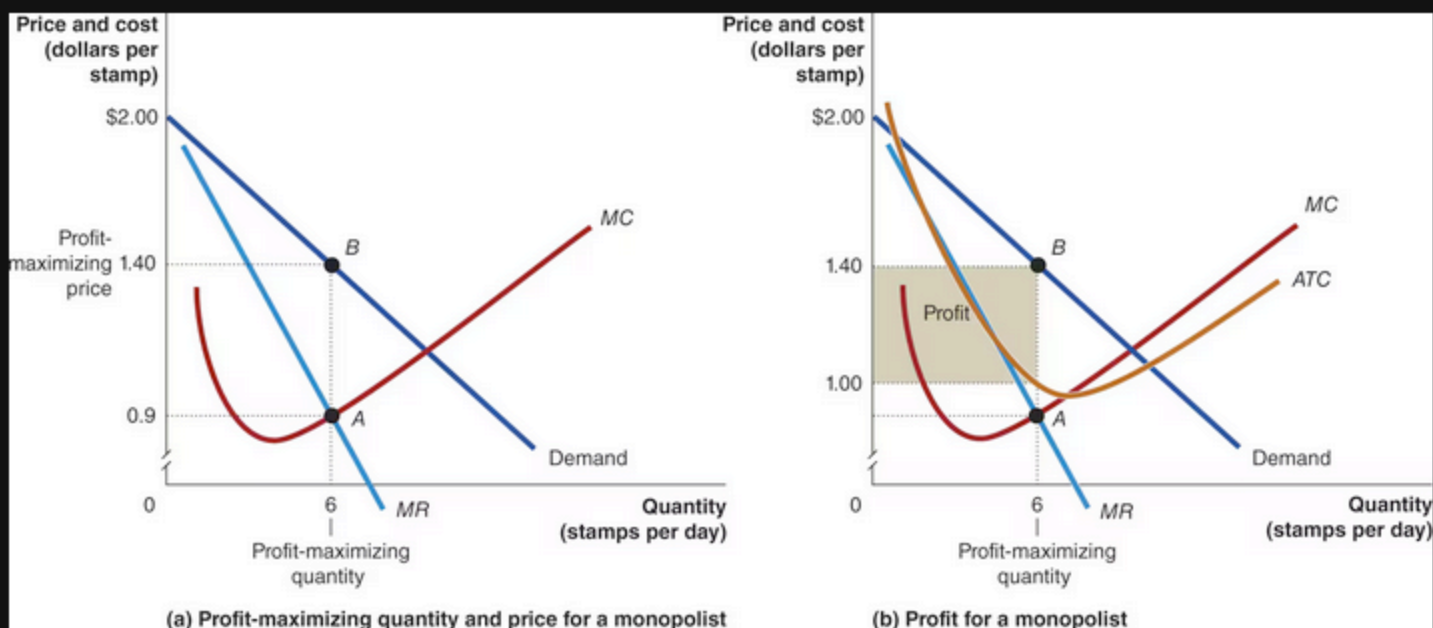
Dell earns the highest profit if it offers a contract price of \$20 per copy and Trulmage accepts the contract. Trulmage earns the highest profit if Dell offers it a contract of \$30 per copy and it accepts the contract. Trulmage may attempt to bargain by threatening to reject a \$20-per-copy contract. But Dell knows this threat is not credible because once Dell has offered a \$20-per-copy contract, Trulmage's profits are higher if it accepts the contract than if it rejects it.

**Figure 13.1**

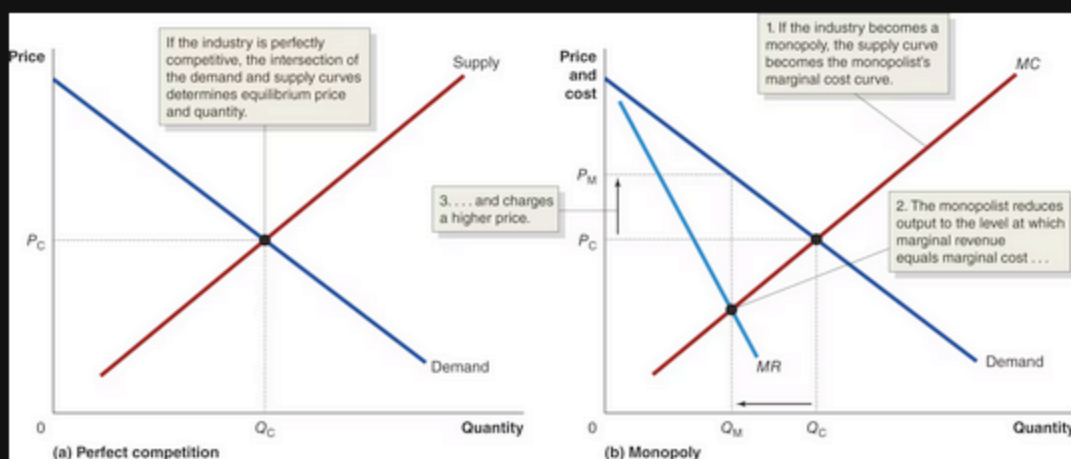
Average Total Cost Curve for a Natural Monopoly



With a natural monopoly, the average total cost curve is still falling when it crosses the demand curve (point A). If only one firm is producing electric power in the market, and it produces where the average cost curve intersects the demand curve, average total cost will equal \$0.04 per kilowatt-hour of electricity produced. If the market is divided between two firms, each producing 15 billion kilowatt-hours, the average cost of producing electricity rises to \$0.06 per kilowatt-hour (point B). In this case, if one firm expands production, it can move down the average total cost curve, lower its price, and drive the other firm out of business.

**Figure 13.3****Profit-Maximizing Price and Output for a Monopoly**

Panel (a) shows that to maximize profit, Canada Post should sell stamps up to the point where the marginal revenue from selling the last stamp equals its marginal cost (point A). In this case, the marginal revenue from selling the sixth stamp and the marginal cost are both \$0.90. Canada Post maximizes profit by selling 6 stamps per day and charging a price of \$1.40 (point B). In panel (b), the green box represents Canada Post's profit. The box has a height equal to \$0.40, which is the price of \$1.40 minus the average total cost of \$1.00, and a base equal to the quantity of 6 stamps. Canada Post's profit therefore equals  $0.40 \times 6 = \$2.40$ .

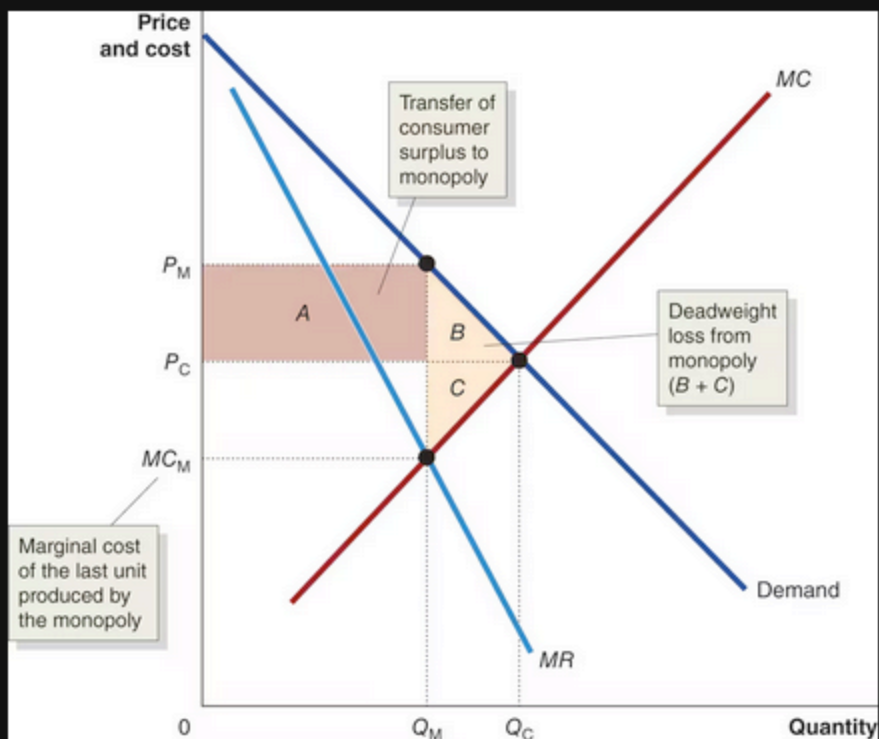
**Figure 13.4****What Happens If a Perfectly Competitive Industry Becomes a Monopoly?**

In panel (a), the market for tablet computers is perfectly competitive, and price and quantity are determined by the intersection of the demand and supply curves. In panel (b), the perfectly competitive tablet computer industry becomes a monopoly. As a result,

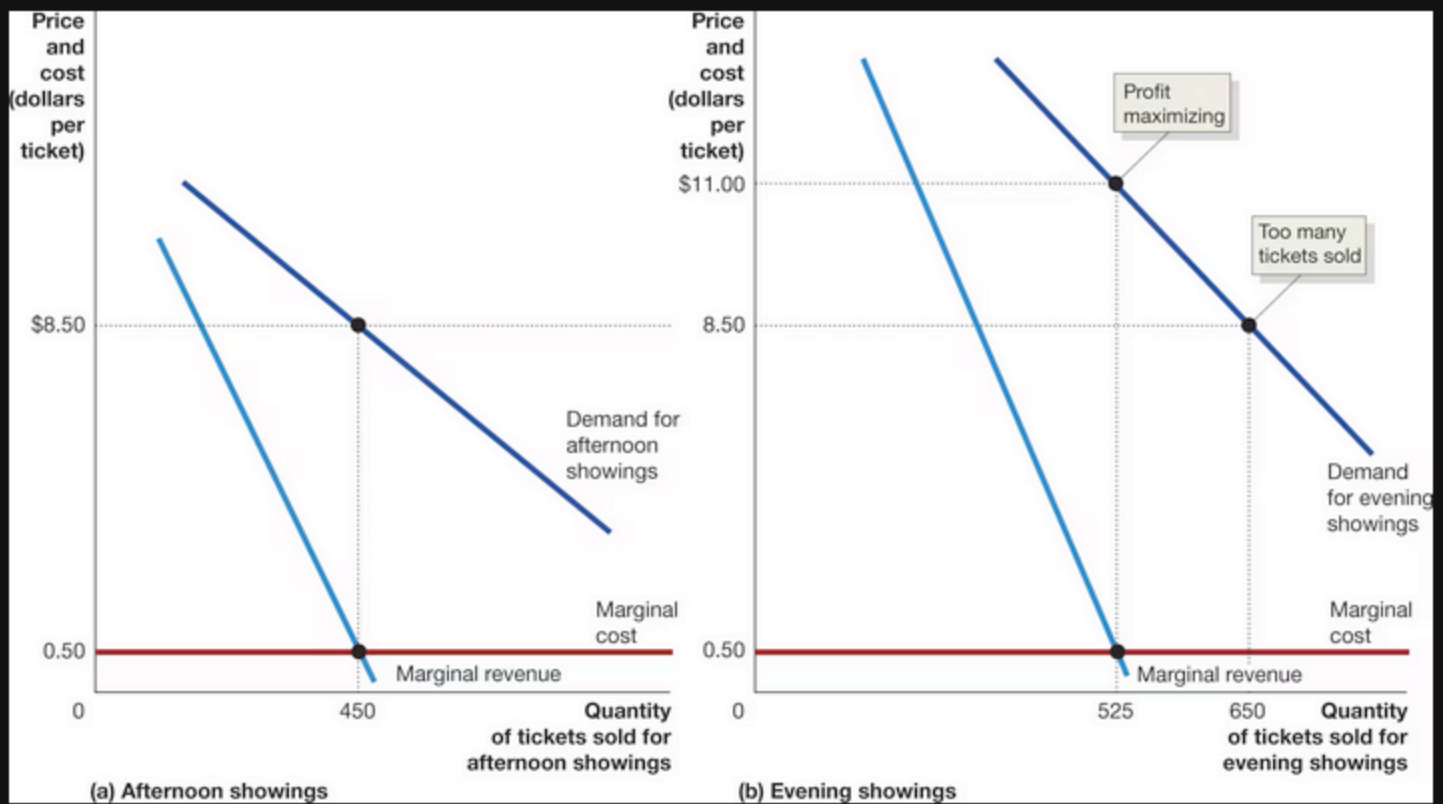
1. The industry supply curve becomes the monopolist's marginal cost.
2. The monopolist reduces output to where marginal revenue equals marginal cost,  $Q_M$
3. The monopolist raises the price from  $p_C$  to  $p_M$

**Figure 13.5**

The Inefficiency of Monopoly



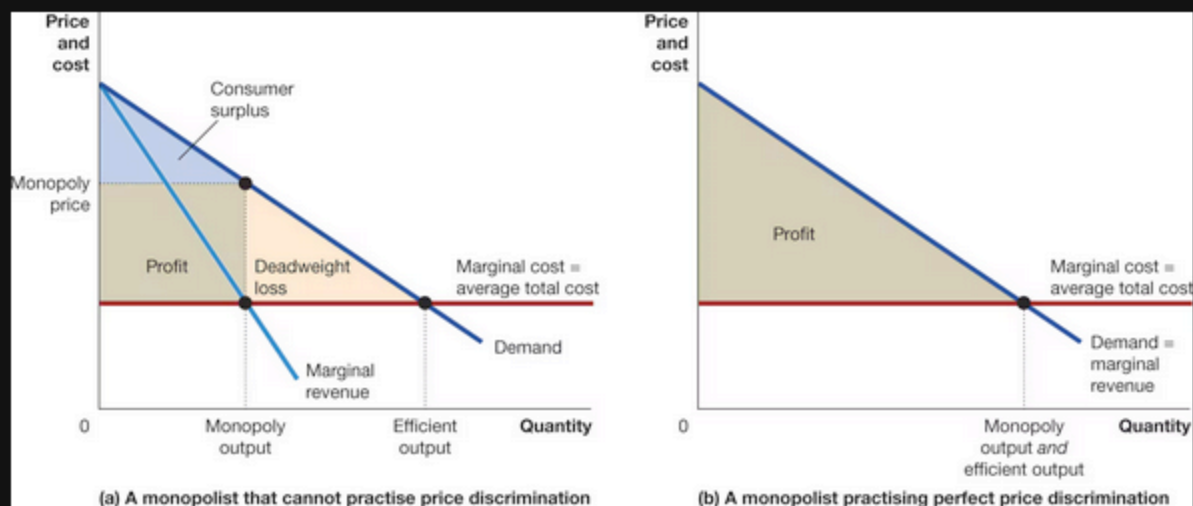
A monopoly charges a higher price,  $P_M$ , and produces a smaller quantity,  $Q_M$ , than a perfectly competitive industry, which charges price  $P_C$  and produces  $Q_C$ . The higher price reduces consumer surplus by the area equal to the rectangle A and the triangle B. Some of the reduction in consumer surplus is captured by the monopoly as producer surplus, and some becomes deadweight loss, which is the area equal to triangles B and C.

**Figure 13.6****Price Discrimination by a Movie Theatre**

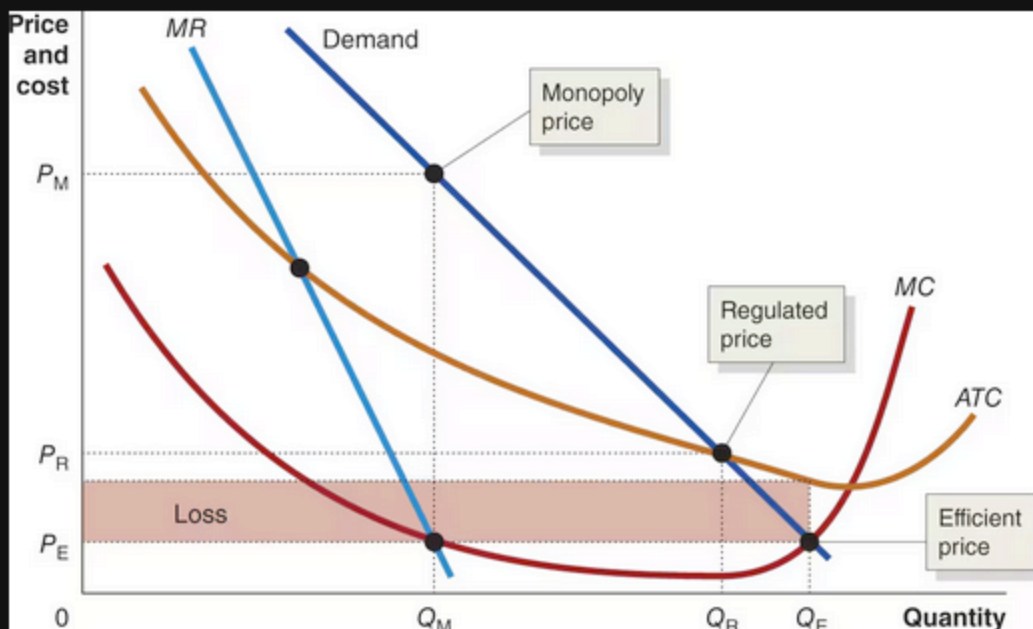
Fewer people want to go to the movies in the afternoon than in the evening. In panel (a), the profit-maximizing price for a ticket to an afternoon showing is \$8.50. Charging this same price for evening showings would not be profit maximizing, as panel (b) shows. At a price of \$8.50, the theatre would sell 650 tickets to evening showings, which is more than the profit-maximizing number of 525 tickets. To maximize profit, the theatre should charge \$11.00 for tickets to evening showings.

**Table 13.1****How Airlines Maximize Profit by Charging Different Ticket Prices**

| Category of Traveller                                                                                    | Schedule of the Traveller                                         | Traveller's Sensitivity to Price | How Do Airlines Maximize Profits?                                       | Example of Price Difference                                                                               |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Business traveller<br> | Inflexible: may need to buy ticket the same week or day of travel | Not very sensitive to price      | Charge business travellers higher ticket prices than leisure travellers | In the spring of 2022, a business-fare ticket between Vancouver and Toronto on Air Canada was \$2519.     |
| Leisure traveller<br>  | Flexible: can buy ticket weeks in advance                         | Sensitive to price               | Charge leisure travellers lower ticket prices than business travellers  | In the spring of 2022, an economy-fare ticket between Vancouver and Toronto on Air Canada was only \$770. |

**Figure 13.7****Perfect Price Discrimination**

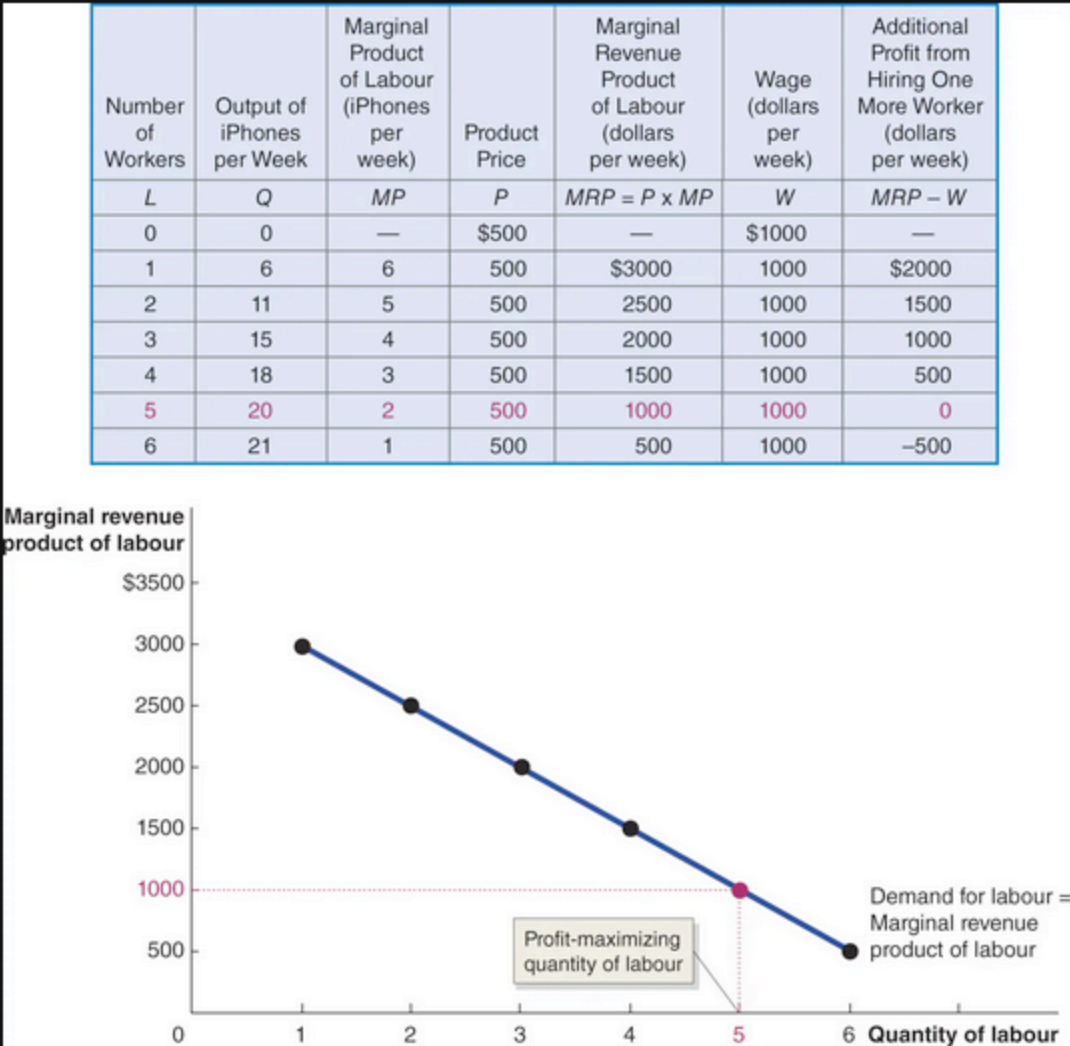
Panel (a) shows the case of a monopolist that cannot practise price discrimination and, therefore, can charge only a single price for its product. The graph shows that to maximize profit, the monopolist will produce the level of output where marginal revenue equals marginal cost. The area of the beige-shaded rectangle shows the resulting profit. Given the monopoly price, the area of the blue-shaded triangle shows the amount of consumer surplus in this market. The economically efficient level of output occurs where price equals marginal cost. Because the monopolist stops production at a level of output where price is above marginal cost, there is a deadweight loss equal to the area of the orange-shaded triangle. In panel (b), the monopolist is able to practise perfect price discrimination by charging a different price to each consumer. The result is to convert both the consumer surplus and the deadweight loss from panel (a) into profit.

**Figure 13.9****Regulating a Natural Monopoly**

A natural monopoly that is not subject to government regulation will charge a price equal to  $p_M$  and produce  $Q_M$ . If government regulators want to achieve economic efficiency, they will set the regulated price equal to  $P_E$  and the monopoly will produce  $Q_E$ . Unfortunately,  $P_E$  is below average cost, and the monopoly will suffer a loss, shown by the shaded rectangle. Because the monopoly will not continue to produce in the long run if it suffers a loss, government regulators set a price equal to average cost, which is  $P_R$  in the figure. The resulting production,  $Q_R$ , will be below the efficient level.

Figure 14.1

The Marginal Revenue Product of Labour and the Demand for Labour



The marginal revenue product of labour equals the marginal product of labour multiplied by the price of the good. The marginal revenue product curve slopes downward because diminishing returns cause the marginal product of labour to decline as the firm hires more workers. A firm maximizes profit by hiring workers up to the point where the wage equals the marginal revenue product of labour. The marginal revenue product of labour curve is the firm's demand curve for labour because it tells the firm the profit-maximizing quantity of workers to hire at each wage. For example, using the demand curve shown in this figure, if the wage is \$1000, the firm will hire 5 workers.

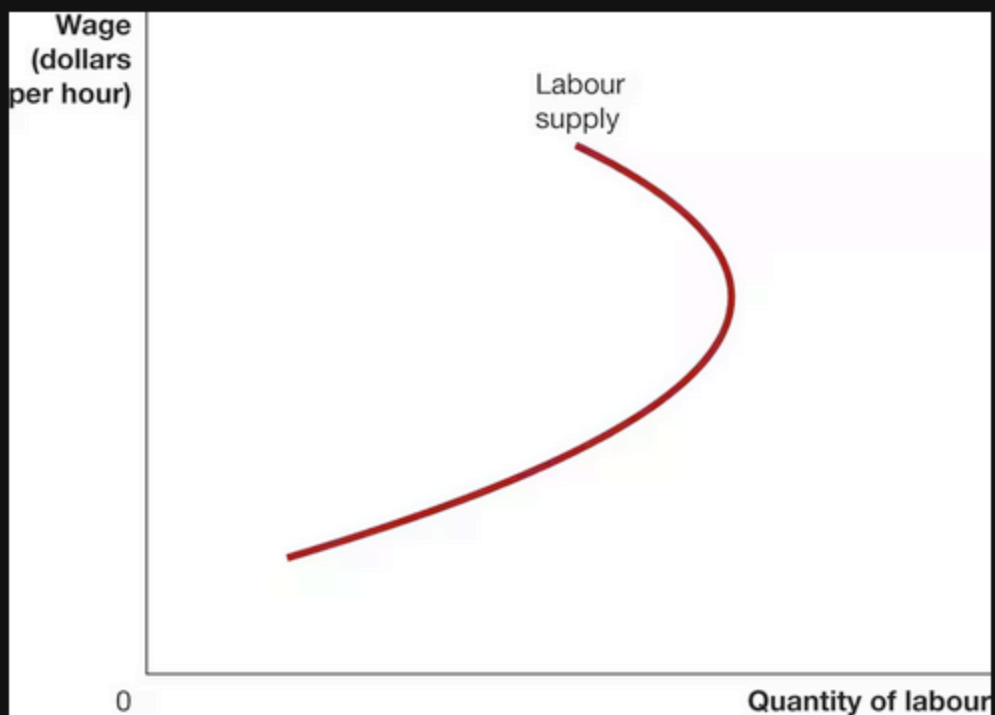
Table 14.1

The Relationship between the Marginal Revenue Product of Labour and the Wage

| When . . .  | the firm . . .                                                    |
|-------------|-------------------------------------------------------------------|
| $MRP > W$ , | should hire more workers to increase profit.                      |
| $MRP < W$ , | should hire fewer workers to increase profit.                     |
| $MRP = W$ , | is hiring the optimal number of workers and is maximizing profit. |

**Figure 14.3**

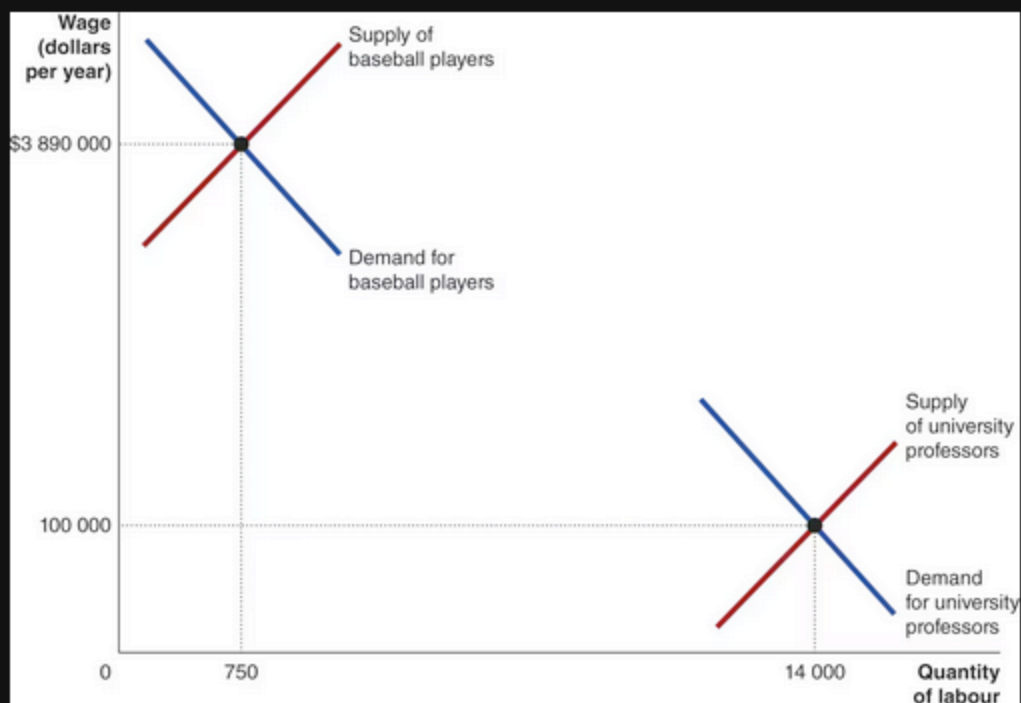
A Backward-Bending Labour Supply Curve



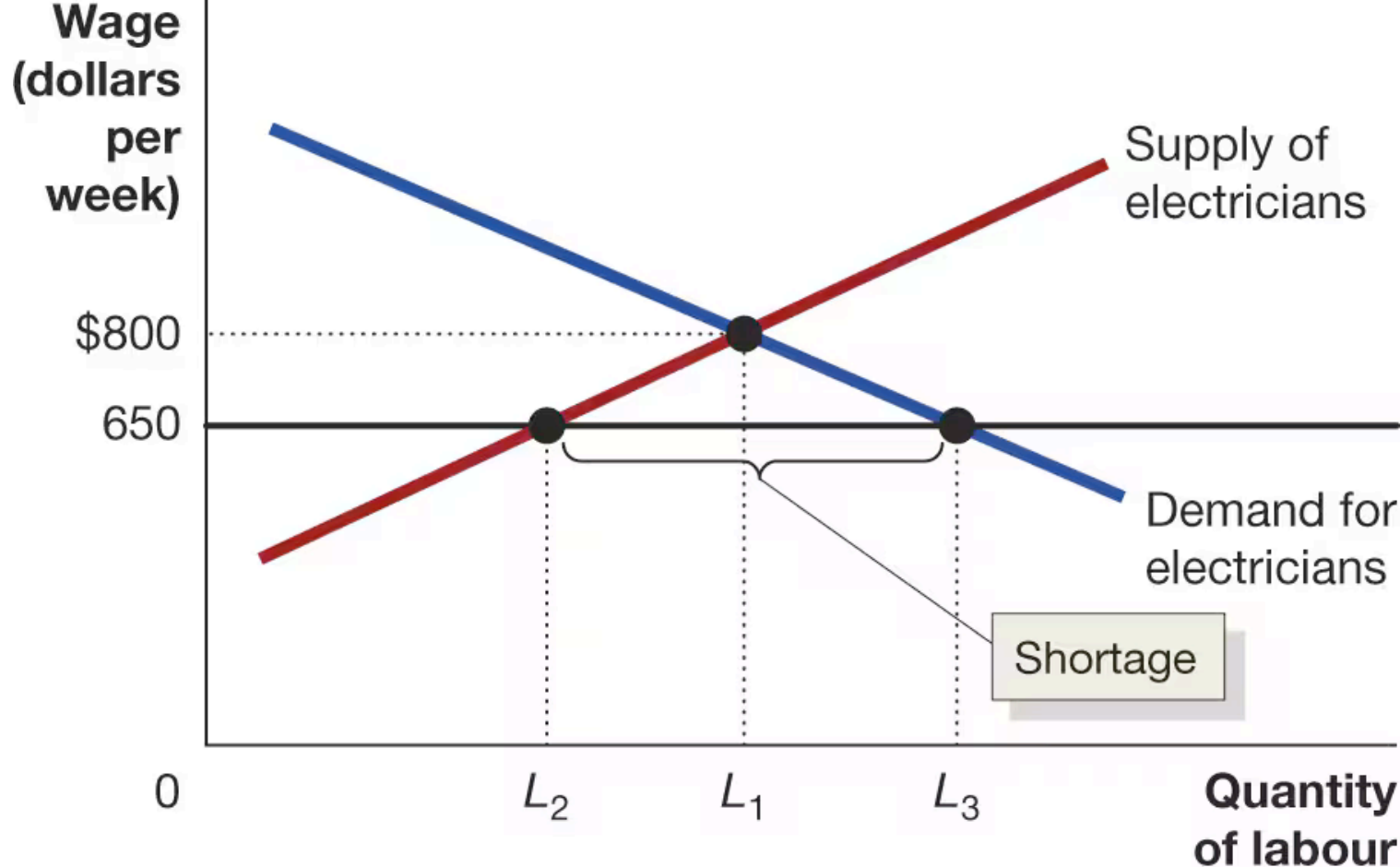
As the wage rises, a greater quantity of labour is usually supplied. As the wage rises above a certain level, the individual is able to afford more leisure even though the opportunity cost of leisure is higher. The result may be that an increase in the wage leads to a smaller quantity of labour supplied.

**Figure 14.7**

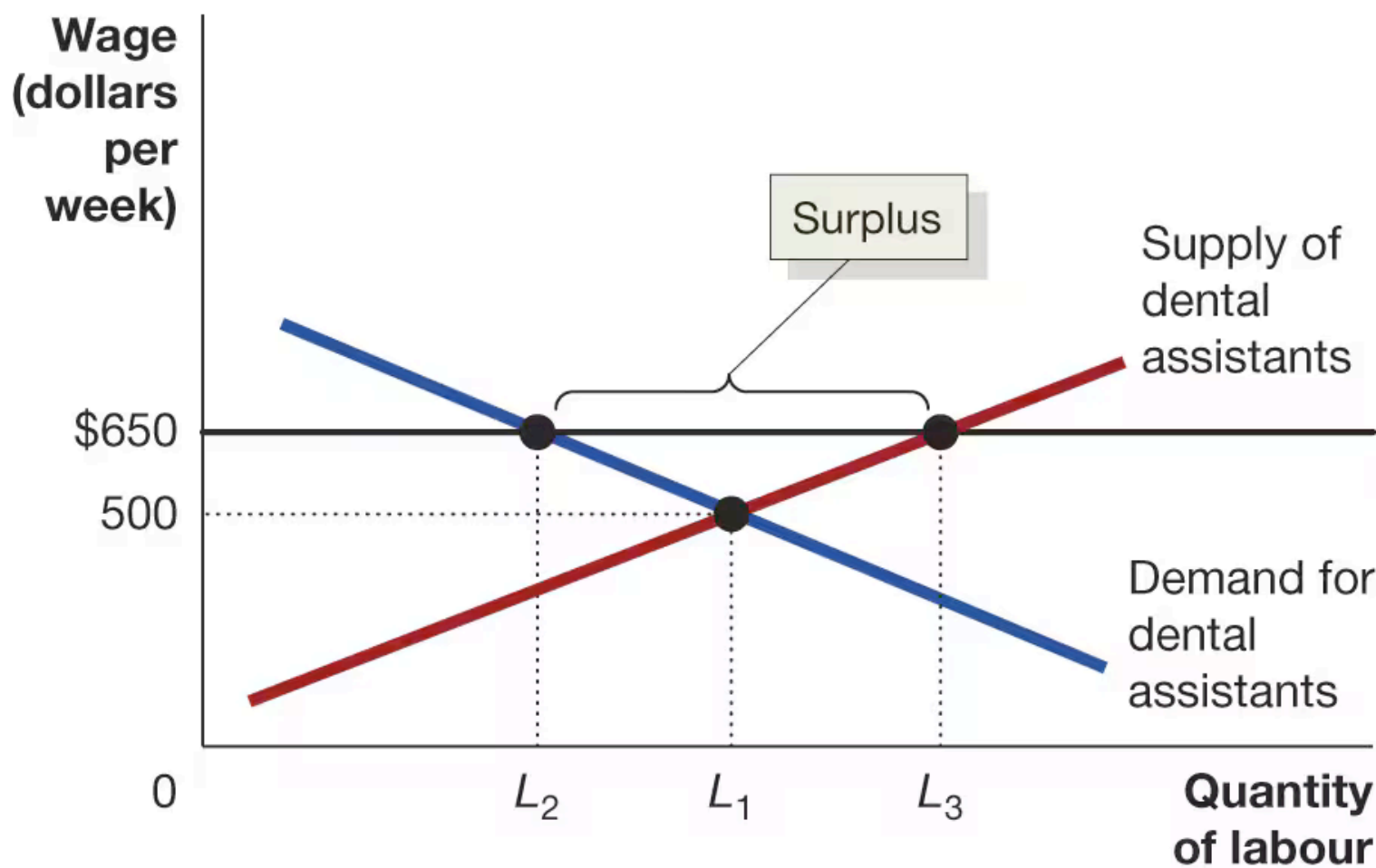
**Baseball Players Are Paid More than University Professors**



The marginal revenue product of baseball players is very high, and the supply of people with the ability to play Major League Baseball (MLB) is low. The result is that in 2020, the 750 MLB players received an average wage of \$3 890 000. The marginal revenue product of university professors is much lower, and the supply of people with the ability to be university professors is much higher. So, in 2020, the 14 000 university professors in Canada received an average wage of \$100 000 (in US dollars), far below the average wage of baseball players.



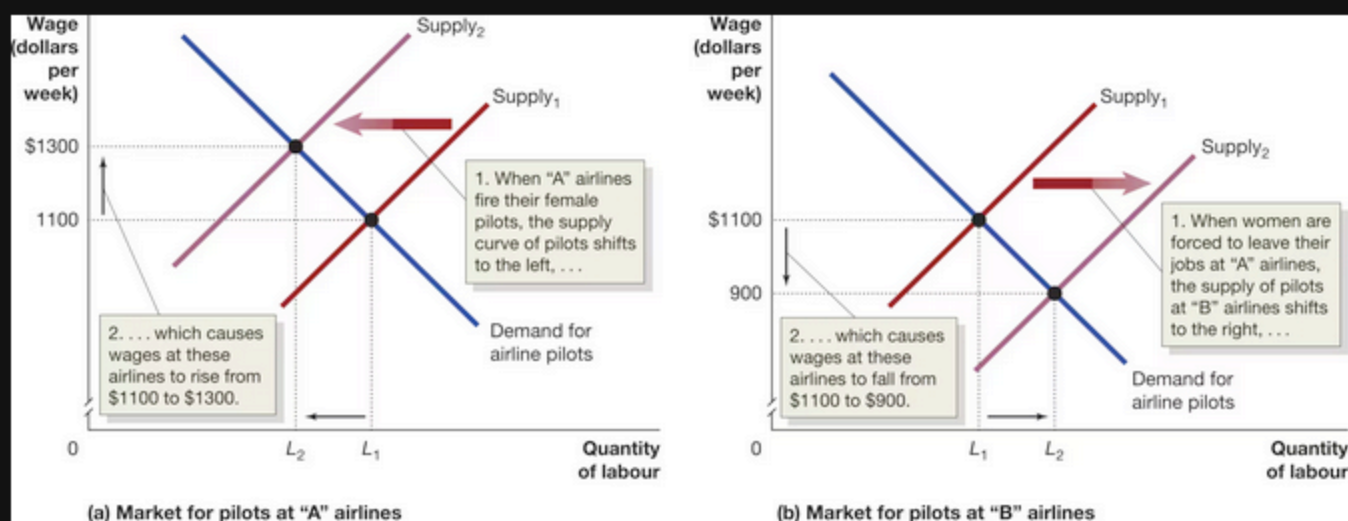
(a) Market for electricians



## (b) Market for dental assistants

**Figure 14.8**

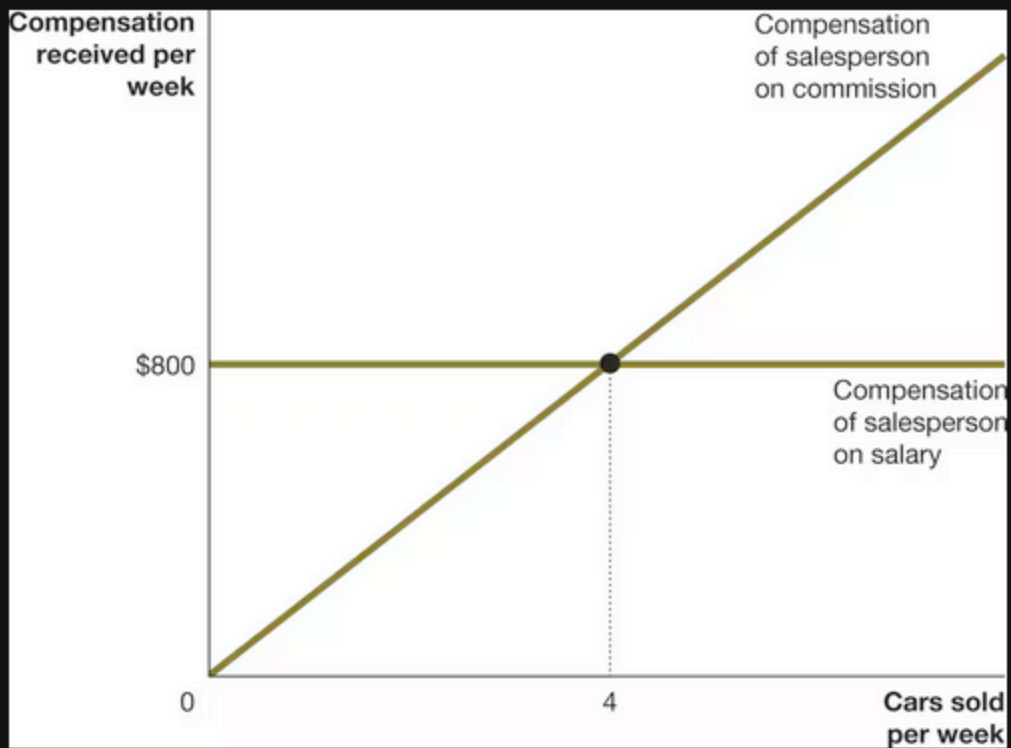
### Discrimination and Wages



In this hypothetical example, we assume that initially neither "A" airlines nor "B" airlines discriminate. As a result, male and female pilots receive the same wage of \$1100 per week at both groups of airlines. We then assume that "A" airlines discriminate by firing all their female pilots. Panel (a) shows that discrimination reduces the supply of pilots to "A" airlines and raises the wage paid by these airlines from \$1100 to \$1300. Panel (b) shows that discrimination increases the supply of pilots to "B" airlines and lowers the wage paid by these airlines from \$1100 to \$900. All the female pilots will end up being employed at the non-discriminating airlines and will be paid a lower wage than the men who are employed by the discriminating airlines.

**Figure 14.9**

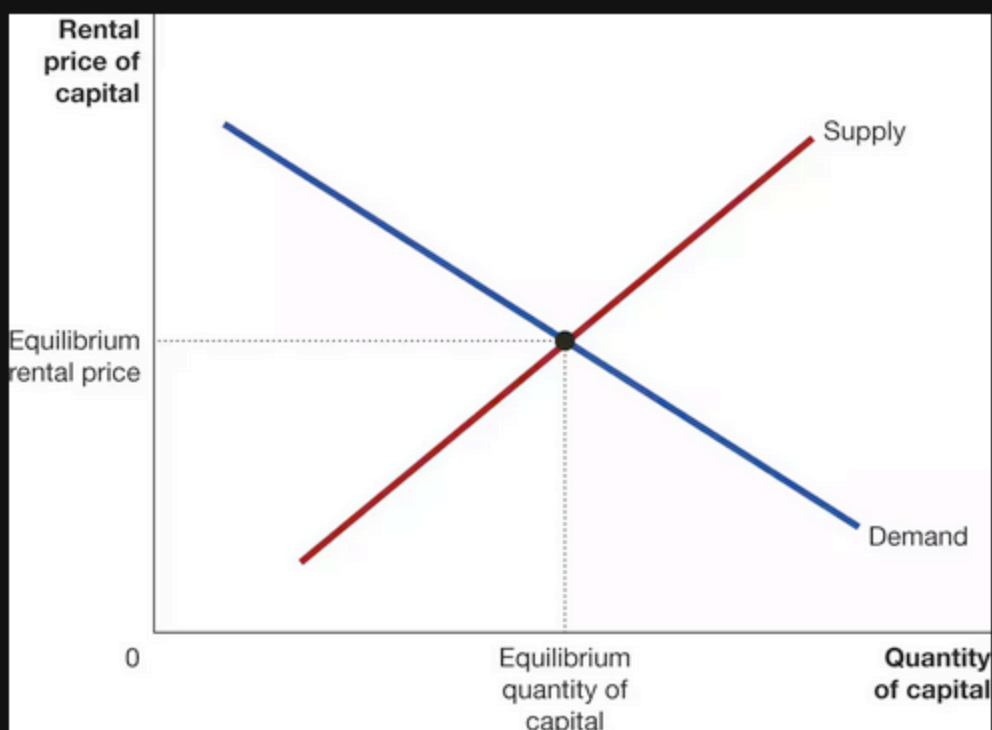
Paying Car Salespeople by Salary or by Commission



This figure compares the compensation a car salesperson receives if they are on a straight salary of \$800 per week and if they receive a commission of \$200 for each car they sell. With a straight salary, they receive \$800 per week, no matter how many cars they sell. This outcome is shown by the horizontal line in the figure. If they receive a commission of \$200 per car, their compensation will increase with every car they sell, as shown by the upward-sloping line. If they sell fewer than 4 cars per week, they would be better off with the \$800 salary. If they sell more than 4 cars per week, they would be better off with the \$200-per-car commission.

**Figure 14.10**

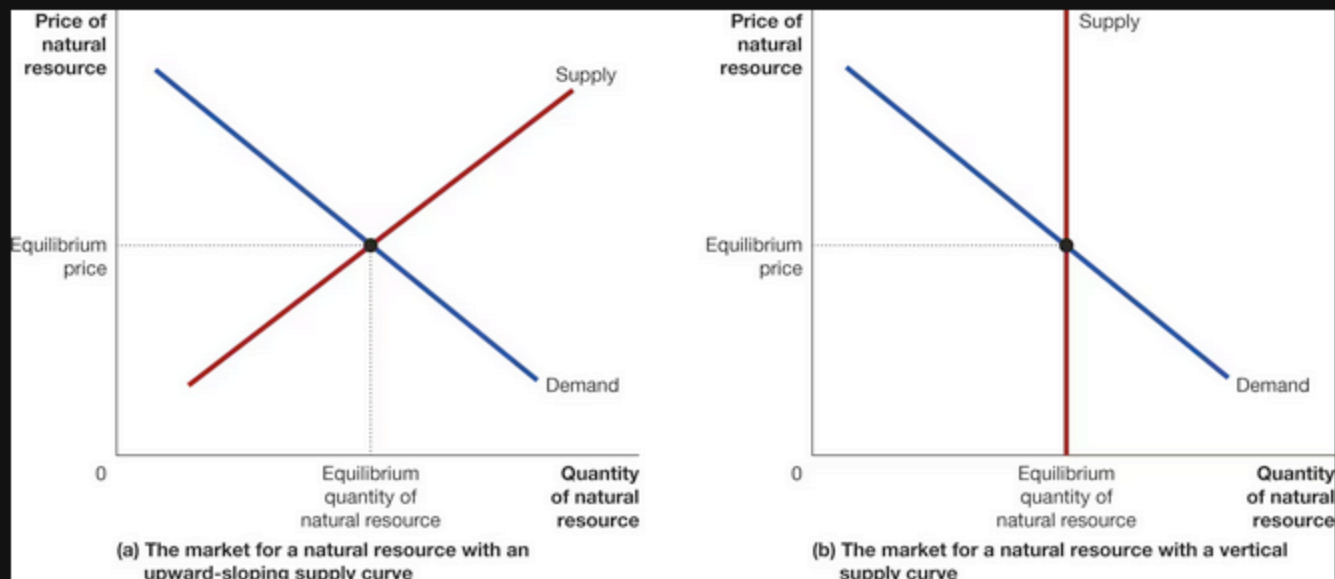
Equilibrium in the Market for Capital



The rental price of capital is determined by demand and supply in the market for capital. In equilibrium, the rental price of capital is equal to the marginal revenue product of capital.

**Figure 14.11**

Equilibrium in the Market for Natural Resources



In panel (a), the supply curve of a natural resource is upward sloping. The price of the natural resource is determined by the interaction of demand and supply. In panel (b), the supply curve of the natural resource is a vertical line, indicating that the quantity supplied does not respond to changes in price. In this case, the price of the natural resource is determined only by demand. The price of a factor of production with a vertical supply curve is called an *economic rent*, or *pure rent*.

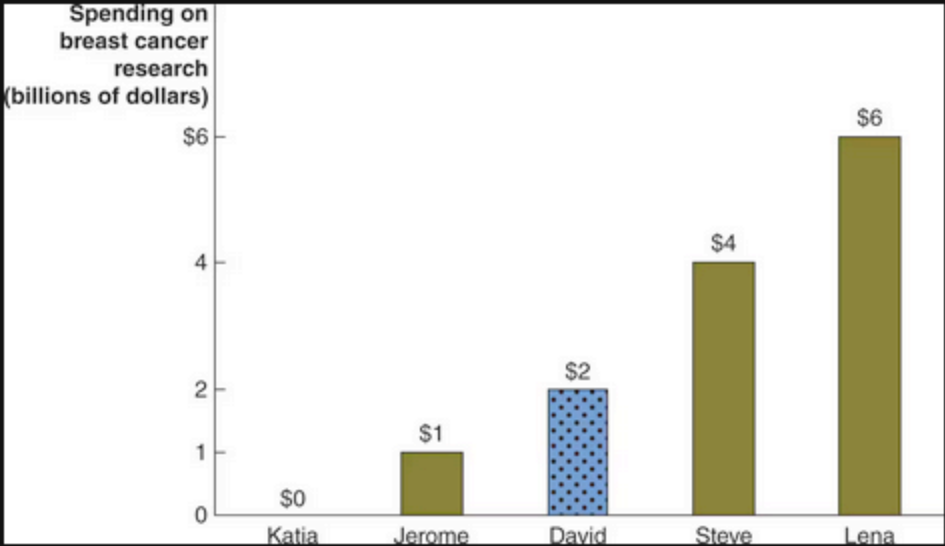
**Table 15.1**

The Voting Paradox

| Policy                                         | Lena | David                        | Katia |
|------------------------------------------------|------|------------------------------|-------|
| Cancer research                                | 1st  | 2nd                          | 3rd   |
| Mass transit                                   | 2nd  | 3rd                          | 1st   |
| Fighting climate change                        | 3rd  | 1st                          | 2nd   |
| Votes                                          |      | Outcome                      |       |
| Cancer research versus mass transit            |      | Cancer research wins         |       |
| Mass transit versus fighting climate change    |      | Mass transit wins            |       |
| Fighting climate change versus cancer research |      | Fighting climate change wins |       |

**Figure 15.1**

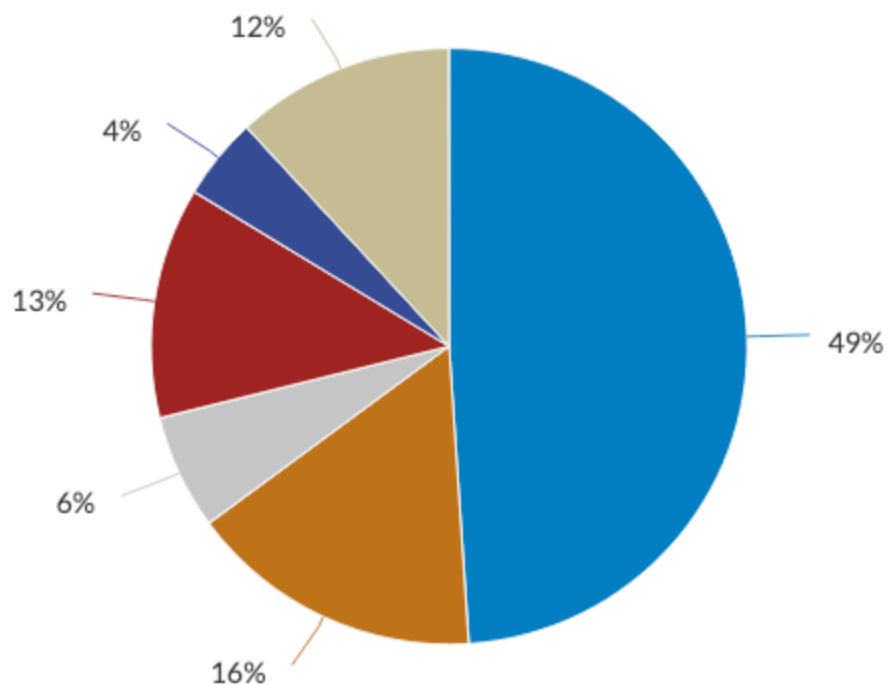
The Median Voter Theorem



The median voter theorem states that the outcome of a majority vote is likely to represent the preferences of the voter who is in the political middle. In this case, David is in the political middle because two voters want to spend more on breast cancer research than he does and two voters want to spend less. In any vote between a proposal to spend \$2 billion and a proposal to spend a different amount, a proposal to spend \$2 billion will win.

Figure 15.2(a) Canadian Government Revenue Sources

Person Income Tax Corporate Income Tax Social Insurance Premiums GST Investment Income Other

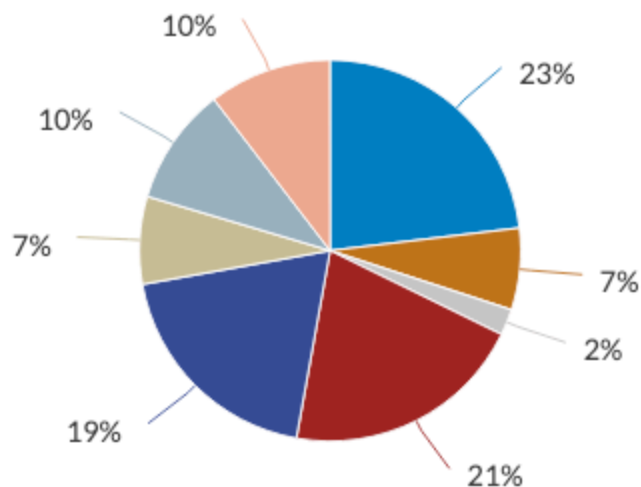


(a) Federal government revenue sources

SOURCE: Data from Statistics Canada, Table 36-10-0477-01. Revenue, expenditure and budgetary balance—General Governments (×1 000 000).

Figure 15.2(b) Canadian Government Revenue Sources

Personal Income Tax   Corporate Income Tax   Social Insurance Premiums   Taxes on Products  
Transfers from Federal Government   Investment Income   Sales of Goods and Services   Other



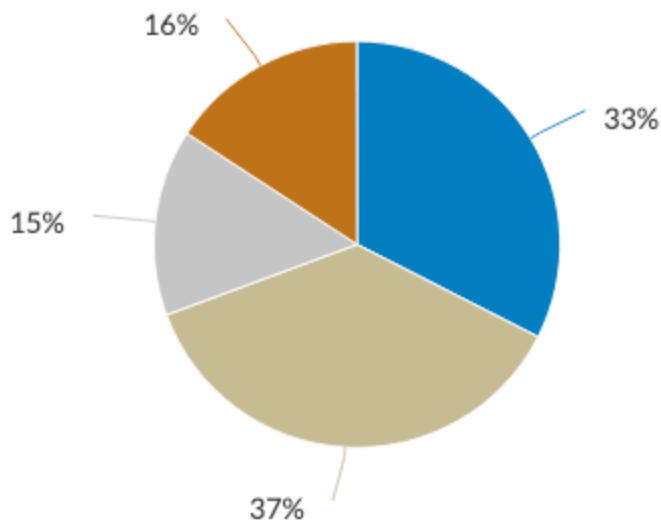
(b) Provincial government revenue sources

SOURCE: Data from Statistics Canada, Table 36-10-0477-01. Revenue, expenditure and budgetary balance—General Governments (×1 000 000).

Social Explorer

Figure 15.2(c) Canadian Government Revenue Sources

Property Tax   Transfers from Provincial Government   Sales of Goods and Services   Other



(c) Local government revenue sources

SOURCE: Data from Statistics Canada, Table 36-10-0477-01. Revenue, expenditure and budgetary balance—General Governments (×1 000 000).

Social Explorer

**Table 15.2**

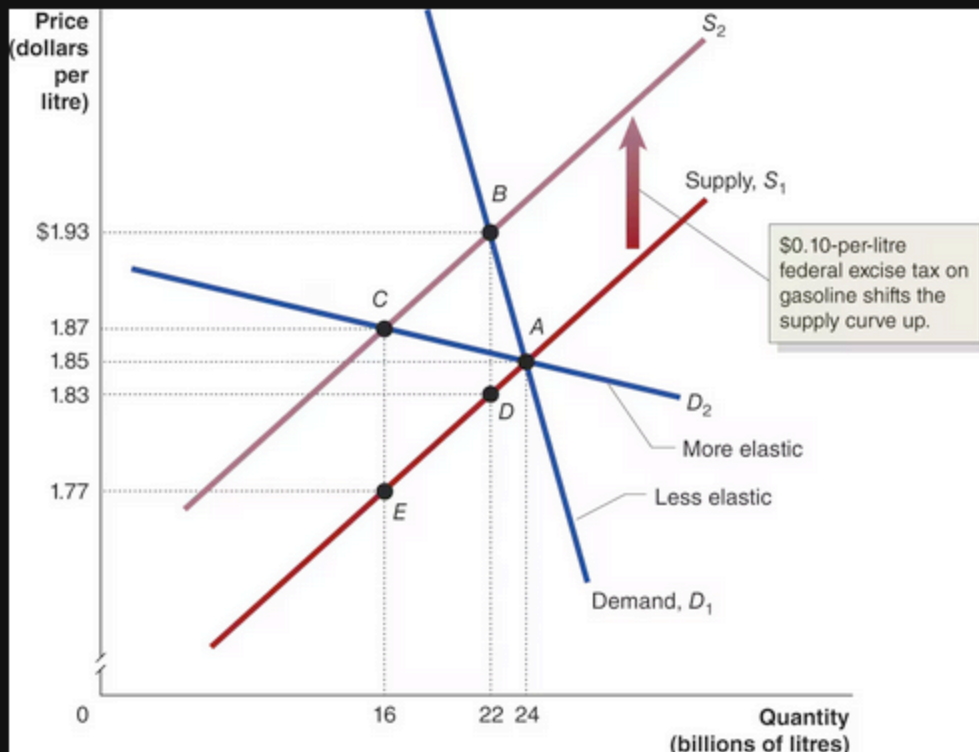
Federal Income Tax Brackets and Tax Rates 2022

| Income (\$)        | Tax Rate (%) |
|--------------------|--------------|
| 0 to 50 197        | 15           |
| 50 197 to 100 392  | 20.5         |
| 100 392 to 155 625 | 26           |
| 155 625 to 221 708 | 29           |
| 221 708 and up     | 33           |

**SOURCE:** Based on Canada Revenue Agency, "Canadian Income Tax Rates for Individuals—Current and Previous Years," <https://www.canada.ca/en/revenue-agency/services/tax/individuals/frequently-asked-questions-individuals/canadian-income-tax-rates-individuals-current-previous-years.html>.

**Figure 15.4**

The Effect of Elasticity on Tax Incidence



When demand is more elastic than supply, consumers bear less of the burden of a tax. When supply is more elastic than demand, firms bear less of the burden of a tax.  $D_1$  is inelastic between point A and point B, and  $D_2$  is elastic between point A and point C. With demand curve  $D_1$ , a \$0.10-per-litre tax raises the equilibrium price from \$1.85 (point A) to \$1.93 (point B), so consumers pay \$0.08 of the tax and firms pay \$0.02. With  $D_2$ , a \$0.10-per-litre tax on gasoline raises the equilibrium price only from \$1.85 (point A) to \$1.93 (point C), so consumers pay \$0.02 of the tax. Because in this case producers receive \$1.77 per litre after paying the tax, their share of the tax is \$0.08 per litre.

**Table 15.5**

2020 Adjusted After-Tax Incomes by Decile

| The upper limit for each decile tells you the maximum after-tax income you could have and still be in that decile. So the lowest income decile includes all those people with incomes up to \$27 700. The share of income indicates the proportion of total income (GDP) people in that decile received after taxes. So the people in the highest decile received 22.0 percent of all the income in Canada after accounting for taxes. |                                |                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-----------------------------|
| Income Decile                                                                                                                                                                                                                                                                                                                                                                                                                          | Upper Income Limit (After Tax) | Share of Income (After Tax) |
| Lowest Decile                                                                                                                                                                                                                                                                                                                                                                                                                          | \$27 200                       | 3.3%                        |
| Second Decile                                                                                                                                                                                                                                                                                                                                                                                                                          | 34 400                         | 5.2                         |
| Third Decile                                                                                                                                                                                                                                                                                                                                                                                                                           | 40 700                         | 6.3                         |
| Fourth Decile                                                                                                                                                                                                                                                                                                                                                                                                                          | 46 800                         | 7.4                         |
| Fifth Decile                                                                                                                                                                                                                                                                                                                                                                                                                           | 53 100                         | 8.4                         |
| Sixth Decile                                                                                                                                                                                                                                                                                                                                                                                                                           | 60 100                         | 9.5                         |
| Seventh Decile                                                                                                                                                                                                                                                                                                                                                                                                                         | 68 300                         | 10.8                        |
| Eighth Decile                                                                                                                                                                                                                                                                                                                                                                                                                          | 78 800                         | 12.4                        |
| Ninth Decile                                                                                                                                                                                                                                                                                                                                                                                                                           | 95 800                         | 14.6                        |
| Highest Decile                                                                                                                                                                                                                                                                                                                                                                                                                         | . . .                          | 22.0                        |

**SOURCE:** Statistics Canada, Table 11-10-0193-01: Upper Income Limit, Income Share, and Average Adjust Market, Total and After Tax Income by Income Decile.

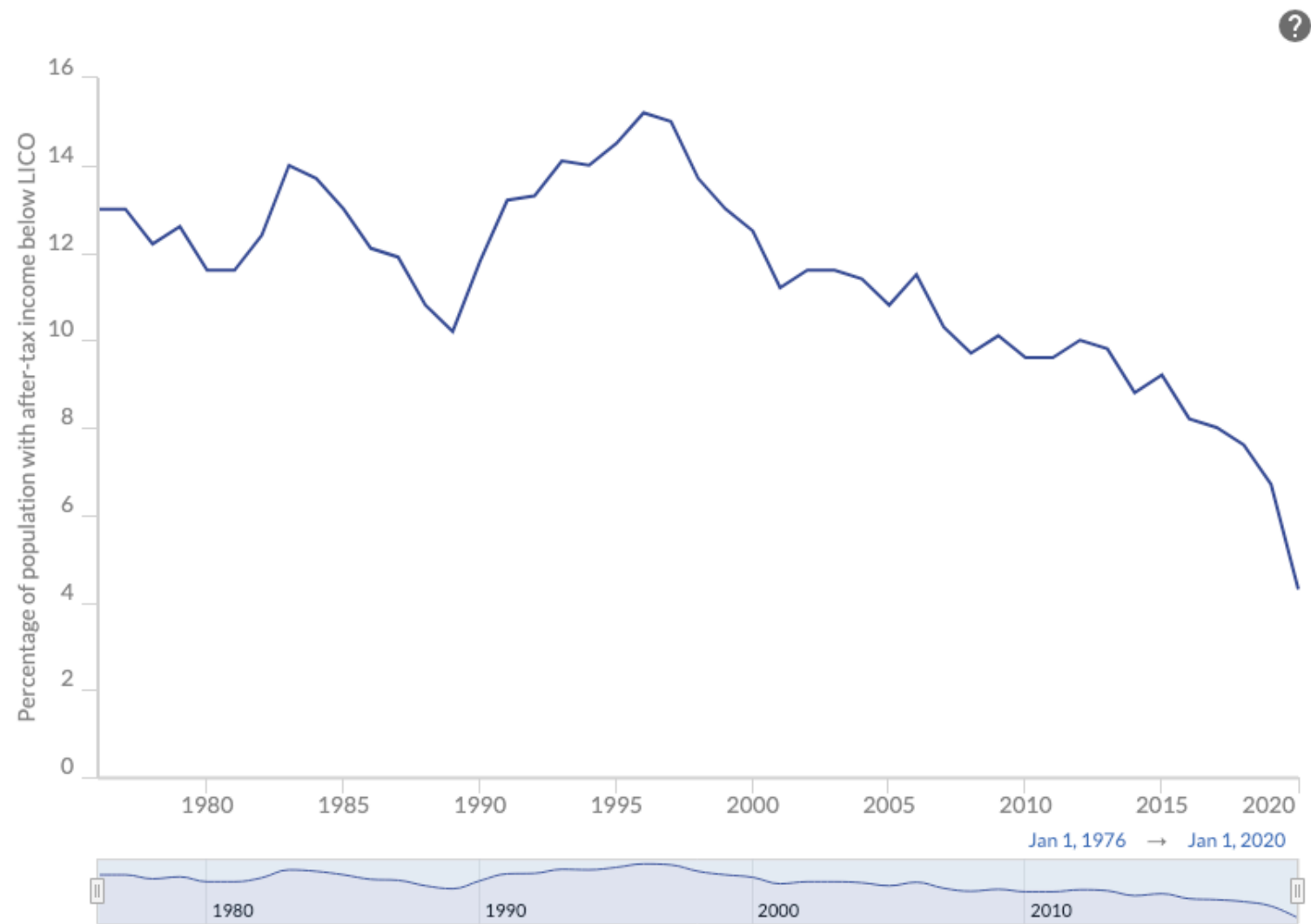
**Table 15.6**

How Has the Distribution of Income Changed over Time? (Percentage of Total Income Received)

| Year | Lowest<br>10% | Second<br>10% | Third<br>10% | Fourth<br>10% | Fifth<br>10% | Sixth<br>10% | Seventh<br>10% | Eighth<br>10% | Ninth<br>10% | Highest<br>10% |
|------|---------------|---------------|--------------|---------------|--------------|--------------|----------------|---------------|--------------|----------------|
| 1977 | 2.5           | 4.9           | 6.4          | 7.6           | 8.7          | 9.9          | 11.2           | 12.8          | 15.1         | 21.0           |
| 1982 | 2.8           | 5.0           | 6.3          | 7.4           | 8.5          | 9.7          | 10.9           | 12.6          | 15.0         | 21.8           |
| 1987 | 3.0           | 5.0           | 6.3          | 7.4           | 8.5          | 9.6          | 10.9           | 12.6          | 14.9         | 21.8           |
| 1992 | 2.7           | 5.0           | 6.2          | 7.4           | 8.5          | 9.6          | 11.0           | 12.6          | 15.0         | 22.0           |
| 1997 | 2.6           | 4.8           | 6.0          | 7.2           | 8.4          | 9.6          | 10.9           | 12.6          | 15.1         | 22.8           |
| 2002 | 2.6           | 4.6           | 5.8          | 7.0           | 8.2          | 9.4          | 10.8           | 12.5          | 15.0         | 24.1           |
| 2007 | 2.7           | 4.6           | 5.9          | 7.0           | 8.2          | 9.4          | 10.8           | 12.5          | 15.1         | 23.8           |
| 2012 | 2.6           | 4.6           | 5.9          | 7.0           | 8.2          | 9.5          | 10.9           | 12.6          | 15.0         | 23.7           |
| 2017 | 2.7           | 4.7           | 5.9          | 7.1           | 8.3          | 9.5          | 10.8           | 12.6          | 15.0         | 23.3           |
| 2020 | 3.3           | 5.2           | 6.3          | 7.4           | 8.4          | 9.5          | 10.8           | 12.4          | 14.6         | 22.0           |

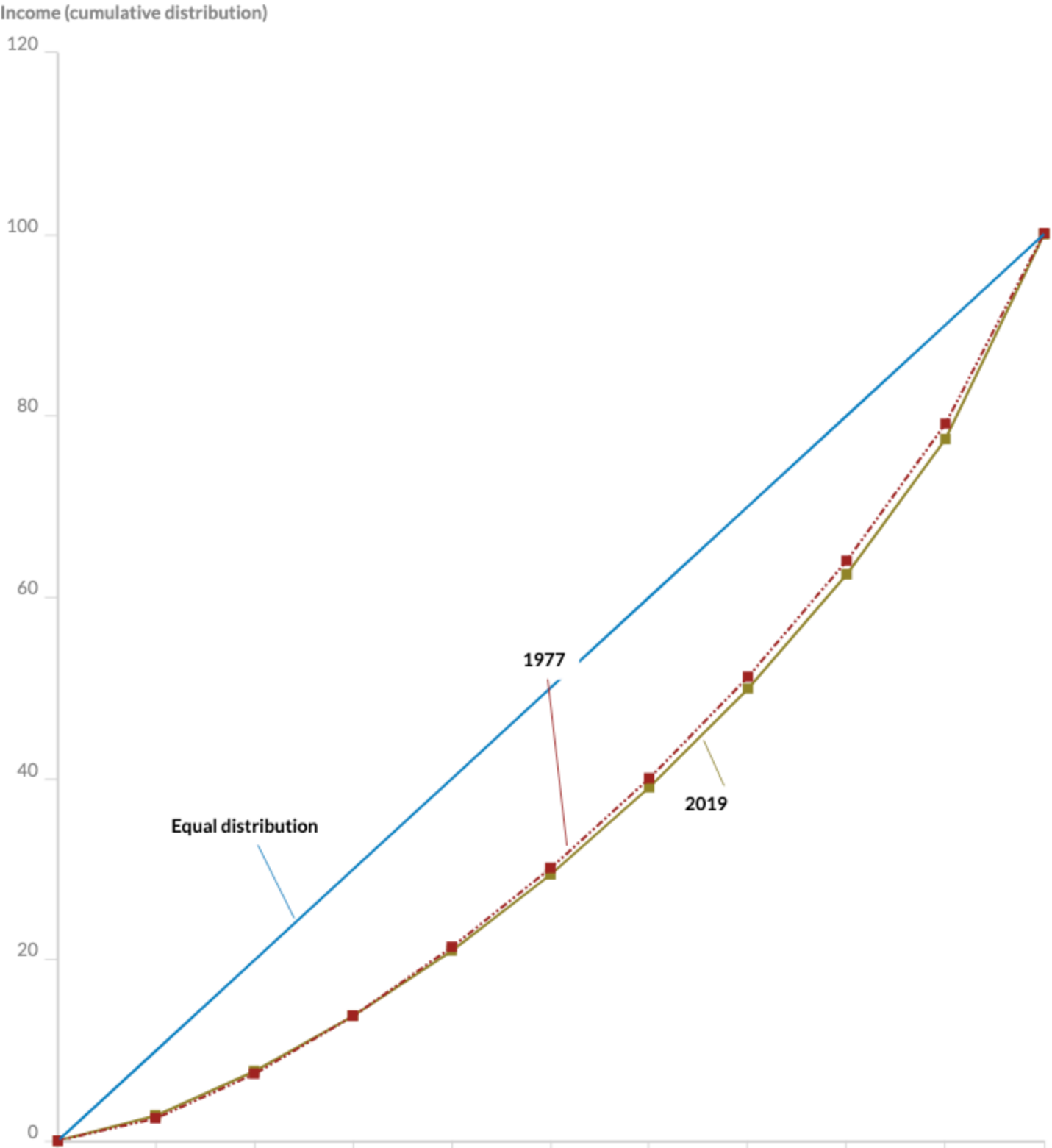
**SOURCE:** Data from Statistics Canada, Table 11-10-0135-01: Low Income Statistics by Age, Sex and Economic Family Type.

Figure 15.5 Percentage of the Canadian Population with Income Below the Low Income Cut-Off



SOURCE: Statistics Canada, Table 11-10-0135-01: Low Income Statistics by Age, Sex, and Economic Family Type.

Figure 15.6 The Lorenz Curve and Gini Coefficient

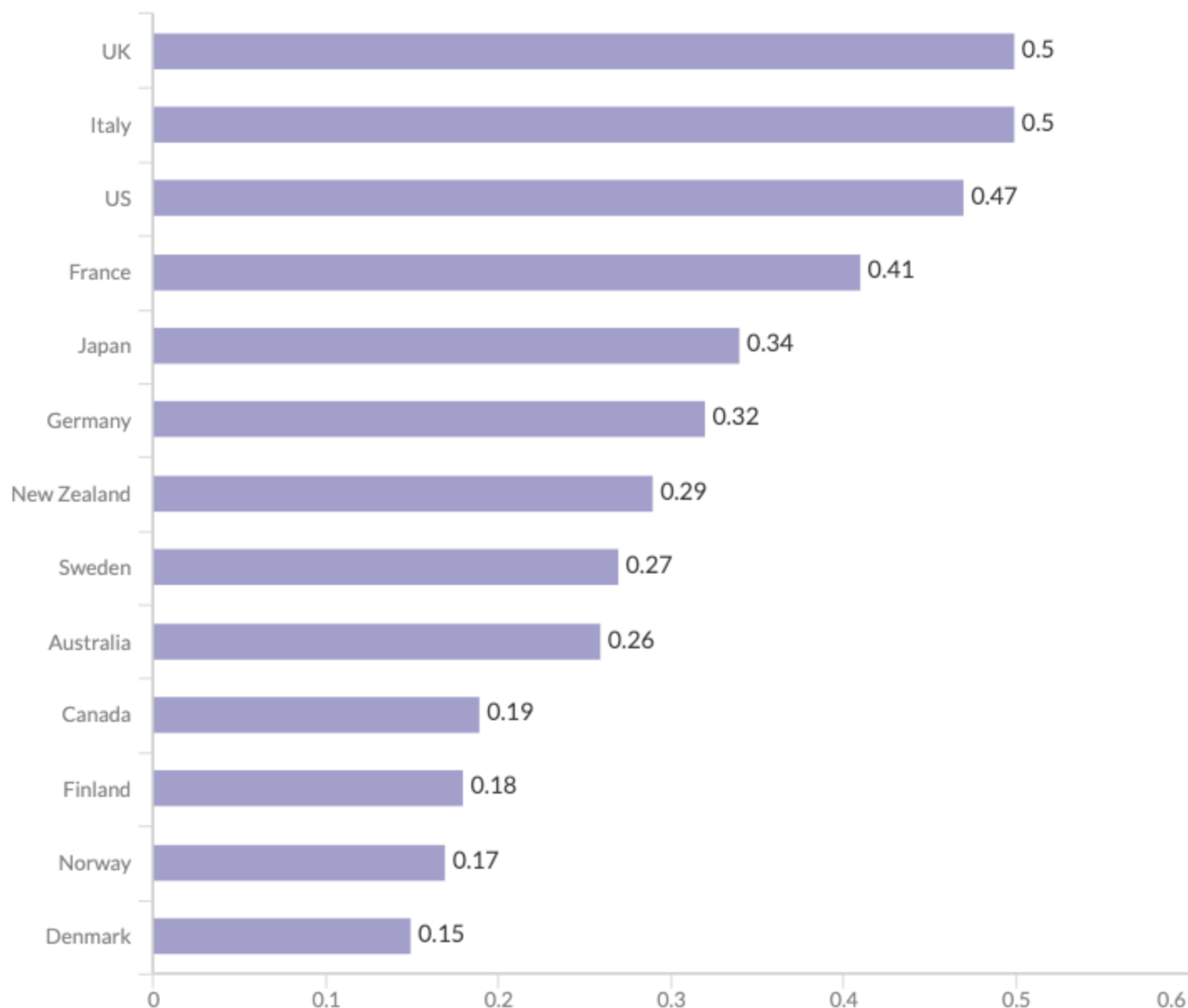


Panel (b) illustrates how to calculate the *Gini coefficient*, which is one way of summarizing the information provided by a Lorenz curve. The Gini coefficient is equal to the area between the line of perfect income equality and the Lorenz curve—area *A* in panel (b)—divided by the whole area below the line of perfect equality—area *A* plus area *B* in panel (b). Or

$$\text{Gini coefficient} = \left( \frac{A}{A + B} \right)$$

If the income distribution were completely *equal*, the Lorenz curve would be the same as the line of perfect income equality, area *A* would be zero, and the Gini coefficient would be zero. If the income distribution were completely *unequal*, area *B* would be zero, and the Gini coefficient would equal 1. Therefore, the greater the degree of income inequality, the greater the value of the Gini coefficient. In 1977, the Gini coefficient for Canada was 0.286. In 2019, it was 0.299, which tells us again that income inequality increased between 1977 and 2019.

Figure 15.7 Intergenerational Income Elasticity in Wealthy Countries



SOURCE: Based on Miles Corak, "Inequality from Generation to Generation: The United States in Comparison."